0. Executive Summary

Stationary battery storage has emerged at large scale as an infrastructural object class. Accelerated variable generation, growing grid constraints, intensifying connection competition, the differentiation of marketing and system-service roles, and digitally guided operating logics together shift the position of the storage asset within the energy system. Under these conditions a large-scale Battery Energy Storage System (BESS) appears as a coupled infrastructure object in which electrical, thermal, control, safety, data, contractual, and commercial layers carry it together. Its quality emerges from the way these layers interact across the real project and operating context.

In the German context this shift is particularly sharp. The stationary storage market reached around 24 GWh of installed capacity at the beginning of 2026, with an annual addition of 6.57 GWh; large-scale storage forms the most dynamic class within that volume and is approaching the effect of the established pumped-storage capacity. At the same time access to the transmission system has become a scarce infrastructure resource. By the third quarter of 2025, the four German transmission system operators were holding connection requests for battery storage of around 211 GW; reserved connection capacities had already reached 51 GW, a magnitude approaching the entire scenario framework of the 2025 Network Development Plan for the year 2037. Under these conditions the quality question of a storage project begins early — at project maturity, technical plausibility, site control, connection concept, and systemic integration.

Against this background, the compendium treats battery storage systems as real technical and operational systems. Its subject reaches from electrical base architecture and connection-side integration, through control, protection, and state management, to evidence, market and grid integration, economic operation, safety, degradation, and structural further development. It covers the system insofar as it becomes effective for the technical, operational, and economic quality of the asset — beyond cell module, container, and grid connection point.

The guiding question is under what technical, project-related, operational, safety-related, evidential, and lifecycle conditions a battery storage system can hold its place over time as a robust infrastructure asset. The concept of system controllability carries this question. System controllability denotes the capacity to keep a BESS as an infrastructure object in legible states across its lifetime — technically, operationally, and institutionally — through active stewardship of the permitted state space, coordination of the technical layers, consistent maintenance of evidence, governance of the contractual interfaces, and preservation of connectivity to changing regulatory and market regimes. In this reading the asset remains technically controllable, operationally robust, demonstrably safe, and economically viable to the extent that its layers act coherently together. The European battery regulatory framework adds a further dimension to that controllability: material origin, carbon footprint, performance, lifetime, recyclability, and later return become technical, commercial, and at the same time regulatorily legible properties of the asset's identity.

The compendium addresses a professional readership in development, engineering, EPCM, operation, asset management, due diligence, and infrastructure strategy. It assumes working knowledge of electrical infrastructure, project realisation, grid- and market-near operating logic, and technical system control. The work draws on more than 25 years of experience in complex energy infrastructures, on direct involvement in BESS-near development and planning contexts in Germany and Spain, and on practical familiarity with EPCM, permitting, and operating logics of large energy installations in the German and European setting. It follows the movement from the early project state to the long-term quality of the asset.

The work reflects the published and accessible state of technical, regulatory, and market developments up to April 2026. It expresses a personal professional view of the author and does not speak for any company, association, or institutional position. Subsequent shifts and divergent institutional or professional readings remain outside the frame of reference.

Part I — System object, positioning, and transformation space

Chapter 1 — The BESS as infrastructure object

1.1 Shift and positioning

Large-scale battery storage systems have left the status of a supplementary technology. A structural shift has taken hold within the object class over recent years that reaches beyond mere maturation. The BESS now appears as an infrastructure object, with its own system boundary, its own operating logic, and its own lifecycle dynamic. Behind this shift stands a real change in the storage asset's position within the energy system. Accelerated growth of variable generation, increasing grid constraints, intensifying connection competition, the withdrawal of conventionally controllable capacity, and the differentiation of market- and grid-near roles together change the function of the storage within the system. To these are added digitally guided operating logics, in which the asset is coordinated in real time across multiple services, markets, and constraints. The storage thus mediates between time, power, energy, grid, operation, and risk — beyond the simple execution of charging and discharging.

Under these conditions the description of a BESS reaches further than installed megawatts and megawatt-hours, and further than arbitrage revenues and balancing-reserve products. Its technical and economic quality shows itself in whether it can hold its place across the duty period under shifting electrical, thermal, market, and regulatory conditions. That is where the infrastructural standing of the subject lies.

1.2 The relevant system body

The expression system body designates the totality of those layers in which technical capability, safety behaviour, operating logic, evidential structure, and economic operation condition one another reciprocally. It serves as an analytical working form for an object whose quality only becomes intelligible when its physical, electrical, control-related, safety-related, data-related, and institutional layers are considered together.

1) *Physical and civil-engineering layer*. Site, footprint, building or container logic, foundations, accessibility, lifting and replacement paths, drainage, fire compartments, separation distances, and route planning belong here. This level reaches into operation, because it determines maintenance reality, segmentation, incident control, and later modification capability. A large-scale BESS is a fixed-location infrastructure object with its own spatial and intervention logic.
2) *Primary electrical architecture*. Beginning with cell chemistry, module and rack composition, string architecture, DC busbars, and isolation concepts, this layer extends through PCS, medium-voltage connection, transformers, protection devices, and auxiliaries to the grid-side interface. Voltage windows, current paths, conversion losses, redundancy, fault selectivity, reactive-power capability, and grid-forming or grid-following properties are decisive. At large scale this layer extends into the real connection body: substation, control technology, protection coordination, and routing reality form part of the effective system shape.
3) *Electrochemical and material layer*. Cell performance, ageing, and safety follow from material, particle morphology, interfaces, conductivities, reaction kinetics, and thermal balance, and from the question of how these properties remain processable in scaled cell and module architectures. The material layer forms part of the system identity of the later asset; it determines performance limits, safety logic, industrial availability, and later lifecycle options.
4) *Control and state-management layer*. BMS, EMS, SCADA, dispatch logic, set-point management, operating modes, reserve windows, and the coordination of competing services act together here. The asset becomes temporally legible at this layer: SOC, SOH, charge and discharge depth, C-rate, temperature window, response speed, and available power appear as operational variables of a regulated state economy. This layer also encompasses the control-engineering properties that shape the asset's behaviour in weak grids, under voltage disturbances, frequency excursions, or phase steps.
5) *Safety and fault-containment layer*. Thermal management, detection, segmentation, shutdown, gas management, propagation containment, alarm signalling, access restriction, restart, and intervention rules belong here. In large-scale storage this layer is a permanent property of the system across normal, fault, and degradation states. It shapes the structural ordering of the asset, the operational release, and the intervention possibilities in event cases.
6) *Data, evidence, and demonstration layer*. Measurement systems, event histories, test protocols, performance indicators, commissioning data, parameterisation histories, warranty records, cycle histories, SOH estimates, and the capacity to record changes to the asset consistently belong here. This layer carries the technical memory of the object. A BESS without continuity of evidence loses legibility, with growing duration, towards investors, warranty partners, regulators, and its own operating team.
7) *Organisational and commercial layer*. Ownership and operator roles, operating responsibility, market access, balance-group capability, aggregator and trader interfaces, securities, insurability, creditworthiness, PPA or tolling structures, warranty boundaries, and the logic of later reconfiguration belong here. This layer determines what technical capability can be translated into monetizable or contractually robust system performance.

The relevant system body, shown in Figure 1, extends beyond the cell module, the container, and the grid connection point. It encompasses the system insofar as it becomes effective for planning, operation, demonstration, and economic viability of the asset. The quality of a BESS arises from the coherence of these layers across the real project and operating context, not from the strength of individual elements.

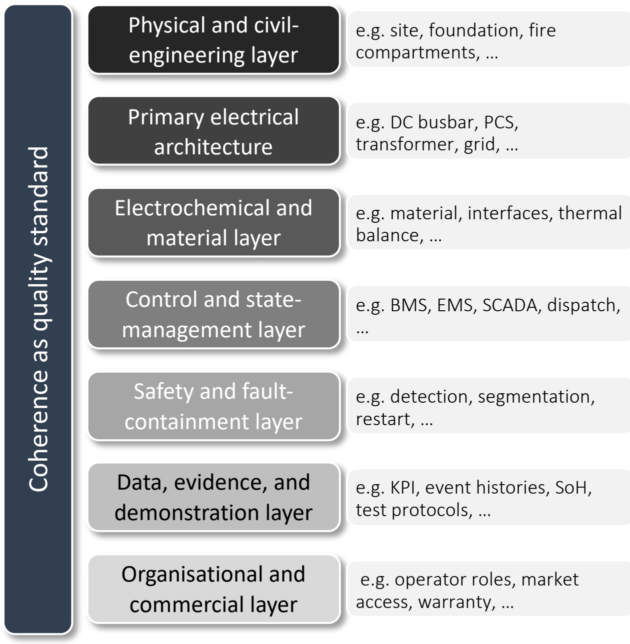

Figure 1. The system body of the BESS as a seven-layer coherence structure. The asset's quality arises from the interaction of the layers, not from the isolated strength of individual components.

1.3 Leading category and assessment standard

The guiding question of this compendium is under what technical, project-related, operational, safety-related, evidential, and lifecycle conditions a battery storage system can hold its place across planning, connection, procurement, commissioning, operation, degradation, expansion, hybridisation, and later return as a robust infrastructure object. What matters is whether the system in its totality remains technically controllable, operationally robust, demonstrably safe, connection-side viable, and economically sustainable across changing project and operating conditions.

The concept of system controllability carries this question as its leading category. It denotes the capacity to keep a BESS as an infrastructure object in legible states across its lifetime — technically, operationally, and institutionally — through active stewardship of the permitted state space, coordination of the technical layers, consistent maintenance of evidence, governance of the contractual interfaces, and preservation of connectivity to changing regulatory and market regimes. The assessment logic begins earlier than many project narratives suppose. Site, connection, site control, technical plausibility, and procurement logic already shape later system behaviour. It also reaches deeper into material and production logics than purely operational accounts capture, and it ends later than a commissioning protocol might suggest. A BESS reveals its real quality in the temporal interplay of dispatch, degradation, evidence continuity — the unbroken inscription of its technical history — retrofit, augmentation, and return.

What matters is the coherence of the system across its layers. Electrical architecture, connection capability, operating-state logic, safety regime, data and evidence structure, market and grid integration, degradation management, and structural adaptability must act together so that technical possibility becomes a robust operating object. Where this coherence is absent, real investment and operating risk emerges.

Chapter 2 — The German transformation space

2.1 Connection scarcity and the maturity-tier process

The German storage market forms an analytically informative transformation space because several developments converge in unusual density: a very high build-out of variable feed-in, a growing demand for short-term and intraday-capable flexibility, rising connection competition in transmission and distribution networks, a regulatory reorganisation of grid access and grid charges, and shifting technical expectations of large-scale batteries in grid operation. The empirical ground for this observation is supplied by the Market Master Data Register (Marktstammdatenregister, MaStR) of the Federal Network Agency (Bundesnetzagentur, BNetzA), in which all stationary battery storage systems have been subject to mandatory reporting since January 2019; the monthly evaluations published by RWTH Aachen as Battery Charts draw on this register and provide a methodologically transparent market reading. On that basis, the German stationary storage market reached around 24 GWh of installed capacity at the beginning of 2026, with annual additions of 6.57 GWh in 2025 — roughly eight per cent above the previous year. The conventional segmentation distinguishes home storage systems (HSS, up to 30 kWh), industrial and commercial storage systems (ISS, 30 to 1,000 kWh), and large-scale storage systems (LSS, above 1,000 kWh). Home storage holds around 80 per cent of the capacity but is declining in annual additions; commercial storage grows steadily; large-scale storage forms with just under one-sixth of the installed capacity the second-largest class and shows the most dynamic development — in 2025, 83 new registrations added 667 MW and 1,244 MWh, while the reported project pipeline of 460 schemes and a combined 4.9 GW and 10.13 GWh substantially exceeds the previously installed LSS volume.

The position of the BESS within the energy system arises from the interaction of market value, grid restriction, connection regime, system requirement, and industrial availability acting simultaneously on the project. Anyone developing, financing, and operating a storage project in Germany meets these layers simultaneously and under growing time pressure. The market compresses the conditions under which a modular-looking technology becomes a complex infrastructure object and is for that reason well suited as an analytical reference space.

The deepest change in the German BESS market currently lies in the connection question. By the third quarter of 2025, the four German transmission system operators were holding cumulative connection requests for battery storage of 211 GW. In 2022, the figure stood at 3 GW. The number of applications rose over the same period from 8 to 545. Connection capacities already reserved had reached 51 GW, a magnitude approaching the entire scenario framework of the 2025 Network Development Plan (Netzentwicklungsplan, NEP) for the year 2037. These figures mark the conversion of grid connection into a scarce infrastructure resource.

Under these conditions the first-come-first-served principle lost its capacity to allocate. It privileged early and often speculative applications, tied up scarce switchgear and assessment capacity, and exposed delivery-mature projects to the same queue logic as immature schemes. With the Cabinet decision of 10 December 2025 and the maturity-tier process (Reifegradverfahren) of the four German transmission system operators, which entered into force on 1 April 2026, the situation was institutionally reordered. Decisive criteria are site control and permitting status, the technical concept for plant and connection, the capability of the applicant, and grid- and system-related contribution — the latter in the sense of the conceptual pair developed in 2.3, between grid-supportive and system-supportive contribution. To these are added an application fee of €50,000 per application and a realisation deposit of €1,500 per megawatt.

For the logic of this compendium, the consequence of the process is more telling than its mechanism. A storage project decides itself at project maturity, site control, technical concept, procurement path, and connection readiness — long before construction and commissioning. The maturity-tier process makes visible what has long held in project terms: a BESS becomes a robust scheme now as it can credibly cross the real entry threshold into the grid infrastructure.

2.2 Shifting technical system expectations

In parallel with the regulatory compression, the technical expectation of large-scale batteries is shifting. With the growing share of converter-coupled feed-in — wind, photovoltaic, and battery installations whose grid coupling runs through power electronics — and with declining systemic inertia from rotating masses, properties gain importance that were treated as marginal in earlier storage texts: voltage support, reactive-power capability, behaviour in weak grids, response to frequency deviations and phase steps, and stability in interaction with other converters.

The system role of the BESS is enlarged by this shift. Large-scale batteries appear increasingly as grid-qualified installations with effects on voltage, frequency, and load-flow regimes, beyond the function of a flexibility resource for arbitrage and control reserve. Properties such as grid forming — the capability of a converter installation to set its own voltage and frequency reference, where a grid-following converter would synchronise to the existing grid voltage (developed in 6.2) — reactive-power provision, and robust behaviour in weak grids belong in the early design decisions and shape the asset's later system role. PCS architecture, control philosophy, ramp behaviour, and connection concept already determine in the planning phase what grid-technical qualification the asset will later be able to sustain.

The technical expectation thus shifts from the question whether a storage system can supply power and energy to the question under what grid conditions it can supply this power in systemically qualified form. System-supportive capability belongs to the architecture; it should not be treated as a downstream operational addition.

2.3 Regulatory intermediate position, system-supportive and grid-supportive contribution

Battery storage occupies an intermediate position within the German regulatory framework. It occupies an intermediate position between the established categories — generator and consumer, grid asset and market object — without falling fully into any of them. This intermediate position opens new functions and at the same time obstructs clean assignment of grid charges, connection contributions, support logic, and the regulatory recognition of systemic services. The practical effect is considerable: a storage system can be deployed in a grid-supportive way and still pay full connection charges; it can deliver redispatch performance without being regulatorily relieved. With the Federal Court of Justice decision of 15 July 2025 (Bundesgerichtshof, BGH; case reference EnVR 1/24), which broadly upheld the calculation of a connection charge for battery storage according to the capacity-price model, this intermediate position was further sharpened. For projects with tightly calculated arbitrage or aFRR revenues, this is a central economic factor.

For planning and assessment, the consequence is that regulatory uncertainty is part of the design problem itself and must not be treated as an external disturbance to be considered at the end of a project. Connection model, deployment regime, revenue assumptions, data obligations, and capacity logic must from the outset be framed under an institutionally mobile horizon. A BESS whose economic viability holds only under a narrow regulatory scenario remains structurally fragile.

Within this frame two terms become load-bearing that are often used loosely in the German discussion. System-supportive contribution designates the positive contribution of a BESS to the function and cost structure of the electricity system as a whole; it covers market-side, grid-side, and system-stabilising effects — price moderation in scarcity hours, reduction of redispatch effort, congestion relief, voltage support, the provision of grid-forming properties, the substitution of instantaneous reserve, and participation in grid restoration. Grid-supportive contribution is more narrowly bounded; it refers to those effects that arise physically in the grid — grid costs, congestion, voltage maintenance, and grid-related system services — and that are only incompletely internalised through price signals.

What matters analytically is that the two categories cannot be cleanly separated along a single boundary. The same dispatch decision can be commercially rational and grid-loading; a grid-relieving behaviour can remain economically under-incentivised where remuneration and charging regimes do not reflect system value. The BESS acts in the electricity system in market and grid simultaneously, and this double effect makes its assessment demanding. The two terms mark a foundational tension of the later operational and assessment logic. System-supportive contribution arises from the interplay of technical behaviour, deployment logic, and institutional frame; market success on its own does not establish it. Grid-supportive contribution likewise arises beyond the physical connection itself, in the actual grid effect of the operating mode. What matters is how technical behaviour, deployment logic, and institutional frame act together in the concrete case.

Part II — Project maturity and technical conception

Chapter 3 — Grid connection as a scarce infrastructure resource

3.1 The allocation question and the mechanics of scarcity

The grid connection of large-scale battery storage has lost the character of a downstream technical compliance check. It has become an allocation question. This shift follows from the simultaneity of several developments: rapid build-out of variable generation, a rising number of large-scale connection requests from storage, data-centre, electrolysis, and electrification projects, the physical limitation of available switchgear bays and substations, and the fact that real grid loading arises from spatially and temporally compressed load and feed-in situations, while balance-sheet annual energy quantities do not capture the loading.

Grid connection therefore belongs to the constitution of the object itself. A storage project can appear economically plausible, technologically available, and electrochemically controllable, and remain infrastructurally underdeveloped where site, interconnection, switchgear-bay availability, substation, protection concept, routing, and technical demonstrations are not brought together in robust form.

Within this frame, *Anschlussfähigkeit* designates more than the simple electrical interconnectability of a project. The German term carries three intertwined meanings, which the English text sustains as a triadic family. Connection capability in the technical-infrastructural sense designates the system's electrical and physical fitness for grid coupling — voltage compatibility, switchgear-bay availability, protection-coordination compatibility, and routing reality. Connection readiness designates the project-side maturity required for entry into the maturity-tier process — site control, permitting status, technical concept, applicant capability, procurement substance, and the alignment of these elements into a credible delivery path. Connectivity designates the lifecycle-institutional capacity of the asset to remain coupled to changing regulatory, market, and grid regimes across its operating life. The three readings act as a single conceptual hand; a project becomes *anschlussfähig* in the full sense only when all three are met. In the body text the variants appear as the context disambiguates.

The scarcity within the German transmission system is a physically bound infrastructure restriction. Its bottleneck dimension lies in the limited number of free switchgear bays in substations, less in the available aggregate capacity. Switchgear bays are not divisible. They cannot be allocated proportionally, nor retrofitted in arbitrary number on short notice. Once reserved, the capacity is bound for long periods and is unavailable to other projects during that time.

The scale of this bottleneck can be named concretely. At 50Hertz alone, by the third quarter of 2025, around 65 GW of connection requests for battery storage were on file. Across Germany, requests summed to 211 GW, while reserved capacities had reached 51 GW. At the same time, the project-side resources of the transmission system operators are themselves limited: grid studies, assessment processes, allocation procedures, and internal engineering capacity do not scale linearly with the number of incoming applications. To these are added supply bottlenecks for switchgear, transformers, and protection equipment. Scarcity arises in the physical grid, in the assessment capacity, and in the industrial delivery chain simultaneously.

The maturity-tier process intervenes at this point. It bundles applications into structured procedural rounds, allows clustering by site and grid connection point, and makes assessment effort plannable. More telling than the procedural form is its substantive logic: a grid connection point acts as a scarce infrastructural entry resource that requires substantive demonstration of project capacity. A claimant must demonstrate that the proposal can become a real project.

3.2 The maturity-tier process as a quality grid

The maturity-tier process makes explicit what substance a storage project must have developed before formal application. The four assessment criteria — site control and permitting status, the technical concept for plant and connection, the capability of the applicant, and grid- and system-related contribution — together form an early-acting quality grid that reaches deep into the pre-development phase.

Site control demands as a minimum entry threshold a legally robust exclusivity agreement with the landowner, valid beyond the entire procedural period. More advanced projects demonstrate notarised preliminary purchase contracts or already-existing leases. The technical plant and connection concept must be available in the form of a complete project plan, comprising system descriptions of the main components, a site layout, a project schedule through commissioning, and — as a minimum requirement of assessment tier B — a completed VDE application form, a quarter-by-quarter power ramp-up forecast, and a robust draft single-line diagram. Higher maturity-tier scores require complete plant drawings of the primary technology and a completed planning of the secondary technology. The routing of the connection cable forms an independent assessment dimension; from the simple routing corridor with spatial-resistance analysis to the complete in-rem securing of the route through land-registry entries and present plan-approval decisions, a four-tier assessment scale awards maturity-tier points for each demonstrated step. The capability of the applicant is assessed via creditworthiness and corporate substance, evidence of orders for time-critical components of the connection technology, and proof of financing through to full project security. The realisation deposit of €1,500 per megawatt sets an immediate financial seriousness threshold.

The fourth category, grid- and system-related contribution, contains in the current procedural version only the criterion of project hybridisation — that is, the co-location of generation, load, and storage at a common connection point. Further aspects of grid-supportive contribution — black-start capability, reactive-power provision, instantaneous reserve — are explicitly named as development needs for the process but could not yet be anchored as allocation criteria within the current legal frame. Their integration into later procedural versions is foreseeable and strategically relevant for systemically effective storage projects.

3.3 Connection body and system-supportive integration as design variables

With growing power class and rising voltage level, the real connection body of the BESS emerges as an independent technical object. It comprises the substation, the medium- or high-voltage interconnection, protection technology, primary-technology layout, secondary technology, control-system interconnection, metering concept, and routing logic. The asset appears at this point as an infrastructural composite system meeting real system conditions, no longer as a largely self-contained container solution. The single-line diagram, protection philosophy, metering and control technology, the GIS or AIS decision, the steady-state operating logic, and the primary-technology layout determine the asset's later operating path. Under constrained spatial conditions, in industrialised settings, or where requirements for availability and weather robustness are high, the choice of switchgear can already become architecturally decisive. The more strongly the grid is shaped by power electronics, the more deeply the connection question reaches into the later operating space; voltage support, reactive-power capability, behaviour in weak grids, response to frequency deviations and phase steps, and stability in interaction with other converters therefore belong to the early conception.

System-supportive integration takes this logic up at the site level. A large-scale battery storage system unfolds its value in dependence on grid location, duration, power, control capability, and interaction with local load and generation profiles. The fit between location, connection body, deployment regime, and technical system role determines whether technical capability becomes systemic added value. From the grid perspective, system-supportive integration means participation in the optimisation of system costs across site and deployment decisions — including generation costs, congestion management, other system-service costs, and local grid effects. A perfect site does not exist; a storage system developed exclusively under the lens of local economics may, despite positive project economics, remain systemically tension-laden, and a grid-relieving property is not in every case economically viable. The BESS acts in the electricity system in market and grid simultaneously.

The maturity-tier process sets a first structural incentive through the hybridisation criterion. Its full steering effect will only emerge once grid-supportive criteria are more strongly operationalised in later procedural versions. Already under the current procedural version, the systemic added value of a project — the contribution it makes at a concrete site under a concrete deployment regime to the function and cost structure of the system — becomes a pre-design variable. It must be built up so that it is visible in the process and later monetizable. In this advance clarification it shows whether site choice and technical concept aim at the same object.

Chapter 4 — Project maturity, site, and spatial plausibility

4.1 What project maturity means

Project maturity designates the early consolidation of those properties that turn a proposal into a system capable of being realised and later operated robustly. They include site control, permitting logic, technical pre-planning, connection concept, the deliverability of critical components, time-related realisability, economic viability, and the organisational capacity to channel these elements with discipline into a consistent development path. The term thus reaches well beyond the notion of a formal pre-stage of construction; it designates the early substantive build-up of the later asset. What is measured is the degree of embodiment of technical, spatial, and institutional substance, independent of the proposal's market promise.

For the BESS this is relevant in a particular way, because the technology can appear modular, industrially standardised, and quick to realise compared with other large-scale installations. This perception underestimates the density of grid-side, permitting-related, and logistic-operational preconditions, as well as the depth of technical interfaces at which later robustness is decided already in the early phase. Where substation, switchgear bay, interconnection, protection concept, routing question, fire safety, maintenance access, data architecture, and lead times of critical components are not carried along early, the resulting deficits act structurally into the later asset and reach deep into architecture, safety, and operation. Project maturity is therefore to be understood as the first form of consolidation of the later system body. A project reaches maturity when its description establishes a robust relation between purpose, system boundary, main functions, interfaces, design assumptions, and verification paths. Project maturity in this sense is a quality state; the achieved degree of substance of the proposal determines it, independent of any imposed time mark. It forms the earliest standard by which to judge whether a proposal in the form in which it is conceived is project capable.

4.2 Site dependence and spatial plausibility

Battery storage is often considered comparatively free in placement. That holds in a limited sense. A BESS in placement stands free of the natural-resource bindings that fix a pumped-storage plant to its geology or a generation park to its wind or solar resource. Its spatial anchoring is nevertheless highly effective. It concerns the distance and integrability to the grid connection point, the structural and safety-related suitability of the site, maintenance and logistical reality, potential extension areas, emission and fire protection, hydrological and geotechnical risks, safety distances, surrounding land uses, and the durability of legal control over the site.

Spatial plausibility arises on several levels at once. First, the site must be physically suitable for construction, operation, safety, and later intervention; this includes ground conditions, settlement sensitivity, flood-risk position (HQ100 and HQ extreme according to the requirements of the federal states), groundwater level, and seismic classification according to DIN EN 1998 (Eurocode 8). Second, the connection must be plausible electrically and in routing terms; line length, line rights, permitting effort, and procurement logic for the connection cable belong here. Third, the asset must remain logistically accessible, maintainable, and capable of intervention in the event of an incident at its planned power class; access routes for heavy goods traffic during transformer delivery and replacement, crane setup areas, fire-water availability, and drainage of contaminated firewater are concretely to be assessed. Fourth, the question of structural durability arises: the site must remain technically and legally clear in the case of later hybridisation, augmentation, or operator change.

With growing power and rising voltage level, these requirements increase markedly. A large-scale BESS is a fixed-location infrastructure object with its own topological, structural, and safety-relevant spatial logic; the conception of an arbitrarily transplantable set of standardised modules falls short for this power class. The site is one of the determinants of lifecycle viability, extensibility, and intervention reality — properties that show themselves in operation and are already laid down in the site decision.

4.3 Site control as an early system act

The maturity-tier process treats site control as one of the two minimum entry thresholds at all. Without a legally robust exclusivity agreement, an application is inadmissible. The requirement is well grounded in substance: a storage project without a secured site remains a declaration of intent and crosses the threshold to a robust scheme only with this binding. The process thereby sets an early quality signal; a grid connection reservation — and with it access to a scarce infrastructure resource — is opened only against substantive prior contributions.

The grading of site control — from exclusivity declaration through notarised preliminary purchase contract to lease or ownership — reflects the maturity of a proposal in its legal and economic binding. Each tier demands a different degree of bindingness, planning horizon, and capital prior contribution. Site control appears from this perspective as an early system act through which the object is brought into a robust form; its weight reaches well beyond an administrative residual condition at the end of a site assessment.

Hydrological and geotechnical risks, safety distances to residential and commercial uses, fire compartments, and escape route concepts belong in the same early assessment round. A secured site whose assessment for these aspects occurs only later may prove legally robust but technically or in permitting terms unsuitable. Site assessment demands a simultaneous consideration of land law, building law, fire protection, grid connection, and logistics; in this simultaneity lies the difference between area and site. A fire-protection concept in line with the building code of the respective federal state and the VdS guidelines (Verband der Sachversicherer, the German insurers' association) for lithium-ion storage must integrate distances between containers, fire-water provision, access routes for the fire brigade, and the discharge of contaminated fire-water into collection vessels; these requirements set structural minimum dimensions that must be considered in early site planning.

4.4 Power-station-adjacent sites and the logic of infrastructure pre-shaping

Former power-station sites are gaining a particular quality in the German BESS market. They unite properties that are structurally advantageous for large-scale storage projects: existing or extensible high-voltage interconnection, available transformer and switchgear infrastructure, secured and often large-area sites, energy-industry pre-shaping in planning law, and established relations with authorities and local operating organisations. This pre-shaping can considerably shorten the development path; it does not relieve the project of a complete technical due diligence.

The advantage arises from several simultaneously effective reliefs. Site control is often prepared, the connection concept can build on existing grid infrastructure, the routing question simplifies or becomes redundant, permitting authorities know the use history and surrounding relations. Technical assessment remains indispensable: legacy contamination from decades of power-station operation — coal dust, oil and lubricant residues, asbestos in older buildings, polychlorinated biphenyls (PCB) in legacy oil-filled transformers, radiological residues at nuclear sites — is to be surveyed and remediated; the demolition status of existing installations must be documented; the actual free grid capacity at the existing connection point is to be verified against the paper statement of the network operator; the noise situation under the German Technical Instructions on Noise (Technische Anleitung zum Schutz gegen Lärm, TA Lärm), fire protection considering the neighbourhood of existing installations, and the integration capability of the legacy secondary technology with the requirements of the new system are to be assessed. Infrastructure pre-shaping thereby opens a substantial starting resource; its effectiveness is measured only by the discipline of the subsequent technical verification.

Where the connection takes place at existing high-voltage or transmission grid nodes, the spatial system logic emerges particularly clearly. Switchgear environment, marshalling and construction areas, cable routes, accessibility for assembly and maintenance, and later operation in the context of existing station philosophies belong in the early site assessment. Hybridisation with existing generation or charging infrastructure strengthens this coupling. The site then influences dispatch expectation, reserve management, safety relations, and later lifecycle paths of the entire object, beyond the connection alone.

4.5 Procurement and long-lead logic as a maturity-tier component

The technical quality of a storage project remains inconsequential where it is not anchored in a realisable time logic. Large-scale storage projects currently stand within a doubly compressed procurement situation. Battery cells, PCS, transformers, switchgear, protection technology, EMS/SCADA components, HVAC systems, and safety-relevant peripherals depend on one another in their deliverability, parameterisation capability, and system integration. At the same time, the degree of industrialisation, quality dispersion, supply-chain geography, and project-specific customisation requirements of the battery industry determine which specifications and dates are achievable under real market conditions.

The maturity-tier process accommodates this logic by treating proof of orders for time-critical grid-technical components — transformers, applicant-side switchgear bays, and cables — as an independent assessment criterion. The grading runs from the inventory list with procurement plan through available supplier quotations to signed purchase contracts. Each tier reflects a different degree of real binding security. For power transformers from 40 MVA upwards, lead times in the current market situation lie typically between 18 and 36 months, comparable for high-voltage circuit breakers and GIS bays, and between 9 and 18 months for medium-voltage transformers and switchgear. To these are added high-voltage cables with lead times of 12 to 24 months for greater lengths, and the requirement of specialised laying capacities. On the battery side, cell availability, module standardisation, quality dispersion, and project-specific integration are added; lead times from order to delivery here lie typically between 9 and 15 months. These strands must be synchronised early.

Where this synchronisation fails, uncertainty shifts from the pre-development phase into those phases in which re-engineering, re-certification, or delayed commissioning generate the highest costs. Procurement security therefore belongs to the technical plant concept and to project maturity; relegating it to a downstream commercial follow-up underestimates its effect on later system quality. Procurement reveals whether a project carries its time logic consistently or comes to rest in a list of components.

Chapter 5 — Technical plant concept

5.1 The plant concept as first consistent system description

The technical plant concept is the first consistent description of a later operating object. It reaches well beyond a catalogue comparison between cell chemistries, and far beyond the choice of an inverter. Within it, purpose, power, duration, architecture, operating regime, connection logic, safety structure, demonstration capability, and industrial availability are brought into a common form. It thereby determines which object will arise at which site under which technical and temporal functional logic, and at the same time which solutions for this project lie outside the permitted space.

The starting point is the application intent. A storage system for very fast control reserve, a grid-side bottleneck or voltage buffer, a diurnally working arbitrage storage, a co-location system at wind or photovoltaic installations, a storage at a former power-station site, or an asset designed in perspective for grid-forming behaviour in the context of future system services differ in their economic deployment and at the same time in reserve management, duration, oversizing, SOC management, power density, thermal loading, data architecture, PCS dimensioning, and demonstration regime. The application intent narrows the solution space before the first main component is selected.

The maturity-tier process demands at this early stage a complete project plan with technical outline, site layout, project schedule, VDE application form, quarter-by-quarter power ramp-up forecast, and a robust draft single-line diagram. These requirements describe in their sum what a technical plant concept must achieve: to bring the later operating object into a form that serves grid operators, permitting authorities, investors, and suppliers as a robust project basis. The robustness of a plant concept is measured by whether the design intent has been carried over into a concretely addressable system.

5.2 Cell chemistry and chemistry path as a system question

At present the lithium-ion family dominates the stationary large-scale storage segment. Within that field, the LFP chemistry — lithium iron phosphate — has reached a particular standing for many stationary applications. It combines thermal robustness, long cycle life, lower dependence on cobalt and nickel, and a safety behaviour that is, in many applications, more favourable than that of NMC systems. For large-scale storage with high cycle counts and long operating duration, this path is currently often well-grounded both technically and economically. For the plant concept, however, a simple comparison of established chemistries falls short. Material and cell concepts shape power density, energy density, thermal robustness, degradation behaviour, safety architecture, manufacturing effort, recyclability, raw-material demand, and cost path of the system in coupled form. The chemistry question therefore determines power limits, safety logic, industrial availability, and later lifecycle options, and reaches well beyond a technology preference.

Beyond the established LFP and NMC paths, three development directions are already being discussed today on technical and raw-material grounds, whose stationary relevance stands at different degrees of maturity and demands disciplined positioning. Solid-state batteries replace the liquid electrolyte with a solid conductor — typically oxide- or sulphide-based ceramics or polymer electrolytes — and promise higher energy and power densities together with an altered safety profile. Model-based system analysis indicates that a high-performance solid-state battery with composite electrodes can deliver, against current liquid-electrolyte cells, an energy-density increase of around 100 per cent and a power-density increase of up to 600 per cent, provided that specific threshold values are met: a solid-electrolyte separator of around 10 µm thickness, an ionic conductivity of the solid electrolyte above 10^{-3} S/cm, specific charge-transfer resistances below 100 Ω·cm², and active-material particles smaller than 5 µm. Cells demonstrated to date — both on sulphide-based solid-electrolyte systems and on polymer-based systems with typical conductivities in the range of 10^{-4} S/cm at elevated temperature — still fall short of these thresholds in central parameters. The path is technically plausible, and for stationary large-scale applications in the next five to ten years not yet a deployment-ready alternative. Sodium-ion cells share their structural logic with lithium-ion cells and largely dispense with critical raw materials; they currently reach energy densities below LFP, but for stationary applications with moderate duration and cycle requirements they may suffice. First commercial cells have been available since 2023/2024; in stationary large-scale applications they stand at the beginning of market penetration. LMFP — lithium manganese iron phosphate — acts as an incremental development of the LFP line with higher energy density at a similarly favourable raw-material basis and emerges as the first practically deployment-ready extension path.

A future chemistry shift does not displace the system character of the BESS. Independent of the concrete path — LFP, NMC, sodium-ion, LMFP, or a later alternative — the asset remains a coupled object of material, conversion, control, connection, evidence, and operation. The chemistry path is a constitutive part of the system architecture and therefore belongs in the technical plant concept; relegating it to a later procurement specification shortens the design logic.

5.3 MW/MWh ratio, duration, and operational state space

Alongside chemistry and cell path, the technical conception includes the system duration, the chosen MW/MWh ratio, the planned C-rate, the relation between installed and dispatchable power, the handling of reserve windows, the assumptions on SOC management and degradation discipline, and the limits arising from temperature, charge and discharge rates, operating modes, and safety requirements. These quantities together determine the operational state space of the system.

Nominal energy is to be distinguished from the energy available in real operation. Between installed nominal capacity and AC-side dispatchable energy lies a restriction space of SOC windows, efficiency losses, reserve commitments, auxiliary-system loads, safety limits, and grid requirements. A two-hour system on nominal terms can, under reserve commitments, parasitic loads, temperature restrictions, and reduced AC power, exhibit a markedly shorter dispatchable duration in real operation. A description in megawatt-hours initially gives a geometric maximum; operational relevance emerges where power, energy, and restrictions are read together.

The C-rate determines the intensity with which the electrochemical inventory is loaded, and acts directly on degradation rate, heat input, and power characteristic. Asymmetric power limits between charge and discharge sides regularly occur in real operation; for LFP cells of current large-scale storage systems, permissible continuous charge rates lie typically around 0.5 C, discharge rates at 1.0 C, and at low temperatures below 0 °C the upper charge limit additionally drops markedly to avoid lithium plating. Reserve windows are equally significant. As soon as a BESS addresses several services — arbitrage, FCR, aFRR, voltage support — a portion of the SOC window remains permanently committed. The plant concept must make this multi-service logic visible early. A storage system that theoretically offers everything without robustly underpinning service quality remains contractually fragile and economically hard to sustain.

5.4 System boundaries, efficiency chains, and auxiliary systems

The technical plant concept must hold the AC/DC system boundary explicit. Depending on the reference point — cell boundary, DC busbar, conversion boundary, AC plant boundary, or market system boundary — efficiency, losses, auxiliary-system loads, and power restrictions are evaluated differently. The choice of system boundary determines what counts as available power, which losses remain in the system, how guarantee values are read, and on what basis contracts and revenue models are built.

The efficiency chain of a BESS runs across several conversion stages: cell efficiency, DC-side distribution losses, PCS efficiency, and transformer losses. Modern PCS achieve efficiencies of 97 to 98.5 per cent per direction with load-point-dependent characteristics; medium-voltage transformers lie between 98.5 and 99.5 per cent with continuous no-load losses. To these are added parasitic auxiliary loads from air-conditioning, heating, lighting, control systems, and where applicable pumps or ventilation equipment. In reserve, standby, and holding phases their relative share can become considerable; a system kept on standby for hours consumes energy without immediately generating revenue from it. These loads belong to the economic and operational fundamental behaviour of the asset and reach in their effect well beyond a mere technical marginal quantity.

Particular attention is due to the thermal management. Temperature influences internal resistances, capacity, ageing rate, and safety behaviour of the cells in coupled form. The dimensioning of the air-conditioning reaches well beyond a downstream building-services question; cooling capacity, control strategy, redundancy, and self-consumption shape the permitted operating window, the lifetime of the electrochemical inventory, and the safety characteristic of the system under real ambient conditions. In hot summer phases with ambient temperatures above 35 °C, the HVAC load can reach a relevant single-digit percentage of rated power; the thermal dimensioning co-determines whether the system can fully call up its rated power even in those phases. At this point it shows itself whether the plant concept governs the storage as a thermal-electrical operating system.

5.5 Grid Code conformity and prequalification as design requirements

Grid Code requirements — the technical rules that define the behaviour of generation, load, and storage installations on the public grid — and prequalification conditions — the technical and formal proofs of suitability for participation in control reserve markets — enter the technical plant concept before components are selected and suppliers commissioned. Which response times, ramp rates, reactive-power ranges, voltage and frequency tolerances, and behaviour requirements during grid disturbances apply shapes PCS architecture, control parameterisation, measurement systems, and communication interfaces from the outset. The governing frameworks are VDE-AR-N 4110 for the low and medium voltage grid and VDE-AR-N 4120 for the high voltage grid, supplemented by the network-operator-specific connection conditions and the prequalification-relevant specifications of the transmission system operators.

For prequalification to FCR (Frequency Containment Reserve) in the continental European synchronous area, full activation within 30 seconds and a sustained capability of at least 15 minutes in both directions apply; symmetric power provision with corresponding SOC reservation per direction is part of the requirement. For aFRR (automatic Frequency Restoration Reserve), full activation within 5 minutes is required, with a sustained capability extending beyond the quarter-hour; aFRR is marketed asymmetrically in Germany. mFRR (manual Frequency Restoration Reserve) operates in 15-minute time slots with longer activation times. Black-start capability sets yet different requirements on PCS architecture, control logic, and the capability for frequency and voltage formation without an external voltage reference; it typically demands grid-forming-capable power electronics, sufficiently dimensioned self-supply of the auxiliary systems, and a defined energy reserve in a defined SOC window. These services require different system designs that do not automatically complement one another and that in part compete with one another.

The consequence for the plant concept is direct. The intended market and system-service roles must be named early and translated fully into architecture, demonstration, and procurement. A concept that describes services as desirable without inscribing the technical preconditions for their delivery remains below the threshold of a robust system specification.

Chapter 6 — Power conversion, DC coupling, and grid behaviour

6.1 The power electronics as operational system coupling

The power electronics form the mediation between the electrochemical inventory and the electricity system. They are the operational heart of the system coupling and act well beyond the role of an interchangeable auxiliary strand. Through the PCS and the conversion chain, active power, reactive power, control quality, response time, ramp behaviour, overload capability, voltage and frequency behaviour, grid support, fault response, and the AC-side usability of the stored energy are determined. Which PCS architecture is chosen and in which operating regime it works shape the asset's later system behaviour more deeply than nearly any other single-component decision.

The technical quality of a BESS depends substantially on how this mediation is conceived. A storage system designed for high market flexibility demands a different balance of reserve management, set-point tracking, short-term overload, and reactive-power management than a storage system intended primarily for grid-near congestion management, voltage maintenance, or, in perspective, grid-forming deployment. These differences are architectural decisions and bind inverter topology, control parameter space, filter design, protection coordination, and certification path; they therefore evade late reconfiguration at the end of planning. The hardware basis is today formed by IGBT modules (insulated-gate bipolar transistors) for power classes up to the double-digit MVA range, increasingly complemented by silicon carbide (SiC) semiconductors with higher switching frequencies and efficiencies; two-level and three-level NPC topologies (neutral-point-clamped) with LCL filters for grid coupling are the current state of the practice.

6.2 Grid-forming and grid-following — a technical distinction with system consequences

The distinction between *grid-forming* (GFM) and *grid-following* (GFL) inverters is for the BESS one of the most consequential architectural questions. It determines which role the asset can take in the grid and under which grid conditions it operates stably.

A GFL inverter behaves as a controlled current source behind an impedance. It synchronises to the existing voltage reference at the grid coupling point through a phase-locked loop (PLL) and feeds in the prescribed current. The PLL continuously compares the measured grid voltage with an internally generated reference and tracks the phase angle accordingly. The operating space of a GFL system depends directly on grid stiffness at the connection point. This stiffness is measured by the short-circuit ratio (SCR), the ratio of short-circuit power at the coupling point to the rated power of the connected installation; values above 3 are regarded as a strong grid, values below 3 as weak, and values below 2 as very weak. In strong grids with a stable voltage reference and an SCR above 3 to 5 the principle works reliably. In weak grids, under steep frequency gradients, or in the presence of high shares of other converter-coupled installations in the electrical neighbourhood, the PLL can become unstable and lead the system into oscillations or loss of synchronism.

A GFM inverter behaves as a controlled voltage source behind an impedance. It provides a voltage reference itself, against which other installations synchronise. It can thereby operate islanded networks, stabilise frequency and voltage independently, and provide forms of synthetic or virtual inertia by responding to frequency gradients with rapid active-power adjustment. The control structures range from classical frequency-power droop and voltage-reactive power droop, through virtual synchronous machine approaches, to matching control and dispatchable virtual oscillator control. Virtual inertia constants lie typically in the range of 2 to 6 seconds, comparable with the inertia of mid-sized synchronous generators. GFM systems are capable of leading black start processes and rebuilding grid segments without an external voltage reference.

In practice, transitions and hybrid forms exist between the two approaches. For the technical plant concept, the relevant question is function-related: which minimum measure of grid-forming properties — voltage stabilisation, instantaneous reserve, or black-start capability — is required for the intended deployment regime, and which investment, demonstration, and certification consequences follow from that. In grids with a high share of converter-coupled feed-in the answer shifts visibly towards GFM. Large-scale batteries are read in this context increasingly as grid-technically qualified installations, beyond the role of a flexibility resource.

6.3 DC coupling as a system-configuration question

Battery storage has traditionally been integrated into the grid via AC connection points. This configuration is well proven and produces several conversion stages. *DC coupling* designates the direct integration of battery storage with other DC components on the direct-current side ahead of the inverter. In its simplest form, a photovoltaic installation and a battery storage share a common PCS. In more complex multi-port topologies, several DC sources and sinks are connected to a common DC bus that is centrally coupled to the grid.

The advantages are technically concrete. Compared with purely AC-coupled solutions, system efficiency can rise because one conversion stage is removed; the efficiency gain in PV-storage hybrids lies typically at one to three percentage points. In shared-inverter systems, the specific inverter investment per megawatt of storage power decreases in many cases. Grid-side reactions, ripple, and the coordination of hybrid operating modes can be better governed in certain applications. For PV-storage hybrids or later couplings with electrolysers, DC coupling can be systemically superior.

The advantages arise under additional requirements. DC arc interruption is technically more demanding than AC interruption because the current in a DC fault is not interrupted by a natural zero crossing; specialised DC switches with arc-quenching chambers, fuses with sufficient breaking capacity for the prospective DC short-circuit current, and in part solid-state switches on a semiconductor basis are required. The dimensioning of the common DC bus and the voltage windows of the PV strings and the battery must be coordinated; where the battery voltage deviates from the maximum power point (MPP) voltage of the PV strings depending on SOC, mismatches arise that must be compensated either by additional DC/DC converters or by an aligned cell count of the battery. Protection and control concepts become more complex because protection coordination must account for the DC currents from both sources; the relevant standards are IEC 60364-8-1 and IEC 62548 for DC-side protection, together with the product-specific standards for DC switching devices. Normative and regulatory frameworks are in part still developing. To these are added ownership, responsibility, and contractual questions where several assets or operators are coupled through a common PCS. DC coupling is therefore an early architectural decision with consequences for the protection concept, permitting logic, operational responsibility, and economic ordering of the project.

6.4 Reactive power, weak-grid behaviour, and grid-side reactions

Reactive-power capability ranks among the systemically most valuable properties of a BESS. A PCS can in principle provide reactive power independently of the state of charge, as long as the apparent-power limit $S = \sqrt{(P^2 + Q^2)}$ is not exceeded. Active-power and reactive-power space compete for the same apparent-power maximum; provision of reactive power reduces the remaining active-power room. The relevant grid connection rules VDE-AR-N 4110 and 4120 require for the static and dynamic reactive-power range typically a power factor cos ϕ between 0.90 underexcited and 0.95 overexcited, where applicable with a voltage-dependent $Q(U)$ characteristic. An installation that is to deliver active-power and reactive-power services simultaneously requires a robust prioritisation logic.

Behaviour in weak grids is a critical test of the system's quality. GFL systems can, in environments with low SCR, high impedance, and low inertia, tend towards PLL-induced instabilities, resonance phenomena, or unwanted interactions with other converters. GFM systems behave more robustly in such constellations; under inadequate parameter design they can produce stability problems of their own, particularly in parallel operation of several grid-forming units, where phase coordination and virtual impedance between the units must be carefully tuned. Weak-grid behaviour belongs for future large-scale batteries within the core area of design and demonstration questions and reaches well beyond special scenarios of individual research studies.

Grid-side reactions through switching events, harmonics, and voltage distortion form a further dimension of grid behaviour. The relevant limits for harmonic distortion follow IEEE 519 and the relevant national specifications: total harmonic distortion (THD) of the voltage at the coupling point typically below 5 per cent, total demand distortion (TDD) of the current depending on the short-circuit ratio between 5 and 20 per cent. Filter design, switching frequency (typically 2 to 20 kHz), transformer configuration, and earthing concept influence the degree of these reactions. In grids with sensitive customers or high prior harmonic loading, corresponding demonstrations are to be inscribed early into the system design.

6.5 Ramp logic, overload capability, and operational quality limits

Power in the BESS is gradient-bounded. The maximum rate of change of delivered or absorbed power is determined by inverter topology, cooling design, protection parameters, and electrochemical loading limits. Typical ramp rates of modern large-scale storage PCS lie at several hundred per cent of rated power per second; the usable rate is governed by grid requirement and plant-internal coordination. Ramp restrictions act on reserve capability, schedule changes, and grid compatibility; excessively steep power ramps can trigger voltage fluctuations in the local grid and load transformers and other components. Ramp logic therefore belongs to the grid and system quality of the asset and reaches well beyond a detail of the control software.

Short-term overload capability describes the ability to deliver power above thermal continuous operation for limited periods. It is particularly relevant for FCR, voltage support, and certain disturbance scenarios. The permissible overload follows a thermal logic: IGBT modules and SiC semiconductors can be operated for seconds to a few minutes above their continuous loading limit, so long as the junction temperature does not exceed its permissible maximum (typically 150 to 175 °C for silicon IGBTs, 200 °C and above for SiC); the thermal capacity of the modules, the coolant mass-flow density, and the prior loading together determine the available overload window and the subsequent recovery time. Typical overload factors lie at 110 to 120 per cent of rated power for a few minutes, up to 150 per cent for sub-second ranges in correspondingly designed systems. For the technical plant concept, overload capability is to be specified as a guaranteed system quantity with duration and recovery requirement; specification as a loose manufacturer property without system reference falls short of the demonstration and contractual logic.

The usable system power of the BESS is the result of an AC/DC-mediated, state-bound, and control-engineering-limited conversion. It is a function of SOC, cell temperature, ageing state, active services, and grid conditions at the coupling point; a constant property it does not represent. Power commitments to grid operators, marketers, and investors therefore rest on dispatchable power under realistic operating conditions; installed rated power on its own does not establish them. In this difference runs the boundary between nominal value and robust operating value.

Chapter 7 — Substation, switchgear, and connection body

7.1 The connection body as an infrastructural composite system

The connection body forms the grid-side materialisation of the BESS. Its quality arises from the coherence between the electrochemical inventory and grid-side reality and reaches beyond the interplay of battery and PCS. Primary technology, GIS or AIS architecture, secondary equipment under IEC 61850, and routing act in this materialisation as a coupled whole; the quality of the connection body emerges in this coupling.

The connection body is the place at which the planning and operating philosophies of the network operators take effect. Primary technology, insulation coordination, protection philosophy, control-system architecture, and communication protocols follow established conventions. They form binding framework conditions into which the BESS scheme must fit; their character as neutral project options does not hold. Where the connection body is read from the perspective of a container product, this binding remains underdetermined. The maturity-tier process demands at this point a greater substantive depth early on.

7.2 Primary technology — transformers, circuit breakers, and switchgear bays

Primary technology comprises the components that immediately conduct the active-power flow: power transformers, circuit breakers, disconnectors, earthing switches, busbars, current and voltage transformers, surge arresters, and earthing systems. For large-scale BESS connections at medium- or high-voltage levels, transformers with rated powers of 40 to 250 MVA are typical, with transformation ratios that match the PCS voltage level (typically 690 V to 1.5 kV AC on the low-voltage side of modern medium-voltage PCS) to the respective grid level. Short-circuit voltage u_k moves typically between 10 and 16 per cent, no-load losses at the per-mille range of rated power, load losses (measured in the short-circuit test) at the low single-digit per-mille range; these parameters determine the thermal and electrical operating point of the transformer under cyclic load.

For BESS the cyclic load characteristic is relevant. A storage system that switches in daily or intraday rhythm between high utilisation and standstill loads transformers differently from a continuously operated generation or load transformer; thermal cycles act more strongly on the ageing of the insulation, the loading of winding and core changes more frequently, and the overload capability under IEC 60076-7 (Loading Guide) becomes operationally relevant. The choice between oil- or ester-based insulating media and gas-insulated variants follows the interplay of fire-protection philosophy, environmental requirement, and maintenance logic of the site; natural esters such as FR3 or comparable vegetable oils have a higher flash point (above 300 °C against 140 to 170 °C for mineral oil) and are increasingly used at sensitive sites.

High-voltage circuit breakers are today often executed as SF_6 breakers, increasingly available as SF_6-reduced or SF_6-free alternatives with fluor nitriles, CO_2 mixtures, or vacuum technology; the F-Gas Regulation (EU) 2024/573 sets the regulatory frame. The short-circuit power at the connection point determines the minimum dimensioning of the switching devices; typical breaking currents at German 110 kV nodes lie between 31.5 and 63 kA, at 220 kV nodes between 40 and 63 kA, at 380 kV nodes between 63 and 80 kA. Underdimensioning is safety-critical; overdimensioning drives investment and maintenance effort without functional added value. The dimensioning of the primary technology arises from the interplay of grid requirement, protection philosophy, and operating profile; an exclusively battery-side perspective falls short of it.

7.3 GIS or AIS — a site-bound architectural decision

The choice between air-insulated switchgear (AIS) and gas-insulated switchgear (GIS) is a site-bound architectural decision that touches CAPEX, footprint, maintenance logic, and operating philosophy simultaneously.

AIS uses ambient air as the insulating medium and requires large insulation distances between live parts. For a 110 kV double-busbar arrangement with four feeders, the typical footprint amounts to 2,500 to 4,000 square metres. AIS is robust, maintenance-friendly given good accessibility, and as a rule more favourable in investment; it is the standard solution where sufficient area is available, where no special environmental requirements exist, and where weather-related restrictions remain manageable.

GIS uses sulphur hexafluoride (SF_6) or, increasingly, alternative insulating gases with lower global warming potential as the insulating medium. The higher dielectric strength allows insulation distances to be reduced drastically; the footprint of a comparable 110 kV installation drops to 200 to 400 square metres, around one tenth of the AIS solution. GIS is the architecture-determining choice under constrained spatial conditions, in inner-city or industrialised settings, at existing power-station sites with limited extension area, and where requirements for weather robustness or protection against aggressive ambient air are heightened.

The decision has consequences beyond the pure footprint comparison. GIS requires specialised maintenance logic, SF_6 management under the F-Gas Regulation with leak-tightness testing and annual loss accounting, different assembly logistics, and tendentially higher CAPEX with lower OPEX. AIS allows simpler extensibility and is, in the event of a fault, more quickly repairable. For BESS schemes at sites with existing substations, the project usually inherits the philosophy of the existing connection body; whoever encounters a GIS installation at the site connects, as a rule, with GIS, which permits economies of scale in maintenance and spare parts holding.

7.4 Secondary equipment, protection, and control technology under IEC 61850

The secondary equipment organises protection, control, measurement, communication, and monitoring of the connection body. It forms the digital nervous system of the installation and decides on response speed, selectivity of protection action, and information flow to the network operator's control centre. The communication architecture follows IEC 61850, a standard that structures data exchange between protection, control, and measurement devices within a substation as well as between station and control centre. The relevant services are MMS (Manufacturing Message Specification) for client-server communication to the control centre, GOOSE (Generic Object-Oriented Substation Event) for fast event exchange in the millisecond range between protection devices, and Sampled Values for time-critical measurement values; IEC 61850-7-420 extends the data model with Distributed Energy Resources (DER) and is becoming increasingly relevant for BESS installations.

The protection functions of a BESS connection cover several layers. At transformer level, differential protection (ANSI 87T) with response times below 30 milliseconds, Buchholz protection, and overpressure protection for the transformer tank, together with temperature monitoring of winding and oil, take effect. At medium- and high-voltage level, time-overcurrent protection (ANSI 50/51) and earth-fault protection (ANSI 50N/51N) act, complemented by directional components (ANSI 67/67N). For interconnection lines, distance protection (ANSI 21) is applied, which under converter-coupled installations requires special parameterisation owing to the limited short-circuit currents of the PCS. For the grid behaviour of the PCS, voltage and frequency protection (ANSI 27/59/81) acts under the specifications of VDE-AR-N 4110 and 4120. Dedicated functions for DC-side faults in the battery circuit close the protection space.

The grading of these protection levels — which function, under which fault picture, with which response time, trips — is the actual engineering problem of the secondary equipment. Classical overcurrent grading loses its selective action with converter-coupled installations because the short-circuit current contribution of a grid-following inverter is limited to 1.0 to 1.2 times, of a grid-forming inverter to 1.1 to 1.5 times, the rated current, and thus lies far below classical synchronous-generator values. Protection coordination therefore works with distance protection for line sections, differential protection for zones with unambiguous current balance, directional protection in both feed-in directions, and communication between protection devices via GOOSE for zone coordination. Errors in this coordination can either disconnect local faults across an unnecessarily wide area or leave critical fault pictures undetected.

The control technology comprises SCADA interconnection, remote-control capability, event recording, disturbance recorder function, and the communication interface to the grid control centre of the TSO, today usually through IEC 60870-5-104 or, increasingly, IEC 61850. For BESS, requirements for real-time communication with high temporal precision are added, because set-point commands for control-reserve services are in part transmitted with second resolution; time synchronisation through PTP (Precision Time Protocol, IEEE 1588) in the microsecond range for protection and control paths, and through NTP (Network Time Protocol) in the millisecond range for the remaining plant communication, forms a necessary basis. The data architecture separates between operationally critical real-time data and analytical aggregate data; both levels are to be maintained consistently without disturbing one another. The operational readiness of the connection body lies in this consistent stewardship of electrical and digital layers.

7.5 Routing and cable technology

The line interconnection between the BESS site and the grid coupling point is a technical and permitting project within the overall scheme. It comprises route layout, underground cable laying or overhead line construction, cross-section sizing, thermal dimensioning, cable transitions, and cable terminations. The maturity-tier process treats routing as an independent assessment criterion. The range extends from a simple routing strategy with spatial-resistance analysis to detailed route planning and complete in-rem securing through land-registry entries and plan-approval decisions.

Cable design follows several restrictions at once. The cross-section must thermally master the rated current and the short-circuit loading; for 110 kV cables with copper conductor, cross-sections of 630 to 2,500 mm² are usual, depending on transmission capacity and laying conditions. The insulation design follows the voltage level, operating voltage, and test requirements; XLPE insulation (cross-linked polyethylene) is the state of the art for 110 kV and 220/380 kV, and oil-paper-insulated cables are now used only in special cases. Laying depth and surroundings substantially influence the current-carrying capacity; built-up ground conducts heat less well than open land, parallel systems generate thermal interactions, and the specific thermal resistivity of the surrounding soil under IEC 60287 determines the continuous-current limit. On longer routes, cable joints, transitions, and terminations enter as separate quality and test objects; installation quality and partial discharge testing after commissioning are critical inspection points.

The permitting logic of the routing often forms the time-critical path of the overall project. Plan-approval procedures for lines from 110 kV upwards can demand 18 to 36 months. Easements, land-registry entries, and line rights are to be secured plot-by-plot; protected areas under nature-conservation law, built-up areas, or sensitive environmental uses extend the time horizons. Route planning is to be carried in parallel with site development; downstream entry after completed primary-technology design comes too late for the time logic of the project. Project maturity shows itself at this point as the capacity for early simultaneity.

With primary technology, the GIS / AIS decision, secondary equipment, and routing, the grid-side materialisation of the BESS is captured in its principal features. Within this composite system, the assumptions of the plant concept meet the regulatory, physical, and organisational realities of the German transmission grid. The connection body is the first real test of system quality; incoherences inscribed at this point act into later operation and can be absorbed by operational discipline only to a limited extent. Technical design, procurement path, routing logic, and operating philosophy yield at this point a coherent object — or remain fragmented; the early coherence of the connection body acts on operational robustness, demonstration capability, extensibility, and economic stewardship of the asset across its full lifecycle.

Part III — Operating architecture and system integration

Chapter 8 — Operation as the management of a bounded state space

8.1 Operation as a selection problem

The operation of a battery storage system reaches well beyond the alternation between charging, holding, and discharging. The asset moves within a bounded state space, the shape of which is determined jointly by electrochemical, thermal, power-electronics-related, grid-technical, safety-related, regulatory, and commercial boundary conditions. These boundary conditions act in coupled form. What appears economically meaningful under a market or grid signal can be degradation-critical from the perspective of cell chemistry, restrictive from the perspective of the power electronics, inadmissible from the perspective of the safety architecture, or no longer callable from the perspective of an existing prequalification. Real operation therefore carries throughout the character of a selection problem under multiply bound restrictions.

The perspective thereby shifts from the name of an operating mode to the structure of the permitted behaviour. SOC, SOH, temperature, charging and discharging power, response speed, reactive-power share, reserve holding, schedule binding, maintenance windows, and safety permissives together form an operational geometry of the permitted. A storage system becomes intelligible as an infrastructure object only once this geometry becomes visible as the object of management.

The current grid context strengthens this view. With the increase of converter-coupled feed-in, declining systemic inertia, growing demand for reactive power, higher sensitivity to voltage and frequency events, and increasing load-flow complexity, storage systems are drawn more strongly into voltage, frequency, and load-flow regimes. Operation is therefore at once the commercialisation of energy and the ongoing answer to an electrical environment whose stability is shifting.

8.2 The permitted state space as a geometric object

The permitted state space of a BESS can be described formally as the intersection of several restriction surfaces. The electrochemical limits — maximum and minimum state of charge, permissible cell voltages, temperature corridors, C-rate upper limits — define the first restriction surface. The power-electronics limits — maximum AC power, ramp rates, reactive-power ranges, protection trip limits — form the second. The grid-technical limits — export capacity, import capacity, frequency and voltage windows, grid-side reaction limits — form the third. Safety-related limits, regulatory bindings, and commercial obligations layer over these.

The available power at any given moment is the minimum of these competing limits. For operational practice it follows that the restriction space is to be treated as a time-dependent object that shifts with temperature, state of charge, ageing state, active services, and grid conditions. An operational management that knows the state space only in its nominal limits works with a systematically too wide management space; the actual limits are narrower, time-varying, and situationally coupled.

For operational practice, three state spaces are analytically separable. The *technical state space* comprises all states permissible from the perspective of cell chemistry, power electronics, and safety. The *operational-strategic state space* narrows this further to those states in which intended services — reserve holding, black-start readiness, voltage support — remain callable. The *situative state space* is the configuration available at the present moment under current grid conditions, running services, and the current system state. Only in the simultaneous management of these three layers does the BESS become controllable.

8.3 Management as ongoing optimisation under uncertainty

The operational management of a BESS unfolds as an ongoing optimisation under several simultaneous uncertainties. Market prices for energy and system services are forecast, not known. Grid conditions change with feed-in and load patterns, which in turn vary with weather, weekday, and season. The asset's own state — cell temperature, internal-resistance distribution, local ageing gradients — can be captured only indirectly. Set points from marketers, network operators, and the EMS arrive with different temporal resolutions and different degrees of bindingness. Under these conditions operational management is a question of prioritisation. Which services are served, when, with which share of the available state space? Which reserve windows are kept clear for committed products, which shares remain available for opportunistic commercialisation? Which degradation costs are knowingly accepted to realise which revenues? These questions belong to normal operation; they are typically taken in second-by-second to quarter-hour cadence and shape the economic outcome of the asset more strongly than the one-time choice of cell chemistry or PCS architecture.

The quality of this optimisation rests on two preconditions. The first is the quality of the available state information: how precisely the system knows its own state and how reliably forecasts for prices and grid states are available. The second is the architecture of the decision layers: how decisions are distributed between BMS, EMS, and the higher-order commercialisation logic, how conflicts are resolved, and which instance carries the final decision under contradictory signals. The internal hierarchy of these layers is the subject of the following chapter.

Methodologically, a robustly managed storage operation moves beyond deterministic point optimisation. The literature on energy hubs and storage planning works with stochastic optimisation, risk-constrained scheduling, and robust optimisation approaches to absorb uncertainties into the dispatch logic, rather than to comment on them after the fact. The methodological core for infrastructure operation lies less in mathematical formalism than in the insight that reserve holding, opportunity use, and safety discipline must be managed simultaneously under uncertainty assumptions. A robustly managed system therefore deliberately holds degrees of freedom that may appear unused in deterministic hindsight, and whose holding produces the operational viability in the first place.

8.4 Operating regime as the assignment of state space and service

An operating regime assigns a specific service palette to the state space of the BESS. Pure primary control reserve claims a narrow SOC corridor around a defined set-point, demands sub-second response capability, and leaves little room for other services. Arbitrage on the intraday and day-ahead markets uses wide SOC amplitudes, accepts longer response times, and permits superpositions with other services if reserve windows are preserved. Grid-near congestion management or voltage maintenance follows local grid signals and can short-term overwrite market opportunities.

Practically relevant, as a rule, is the question which combination of these regimes is served simultaneously and how the transitions between regime states are organised. A BESS that simultaneously delivers aFRR reservation, intraday arbitrage, and conditional grid-supportive function operates within a layered model in which each layer claims a part of the state space and holds a defined priority against the others. The operational management orchestrates these layers so that no layer systematically blocks the others.

For the technical plant concept, this multi-service use acts back into the design phase, because it determines which PCS power, which battery size, and which reserve windows are to be provided. The feedback between operating strategy and system design is tight: a plant strategically designed for multi-service use can realise revenues above the sum of its individual services, provided that the operational management resolves the conflict spaces with discipline. The architectural multi-use capability remains below its potential without multi-use-disciplined operational management.

8.5 Conflict spaces and priority hierarchy

Conflict spaces arise where different services access the same part of the state space simultaneously. They typically show themselves in four technical dimensions at once: as competing power demands, where one service requires positive and another negative active power; as competing SOC target bands, for instance where one service requires a high state of charge for later discharge while another claims the middle SOC for symmetric callability; as competing response times, where sub-second FCR delivery coincides with quarter-hour schedule binding; and as competing energy budgets across different time horizons, for instance between hourly intraday arbitrage and day-spanning reserve holding. The practical configurations follow this pattern: an arbitrage signal demands high charging power while an aFRR call requires symmetric provision; a grid-congestion situation requires discharge while the market price signals charging; cell chemistry and temperature signal restraint while set-points demand aggressive operation. Such situations belong to the normal case of multi-use operation.

The operating architecture requires an explicit, pre-emptively acting priority hierarchy, as shown in Table 1, for the resolution of these conflicts. Cooperative negotiation in the running cycle achieves neither the required response time nor the unambiguity that is needed in event situations. The hierarchy is at its core a governance decision negotiated between operator, marketer, and where relevant network operator; it becomes technically effective only once it is unambiguously mapped in the control architecture across all layers: as a hardware interlock in the BMS for safety-relevant limits (cell temperature, cell voltage, insulation resistance, smoke and gas detection); as a rule-bound permissive in the PCS controller for grid- and power-electronics bindings (ramp limitation, reactive-power window, frequency and voltage limits, derating curves); as a dispatch logic in the EMS for the orchestration of market-related and grid-related services; and as an override in the higher-order commercialisation system or VPP for contractually committed services. Typical priority levels in descending order are: safety limits and protection action — that is, thermal, electrical, and mechanical triggers from the BMS and protection relays that act immediately and independently of market or schedule signals; regulatorily mandatory services

Priority level	Power + / - P	SoC target range	Response time	Energy budget	Realisation layer
Safety & protection	Override	Lockout zone	immediate	independent	BMS
Regulatory mandatory services	Reserve	Reservation	FCR / aFRR	committed	PCS / EMS
Contractual services	Penalties	promised	SLA	reserved	EMS
Opportunistic commercialisation	Arbitrage	flexibel	Market time	available	EMS / VPP
Degradation at idle	Sparing	Favorable state	slow	optional	EMS

Table 1. Priority hierarchy in multi-use operation. Value stacking remains operationally bounded because safety, reserve, contractual, and market logics claim the same power, energy, and state space.

such as black-start readiness and prequalified reserve (FCR, aFRR, mFRR) with their respective reservation, activation, and verification conditions; contractually committed services with penalties for non-performance, such as tolling obligations, capacity contracts, or bilateral PPA structures; opportunistic commercialisation without firm commitments — that is, spot and intraday arbitrage as well as intraday redispatch opportunities; and optimisation for degradation minimisation at idle, that is, the active holding of a favourable SOC and temperature window during periods of low market activity.

The quality of this hierarchy shows itself in its demonstrability. Every dispatch decision must remain locatable in the evidence layer: with timestamp, assignment to the activated service class, origin of the executed set-point, documentation of the rejected competing demands, and indication of the underlying priority rule. Technically this requires event-based logging at sufficient temporal resolution — typically in the second to sub-second range for PCS and protection events, in the minute cadence for EMS dispatches, and in the quarter-hour cadence for market-related schedule data — an immutable storage of the raw events, a traceable chaining from trigger signal through hierarchy check to executed action, and an unambiguous versioning of the priority rules themselves, so that later audits can reconstruct under which rule state a decision was taken. An operation that resolves conflicts under undocumented or situationally variable rules generates, in event cases, liability and warranty questions that can hardly be answered, whether in a thermal incident, in a non-performance during a reserve activation, or in a grid-side reaction subsequently attributed to the asset. The explicit, documented, and verifiable priority tree therefore acts simultaneously as an operational tool and as part of the asset's evidence architecture; it has direct effect on insurability, refinance ability, and contractual robustness.

Under these conditions the state of charge takes on a double role. It is both an instantaneous operational quantity and an allocated possibility space for future calls. An SOC window forms an operational reserve order, in which activation probabilities of committed services, expected market opportunities, grid-related requirement patterns, and safety-related retention bands are anticipated and prioritised against one another. The available AC-side energy emerges only after the deduction of reserve commitments, parasitic losses, SOC window limits, and further restrictions; the operational management decision translates this composition into the asset's running dispatch.

Chapter 9 — BMS, EMS, PCS — hierarchy and coherence

9.1 Three control layers, one system object

The internal control architecture of a BESS arises from the interplay of the battery management system (BMS), the energy management system (EMS), and the power conversion system (PCS). These three layers are functionally distinguishable and operationally inseparable. Each works in its own time and decision layer, each sees a section of the asset, each takes decisions based on partly overlapping, partly disjoint information. The quality of operational management depends on whether these three layers are designed, parameterised, and operated as a coherent hierarchy.

The BMS protects the electrochemical integrity of the system. It monitors cell voltages, cell and module temperatures, current paths, balancing states, the insulation resistance against earth, contactor states, gas and smoke detectors, and those permissive limits that the downstream layers must observe. At cell level it works in cycles of a few milliseconds, during protection trips it acts in the single-digit millisecond range; balancing decisions and thermal management move in the second-to-minute cadence. The SOC estimation rests in practice on a combination of Coulomb counting, rest-voltage-based open-circuit voltage calibration, and model-based estimators such as Extended or Unscented Kalman filters; the choice of method determines the precision with which SOC and available energy are reported outwards. The spatial resolution is highly local: a large-scale storage system typically contains several thousand to tens of thousands of cells, which in a hierarchical structure of cell-monitoring ICs, module BMS, and rack or string BMS communicate, mostly through CAN-based internal buses, up to a higher-order BMS master with an interface to the EMS.

The EMS organises the plant intelligence at system level. It translates revenue logic, reserve obligations, time windows, grid requirements, forecasts, and schedules into concrete set-points and operating priorities. It operates within those limits set by BMS, PCS, contractual and market specifications, and by the current system state. Methodologically it works typically with rolling optimisation windows: a day-ahead horizon of 24 to 36 hours for schedule preparation, an intraday horizon of a few hours for adjustments, an ongoing real-time alignment in the second-to-minute cadence. In high-performance implementations Model Predictive Control (MPC), mixed-integer optimisation, and stochastic scenarios are deployed to absorb uncertainties in price, grid, and state information into the dispatch logic. The spatial resolution of the EMS is the entire asset, or several assets in a fleet.

The PCS mediates between the DC-side storage state and the AC-side grid behaviour. It determines the real form in which active and reactive power is provided, limited, converted, or refused. The power electronics are today based on IGBT modules, increasingly also on SiC MOSFETs, with switching frequencies typically between 2 and 20 kHz and topologies such as two-level or three-level NPC inverters, which are connected to the medium-voltage transformer through LCL filters. The control level of the PCS works on several time scales at once: the inner current control in the microsecond-to-millisecond range, the outer voltage or power control in the millisecond-to-second range, and the synchronisation and grid behaviour through PLL structures in grid-following operation, or through virtual impedance and inertia models in grid-forming operation. Through the PCS, ramps, control quality, reactive-power capability, short-circuit contribution, fault ride-through behaviour, and a substantial part of the available power become effective.

9.2 The hierarchy as decision and restriction architecture

The hierarchy shown in Figure 2 between the three layers is asymmetric. BMS and PCS set hard limits that the EMS must not exceed; they form the restriction layer. The EMS chooses, within these limits, the concrete set-points and prioritisations; it is the decision layer. This asymmetry protects the asset from economically driven operating modes that would be technically harmful: a pure revenue optimum can produce load profiles that cause lithium plating at low temperatures, local thermal hotspots in mid-string modules, or accelerated ageing of the separators.

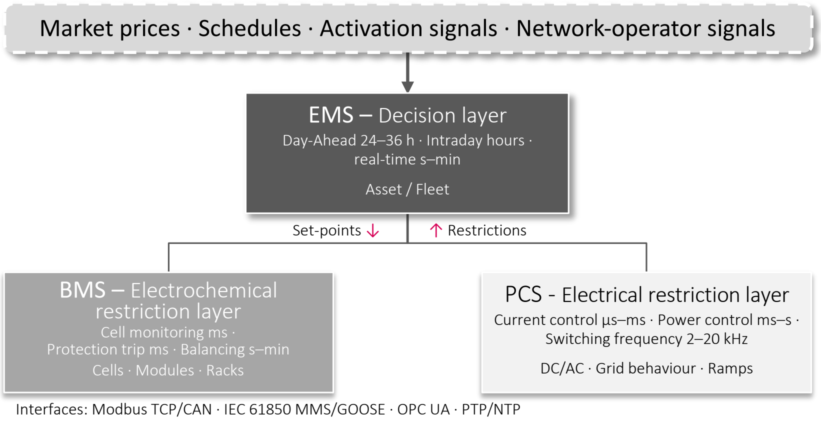

Figure 2. Asymmetric control architecture between BMS, EMS, and PCS. Restrictions ascend from the protection and conversion layers; set-points may only be passed downwards within these limits.

The propagation of restrictions between the layers follows a clear pattern. The BMS reports to the EMS cyclically — typically in the 100-millisecond to 1-second cadence — the currently permissible charging and discharging power, the available energy in both directions, the valid temperature and SOC reserves, the state of balancing and auxiliary systems, and forward-looking warnings about protection limits that could be reached soon. The PCS reports to the EMS the AC-side deliverable active and reactive power under current grid conditions, the achievable ramp rates, the remaining overload capability, and any active deratings due to module, filter, or transformer temperatures. The EMS processes this information together with external signals — market prices, schedules, activation signals, network-operator signals — and returns set-points that lie within the reported limits.

The quality of this communication is operationally decisive. Delays in the propagation of restrictions lead to set-points that, by the time they arrive, are no longer deliverable; the consequences are corrective actions, unwanted oscillations, or control-side trips. Inconsistencies in the state estimates — where BMS and PCS measure the same parameter differently or apply a divergent SOC definition — generate oscillations whenever the EMS switches between the two information sources. The architecture must therefore define clear competences, consistent data interfaces, coordinated time cadences, and unambiguous conflict-resolution rules. Typical interface protocols are Modbus TCP and CAN in BMS–PCS communication, IEC 61850 with its services MMS and GOOSE for integration into the station control technology, and OPC UA for communication between the EMS and higher-order systems. Time synchronisation through PTP (IEEE 1588), or at minimum through NTP, forms a necessary basis for cross-layer event correlation.

9.3 The integration question — monolithic or multi-vendor

A strategic architectural decision for every BESS scheme is the question of whether BMS, EMS, and PCS come from a single source or are integrated from different suppliers. *Monolithic solutions* — a single integrator delivers all three layers as a coordinated package — offer advantages in commissioning, warranty boundary, and fault remediation. The compatibility of the interfaces is validated in advance, parameterisations are coordinated, responsibilities in the event of a fault are unambiguous. The price of this closedness is a high degree of vendor lock-in across the lifecycle, particularly with firmware updates, replacement parts, and later augmentations, the compatibility of which must be confirmed again by the same integrator.

Multi-vendor architectures combine specialised suppliers at each layer — for instance a BMS from the battery manufacturer, a PCS from a leading inverter producer, and an EMS from an independent software supplier. This architecture allows access to best-in-class performance at each layer, increases independence from individual suppliers, and eases later updates or the replacement of individual components. It shifts the integration responsibility and the interface risk to the operator or to a system integrator commissioned by them, and produces a more complex warranty structure in which, in a fault case, the attribution to a specific component must first be clarified.

Standardised communication protocols substantially reduce the multi-vendor risk, without resolving it fully. IEC 61850, with its object-oriented data model of Logical Nodes and Common Data Classes and its services — MMS for client-server communication, GOOSE for fast event exchange in the millisecond range, Sampled Values for time-critical measurement values — forms the basis for communication within the station and to the network operator; IEC 61850-7-420 specifically defines data models for Distributed Energy Resources and is increasingly used for BESS installations. At the BMS–PCS–EMS level, Modbus TCP, DNP3, OPC UA, and increasingly MQTT are widespread. Manufacturer-specific protocols continue to exist, particularly in BMS communication at cell and module level, where the battery manufacturers typically do not fully expose the interface into the rack and module electronics. The decision between monolithic and multi-vendor architecture is at its core a governance decision: which integration risk should the operator bear, which flexibility is important to them across the lifecycle, how can responsibilities in a warranty case be delineated, and which competences does the operator hold internally to manage a multi-vendor environment technically.

9.4 The PCS as carrier of grid-qualified functions

With the growing demand for voltage-supporting and, in certain contexts, grid-forming behaviour, the standing of the PCS within the overall system shifts. Its relevance arises beyond conversion efficiency, from the capability to react to voltage disturbances, frequency deviations, phase steps, and weak grid conditions either inherently or in a control-engineering robust manner. The PCS thereby becomes the carrier of grid-qualified functions that were earlier primarily associated with synchronous generators.

The distinction between grid-following and grid-forming operation is in this context technically decisive. Grid-following converters synchronise to the measured grid voltage through a phase-locked loop (PLL) and feed in according to a current set-point; they presuppose a sufficiently stiff grid and can become unstable under low short-circuit power or a high converter share in the surroundings. Grid-forming converters generate, by contrast, their own voltage and frequency reference and behave towards the grid as a voltage source with defined impedance. The control structures used for this range from classical frequency-power droop and voltage-reactive power droop, through virtual synchronous machine approaches (VSM, VISMA), to matching control and dispatchable virtual oscillator control (dVOC); the variants differ in how inertia, damping, and current limiting under fault are reproduced.

From the perspective of the operating architecture, this functional shift has three consequences. The first concerns parameterisation. Control parameters for grid-forming functions — virtual inertia constants, damping factors, droop characteristics for active and reactive power, voltage control parameters, current limiters for fault cases — must be precisely tuned to the local grid behaviour. Inertia constants set too low produce no effective frequency support; set too high they can lead, under weak coupling, to sub-synchronous resonances or control-side instabilities. The second consequence concerns fault behaviour. Fault ride-through profiles for under-voltage events (LVRT) and over-voltage events (HVRT) under VDE-AR-N 4110, 4120, and 4130 and the relevant grid connection rules, as well as the required feed-in behaviour during and after the disturbance — reactive-current priority, active voltage support, ordered return gradient — determine both the dimensioning of the power electronics and the control design. The short-circuit current contribution of a grid-forming converter is typically limited to 1.1 to 1.5 times the rated current, and thus markedly lower than that of a synchronous generator; this limitation acts into the protection coordination of the connection body, because classical overcurrent grading loses its selective action under these conditions and complementary protection principles become necessary. The third consequence concerns the test logic. Grid-forming functions demand system tests under realistic grid conditions; in practice this is performed through hardware-in-the-loop validation with a real-time grid simulator (RTDS, OPAL-RT) before commissioning, complemented by field tests after commissioning. The demonstration of these functions arises only in this combined system test; isolated component testing remains below the required scope.

For the EMS this development means that grid-state information becomes a relevant input to dispatch optimisation. Voltage deviations, local frequency gradients (df/dt), measured or computed short-circuit power at the connection point, and grid topology changes flow into the decision on which share of the asset is reserved for grid-supportive behaviour. The boundary between operational management and system management thereby shifts; the BESS becomes an active element of grid management that takes its dispatch decisions in coordination with the state of the electrical environment.

9.5 Higher-order layers — VPP, aggregator, fleet

Above the EMS of an individual installation typically lies a higher-order commercialisation or fleet layer. Virtual Power Plants (VPP), aggregators, and fleet operators bundle several assets into a marketable unit, optimise across the portfolio, and take over the interface to electricity markets, control reserve markets, and balance group systems.

The relation between the EMS of the individual installation and the higher-order commercialisation layer follows a similar pattern to that between BMS and EMS. The individual installation reports available power, reserve holdings, and state quantities; the commercialisation layer returns set points that lie within these limits. The governing time cadences arise from the products: 15 minutes for intraday and balance group products, seconds to minutes for control reserve activations, down to the single-digit second range for FCR activation. Communication runs through company-specific interfaces, but increasingly through standardised APIs; OpenADR 2.0b and the more recent OpenADR 3.0 are establishing themselves for demand-response-near products, while communication with the transmission system operators and balance group coordinators continues through established interfaces such as IEC 60870-5-104 and specific market communication formats. Latency requirements are product- and regulatorily dependent: for FCR in Germany the maximum activation time is 30 seconds for full power; for aFRR it is 5 minutes; mFRR products operate in 12.5- or 15-minute cadences.

The portfolio logic decides how activations are distributed across the connected assets. Modern aggregator platforms use optimisation algorithms that, alongside pure distribution, also account for state information (SOC, SOH, degradation budget), grid restrictions per feed-in point, and individual contractual boundary conditions; the individual asset thereby receives a set-point that arises from a portfolio perspective and reaches beyond the individual market signal. The decision whether and through which commercialisation structure a BESS is operated has consequences for the plant architecture. A storage system that is exclusively commercialised locally — for instance in the co-location context with a PV installation — requires different EMS equipment from a storage system in a large commercialisation portfolio that is optimised daily across different products. The portfolio logic acts back on the individual installation: which interfaces, which data quality, which response speeds, which telemetry depth are required? These questions belong to the early system conception; their relegation to the commercialisation negotiation after commissioning comes too late for the architectural design.

9.6 Coherence as a quality property

The technical quality of a BESS depends substantially on whether BMS, EMS, and PCS are designed and parameterised as a coherent hierarchy. An EMS can compute economically attractive operating modes that are inadmissible from the perspective of the BMS, or only restrictedly deliverable from the perspective of the PCS. A BMS can effectively protect the electrochemical inventory and at the same time, in a poorly coordinated overall system, produce a plant whose market and grid behaviour remains unreliable, sluggish, or commercially illegible. A PCS can be technically excellent and still fail to unfold its full potential where EMS set-points are not precisely tuned to its capabilities, or where BMS-side derating reports arrive with delay.

Coherence arises from three elements. The first is the consistent state definition across all layers: SOC, SOH, available charging and discharging power, available energy in both directions, temperature and cell-voltage quantities must be defined and measured in BMS, EMS, and PCS to the same convention; an SOC value referenced to nominal capacity in the EMS together with an ageing-normalised value in the BMS produces systematic misallocation. The second is clean interfaces with clearly defined time cadences, protocols, value ranges, failure behaviour, and fault indicators, together with time synchronisation through PTP or NTP, which alone makes cross-layer event correlation possible. The third is the coordinated parameterisation that carries the philosophy of operational management across all layers — from the protection parameter and the derating curve in the BMS, through the control parameters of the PCS, to the dispatch priority in the EMS; a conservative BMS parameterisation alongside an aggressive EMS strategy produces ongoing conflicts that render the asset inefficient and unpredictable.

This coherence is established in the engineering phase, verified in commissioning through Factory Acceptance Test and Site Acceptance Test, and maintained in operation. Firmware changes, parameter adjustments, replacement of components, or updates to the commercialisation software can disturb the coherence and must be tracked through a disciplined change-management process that comprises versioning of firmware and parameter states, regression testing against documented reference conditions, and a traceable approval chain. Every change at one layer creates a verification claim against the other two, which follows from the coherence logic itself.

Chapter 10 — SOC policy, reserve holding, and available energy

10.1 The state of charge as a strategic variable

The state of charge (SOC) appears in everyday view as the fill level of a container. For the management of a large-scale BESS this image carries only so far. SOC is a calculated, not a directly measured quantity; it is formed from current integration (Coulomb counting), open-circuit-voltage calibration during rest phases, and model-based state estimators, and integrates charging history, efficiencies, thermal management, cell dispersion, balancing state, reserve obligations, and ageing-dependent reference points into a single dimensionless quantity. The same SOC value carries a different operational content under different C-rates, temperatures, ageing states, or service bundles. The nominal capacity at beginning of life and the currently available capacity captured in SOH diverge increasingly across the lifetime; a 100-per-cent SOC report based on BoL capacity designates a different absolute energy quantity from the same report against the aged reference.

To this are added cell dispersion and kinetic effects. In a large-scale storage system with several thousand cells per string, the system SOC is an aggregate, typically formed from a weighted mapping of the cell SOCs. The BMS internally often works against the weakest cell — minimum cell SOC during discharge, maximum during charge — while the report to the outside corresponds to a mean value; the difference between the two views grows with progressing ageing, because cell dispersion and internal-resistance distribution increase over the cycles. Diffusion limitations and relaxation effects additionally generate a difference between electrochemically present and short-term retrievable charge; after a discharge pulse, the SOC reading shows a drift over several minutes that arises from the redistribution of charge carriers within the cell and does not represent a physical recharging.

From this multivalence follows the practical significance of SOC. It is the scarcest resource of the operation. Through it, reserve obligations, arbitrage options, degradation exposure, and non-delivery risks are coordinated in time. The SOC of a large-scale storage system is less a measurement quantity than an object of management; it carries the temporal structure of the commitments the asset has entered towards market, grid, warranty, and safety.

For the energy management system this means that SOC management reaches beyond the summation of individual schedule commands. It is a policy in the strict sense, a rule-based, temporally graded, and priority-based holding of the state space in that form which can absorb the competing claims of the next hours, days, and weeks. At this level the storage becomes legible as a managed infrastructure object.

10.2 The three window types — technical, operational-strategic, situative

The operationally used SOC range always lies within a graduated window system. Three window types interlock and bound the state space at different depths.

The *technical window* arises from electrochemical, thermal, and safety-related limits. It describes the charge range that is in principle reachable and permissible. The full geometric span from 0 to 100 per cent is not used in regular operation. For LFP cells (lithium iron phosphate) the technical window typically lies between 5 and 95 per cent SOC at cell voltages of around 2.5 to 3.65 V; for NMC chemistries (nickel manganese cobalt) the cell voltage limits lie more narrowly at around 3.0 to 4.2 V, with a more strongly non-linear OCV characteristic in the upper range. The window is temperature-dependent: below around 0 °C cell temperature, charging is strongly restricted or prevented for Li-ion chemistries to avoid lithium plating, unless integrated preheating brings the cells into the permissible range before charging begins; above around 50 to 55 °C, deratings or protection trips activate. The boundaries of this window are hard — exceedances lead to switching logic or damage and are negotiable neither economically nor contractually.

The *operational-strategic window* lies within the technical window. It is set on grounds of degradation discipline, warranty conditions, maintenance policy, and long-term asset stewardship. Its boundaries are not physically binding; they follow from a balance between short-term revenue breadth and long-term state quality. The Warranty Envelope is here directly effective. Typical warranty contracts define a maximum number of cycles per year (often 365 to 550 full-load equivalents), a maximum depth of discharge (often 80 to 90 per cent), an average cell operating temperature (often 25 to 35 °C), maximum permissible C-rates and, in some cases, upper SOC limits for longer standstills, because calendar ageing accelerates at high SOC and high temperature. Operation outside these window conditions shifts liability and performance commitments and generates institutional follow-on costs even where the technology would still permit the behaviour.

The *situative window* lies in turn within the operational-strategic window. It arises from reserve obligations, schedule bindings, temperature corrections, near-term safety conditions, and current market positions. It is the narrowest and at the same time the most mobile window; it changes during the day, shifts under regime changes, and is often asymmetric, because different services demand different reserves at the upper and lower flanks.

The symmetric geometry shown in Figure 3 is a snapshot of the regular case; in running operation, upper and lower reserve flanks shift independently of one another, depending on use type and ambient conditions. The decomposition into usable window, reserve window, and dispatchable window forms the operational basis of the dispatch decision: the usable window describes the currently permissible range, the reserve window the shares bound for committed services, and the dispatchable window the remainder, which is freely available for opportunistic commercialisation and ongoing optimisation.

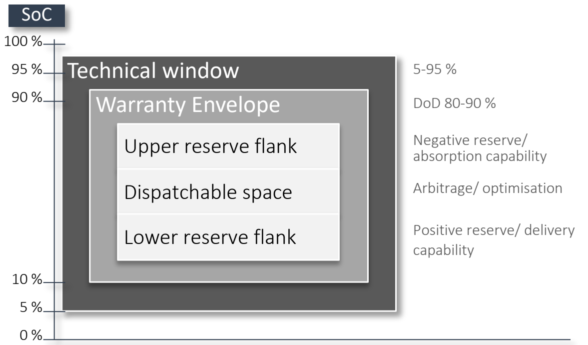

Figure 3. Operational SOC state space of a BESS. Technical limits, warranty logic, and situative reserve commitments narrow the freely dispatchable space; in running operation, upper and lower reserve flanks shift independently of one another, depending on use type and ambient conditions.

In the layering of the three windows the entire SOC policy of the asset finds its operational form. It makes visible why, of a nominal capacity, only a fraction operationally remains as free dispatch space, and it explains why the same installed capacity produces different room for action under different regimes.

10.3 Reserve holding and multi-service operation

As soon as a BESS addresses more than one service, the SOC becomes the common resource of competing claims. Frequency Containment Reserve (FCR), automatic Frequency Restoration Reserve (aFRR), manual Frequency Restoration Reserve (mFRR), intraday arbitrage, reactive-power provision, voltage support, congestion action, and contractually committed capacity each claim sections of the available state space, often simultaneously, often with different time horizons, and often with opposing reserve requirements.

The product logic shapes the reserve geometry in detail. FCR is tendered in the continental European synchronous area as a symmetric product in 4-hour slots, with full activation within 30 seconds and a sustained capability of at least 15 minutes in both directions; for an offered MW block of FCR, the BESS must in arithmetic terms hold at least 0.25 MWh per direction as a symmetric reserve and choose an SOC range that maps this bidirectional capability at all times. aFRR is commercialised asymmetrically in Germany, also in 4-hour slots, with full activation within 5 minutes and a sustained capability that typically extends beyond the bare quarter-hour; mFRR works in 15-minute cadences with longer activation times. Prequalification demands proof that the offered power can be called up under all relevant state conditions, including temperature limits, SOC margins, and active derating states.

Multi-service use in the sense of revenue stacking is real and bounded by operational blocking. Blocking designates the situation in which a part of the state space or capability is already occupied by one service and is therefore no longer available for competing services. The same SOC window cannot simultaneously be held for an activation reserve and claimed for an actively running arbitrage schedule. The same ramp capability cannot simultaneously be available for a primary-control reaction and an intraday-fast schedule change. The sum of contractually committed services therefore lies, as a rule, below the sum of theoretically possible services; the difference describes the space of technical blocking.

Reserve holding is for this reason a planning quantity with its own weight in schedule formation; its relegation to a residual quantity after schedule preparation underestimates its operational effect. It intervenes in the overall regime of dispatch, and its quality decides whether reserve commitments can in fact be activated in the event case. A reserve commitment without robust SOC policy remains a commercial assurance whose operational substance fails in the event case. Prequalification tests generate such events deliberately and systematically; in them, the difference between a disciplined-led and a merely contractually documented storage system shows itself.

The monetisation of several services from a single inventory remains attractive as long as their operational order is disciplined-led. As soon as the policy is missing, the calculation tips: reserve failures, penalties, degradation beyond expectation, warranty conflicts, and the loss of prequalification generate costs that can outweigh the additional revenues. The economics of multi-service use therefore stand only as far as the reserve holding extends.

10.4 Asymmetric reserve commitments

Reserve holding rarely acts symmetrically on the SOC window. *Positive reserve* — the capability to feed energy into the system under grid under-frequency — enforces a minimum charge. *Negative reserve* — the capability to absorb energy under over-frequency — enforces a minimum free capacity at the upper edge of the window. Where both directions are offered simultaneously, the available window shrinks at both flanks at once.

This asymmetry reaches beyond an arithmetic question and shapes the daily management. A storage system that during the day draws from a PV-near co-location and is to be discharged in the evening for arbitrage must place its reserve windows so that charging windows do not block prematurely and discharging windows do not prematurely run dry. Where charging and discharging capability are technically asymmetrically designed, the asymmetry shifts in both directions independently of one another. Typical causes are C-rate asymmetries of the cell chemistry — for LFP the permissible continuous charge rate often lies at around 0.5 C, the discharge rate at 1.0 C or higher — temperature-dependent charge deratings that bite more strongly than discharge deratings, and plant-side conditions such as transformer or filter cooling, which are stressed differently in charging phases than in discharging phases.

Parasitic loads distort the asymmetry further. HVAC systems, thermal-management pumps, the supply of BMS and control computers, PCS auxiliary circuits, and lighting generate in hold mode a base load in the order of single-digit to lower double-digit kilowatts per MW of rated power, which rises markedly with ambient temperature; at high outside temperatures the HVAC load alone can reach several per cent of rated power. In summer, or under high cycle activity, the net charging energy is therefore disproportionately reduced, while the net discharging energy is less strongly affected, because the cooling runs anyway for cell sparing.

The economic effect of this asymmetry is robustly quantified. In a merchant reference case with 10 MW and 20 MWh, 90 per cent efficiency, and a maximum of two cycles per day, an asymmetric limitation of the charging power to 6 MW leads to a revenue decline in the order of 16 per cent; a nightly blocking window from 19:00 to 06:00 reduces revenues by around 39 per cent. The figures show that the geometry of the reserve window and the temporal availability of power translate immediately into the revenue space.

For the SOC policy it follows that reserve windows are to be designed for unfavourable superposition cases, in which several services call simultaneously: simultaneous FCR call in the positive direction and aFRR call in the same direction at high ambient temperature with running arbitrage discharge. A design solely on the average expected value of the claims falls short under such superpositions. In the holding for unfavourable superposition cases lies the difference between robust and nominal reserve management.

10.5 The active policy of available energy

The dispatchable AC-side energy of a BESS arises in a graduated chain from nominal energy, through the energy usable within the technical window, less reserved energy, and after deduction of the conversion losses and parasitic loads. Each stage of this chain emerges from a decision; no stage emerges of itself.

The individual loss sources are quantifiable. The PCS efficiency of modern hardware lies typically at 97 to 98.5 per cent per direction, with a load-point-dependent characteristic that drops noticeably under partial load and particularly below around 10 per cent of rated load. The medium-voltage transformer has, depending on design, an efficiency between 98.5 and 99.5 per cent, but adds no-load losses that arise continuously irrespective of current power. Auxiliary loads — HVAC, pumps, BMS, control computers, control technology, lighting — lie in hold mode typically at 1 to 3 per cent of rated power and rise markedly under high ambient temperature or strong cell activity. In sum, an AC–AC round-trip efficiency arises that lies between 85 and 90 per cent for well-designed large-scale storage systems and can fall to 80 per cent and below under unfavourable dimensioning or a high standby share.

The active policy of available energy has several leading tasks that are related to one another in a structured rule set. It secures the usable energy within permissible windows across changing temperature conditions, partial states, and cell dispersions. It reserves those shares that are required for committed or expected services and holds them defensible against short-term more attractive alternatives. It bounds degradation-accelerating extreme states and keeps the lifecycle economics in view, particularly the combination of high SOC, high temperature, and long standstills, which drives calendar ageing disproportionately. It bounds conversion and standby losses, which can grow unnoticed in hold and reserve windows; an installation kept thermally active in continuous FCR holding accumulates parasitic energy quantities over a year that consume several per cent of gross revenue. And it structures energy availability temporally along real activation, schedule, or price windows; an averaging view does not carry this temporal structure.

The distance between a nominally described and an actively managed storage system shows itself particularly clearly in this policy. Two technically identical installations with the same MW/MWh dimensioning, the same chemistry, and the same connection can differ substantially in dispatchable energy and revenue reality, where their SOC and reserve policies are carried with different discipline. The active policy of available energy thus forms the layer at which operational quality passes into value creation.

Chapter 11 — Degradation management and functional availability

11.1 Degradation as a running operational quantity

Degradation in the BESS appears in many project descriptions as a later accounting — a percentage value that drops from 100 per cent, a curve that fills over time, a number checked against warranty thresholds. This view reaches only a limited distance for the management of a large-scale storage system. Ageing arises within dispatch; it forms a running operational quantity, not a downstream balance-sheet item. Every charge, every discharge, every temperature exposure, and every dwell state inscribe itself into the electrochemical inventory; the sum of these inscriptions forms the real state of the asset.

The electrochemical mechanisms behind this inscription are differentiated. The growth of the solid-electrolyte interphase (SEI) on the anode binds cyclable lithium, raises the cell internal resistance, and is the dominant driver of calendar ageing; its rate scales strongly with temperature and with the potential of the anode, which in turn depends on SOC. Lithium plating occurs at low temperatures and high charging rates, where lithium-ion intercalation into the graphite lattice is kinetically limited; the metallic lithium at the anode surface is in part irreversibly lost, in part forms dendrites with safety consequences. Particle cracking in active materials, through mechanical stresses under volume changes during intercalation, reduces the active surface and acts particularly on NMC cathodes; LFP is mechanically more stable but ages along SEI and electrolyte-decomposition pathways. Transition-metal dissolution from the cathode into the electrolyte, binder decomposition, and electrolyte drying complete the picture. Macroscopically, these mechanisms show themselves in two quantities: capacity fade and internal-resistance rise (power fade).

Between active operation and passive ageing no sharp boundary runs. Even the mere holding of SOC, the apparently inactive standby phase at moderate temperature, the resting over a weekend contributes to ageing. Calendar ageing and cyclic ageing act additively and in accelerating interaction. A purely cycle-based management leaves the calendar shares underexposed; a purely calendar-based view underestimates the effect of aggressive cycle regimes. Degradation management thereby means the disciplined embedding of the ageing dynamic into operating strategy, SOC policy, reserve management, maintenance cycles, and investment assessment; the narrower reading as mere warranty management falls short of the task. In this perspective the storage system becomes legible as an object whose technical truth develops in time.

11.2 Calendar and cyclic ageing

The ageing of a lithium-ion BESS decomposes into a calendar and a cyclic component. Both have their own physical drivers and can be described in formal relations.

Calendar ageing follows in first approximation an Arrhenius-type relationship. The ageing rate rises exponentially with absolute temperature and additionally increases with mean SOC. The temperature-dependent rate can be described through an activation energy E_a and the universal gas constant R as $exp(-E_\mathrm{a}/RT)$, where typical activation energies for SEI growth lie in the range of 20 to 50 kJ/mol. High SOC levels at high temperatures generate the strongest calendar stress; lower SOC levels at moderate temperatures hold ageing in the rest range. In empirical derivation, a temperature increases of around ten kelvins in the relevant operating range between 15 and 45 °C leads to a doubling to tripling of the ageing rate, depending on chemistry and SOC level. The capacity decline through calendar ageing follows over longer periods approximately a square-root-of-time dependency, which reflects the dominant role of diffusion-controlled SEI growth. This sensitivity shapes the entire thermal management of the asset — the HVAC dimensioning, the reserve policy on hot days, the seasonal calibration of the SOC mean-values.

Cyclic ageing follows a logic that approaches the Wöhler-curve analogy. The damage rises disproportionately with cycle depth (depth of discharge, DoD) and grows additionally with C-rate, average cycle temperature, and cycle count. Shallow partial cycles — for instance FCR operation with ten to twenty per cent DoD around a mean SOC — contribute markedly less damage per full-cycle equivalent than deep cycles of eighty to ninety per cent. The relationship can be transferred to real load profiles within the framework of a rainflow-counting procedure originally drawn from material fatigue; the individual partial cycles are identified and weighted with a depth-dependent damage factor. For LFP chemistries in current large-scale storage systems, manufacturers typically specify cycle endurances in the order of 6,000 to 12,000 full cycles to end of life under defined reference conditions (25 °C, DoD 80 per cent, moderate C-rate); NMC lies, depending on generation, between 3,000 and 6,000 full cycles. These values are not service guarantees but reference points in test protocols; the actual cycle endurance in the field depends directly on the superposition of the influential parameters.

A storage system that runs predominantly shallow cycles ages structurally differently from a storage system with deep diurnal cycles. The superposition of both components in practice takes a modified additive form with cross-effects between temperature and cycle regimes. High temperature accelerates both calendar and cyclic rates, high C-rates raise the local cell temperature and thereby indirectly the calendar component during cycle management. For the operational management it follows that the ageing path of the asset is shaped by the parameters of its operation as much as by its chemistry.

11.3 Energy and power SOH

The state of health (SOH) is in project practice often carried as a single number — usually as energy-related residual capacity in per cent of nominal energy. For an asset that carries power and energy in two different markets, this simplification remains too coarse. Two health concepts must be distinguished.

The *energy-related SOH (SOHE)* describes the share of nominal energy that is still extractable under reference conditions within the permissible window. It shapes the temporal duration of the asset — the question of how many hours the system can deliver under rated power — and is the governing quantity for arbitrage, long-running services, and tolling structures. Its estimation in the field proceeds through periodic reference cycles, through partial-cycle inference methods such as Incremental Capacity Analysis (ICA) and Differential Voltage Analysis (DVA), which track characteristic features of the dQ/dV- or dV/dQ-curves over time, and through model-based methods with recursive parameter estimation.

The *power-related SOH (SOHP)* describes the share of nominal power that the asset can provide under defined conditions. It shapes the rapid response and ramp capability and is the governing quantity for FCR, aFRR, voltage support, and, in perspective, grid-forming-near functions. Its governing physical basis is the rise of the cell internal resistance R_i through SEI growth, contact resistances at current collectors, and separator changes; it is typically measured through defined current-pulse tests (HPPC, Hybrid Pulse Power Characterization) or through the evaluation of DC resistance from running operating data. Under certain operating regimes, SOHP ages faster than SOHE, because internal resistance rises act disproportionately on available power, while capacity losses remain more moderate.

SOC and SOH must not be conceptually conflated. Both are estimation problems with different time logics: SOC fluctuates in operation across minutes and hours and is derived from current integration, voltage information, and model assumptions; SOH changes across months and years and describes capacity loss and resistance rise. In Coulomb counting, integration errors accumulate, and the available battery capacity must, precisely on grounds of ageing, not be presupposed as a fixed quantity in operation; equally, an OCV-based SOC estimation is robust only under resting, temperature-stable conditions, a precondition often not met in field operation. An operationally used SOC is therefore not a simple fill-level value but a model-supported working value whose informational content depends on ageing, temperature, C-rate, rest periods, and the quality of the estimation algorithms used. For SOH this holds in still stronger measure, as capacity and internal resistance are not directly accessible in the field but only inferentially.

The distinction of SOHE and SOHP is directly relevant for asset management and revenue planning. A storage system with SOHE at 85 per cent can already lie in its capability at around 75 per cent. A management that exclusively tracks SOHE does not capture this distance and gradually loses access to power-intensive services without reflecting this in the management metrics. A robust management therefore works with both quantities in parallel and additionally keeps cell dispersion in view, because the most strongly aged cells of a string already determine the available power and energy of the overall system before the mean SOH makes this visible.

11.4 Nominal and functional availability

The availability of a BESS belongs to the most misunderstood quantities in storage operation. Formally, it can be described as the share of time in which the system is operational. This *nominal availability* captures the outer shell of the asset — the question of whether the installation can in principle be run.

The *functional availability* goes further. It describes in what extent, and under which conditions the system can in fact deliver a concretely required performance in the required period. It is service-specific. An installation can be fully available for spot-market operation, but only restrictedly available for a particular reserve provision — for instance because the temperature reaches the boundary range, because the SOC reserve has been consumed by a prior activation, because short-term current limits constrain the ramp, or because maintenance windows cap the full power.

Functional availability is therefore a family of availabilities, each relating to a concrete function. For the same installation over the same period, the availabilities for FCR, aFRR, reactive power, voltage support, and arbitrage can fall apart substantially. In measurement terms, it can be described as the share of time in which the system simultaneously satisfies the product-specific conditions — reservation level, response time, sustained capability, ramp capability, temperature and SOC boundary conditions. This share is operationally verifiable and becomes, in event cases, the governing quantity for the attribution of performance and penalty.

The practical consequence is considerable. Contracts that govern availability without functional reference generate grey zones; in event cases it becomes contested whether the asset has fulfilled its commitment. Insurances that measure availability as a global value often recognise degradation-related performance losses with delay. Financiers who rely on nominal availability map the gradual devaluation of the asset only incompletely. Function-related availability sustains the institutional legibility of the BESS across its lifetime.

11.5 Degradation management as operational discipline

The insight that ageing runs along and that availability is to be read functionally leads to a concrete operational discipline. Degradation management requires that operating mode, SOC policy, thermal management, maintenance regime, and revenue strategy are managed under a common order.

The first level of this discipline concerns parameterisation. Charge and discharge limits, temperature windows, C-rate limits, and SOC bands are adjusted across the lifetime, because the ageing behaviour of the real asset diverges from the original model and because the weights between short-term revenue maximisation and long-term substance preservation shift; a one-time fixing under supplier recommendation is too narrow for this adjustment logic. Commissioning marks in this perspective the transition from construction into operation, and at the same time the first robust evidence layer of the asset; from this point on, the technical identity of the system rests on verified states, parameterisations, measurement values, protection coordination's, and demonstrable reactions of the real system, no longer exclusively on design, datasheets, and delivery documents. Only where reference states, test boundaries, deviations, parameter states, and first operating data are cleanly secured does a robust basis arise for later fault analysis, performance evaluation, warranty attribution, optimisation, and lifetime management. Interoperability and test standards such as IEEE 2030.2, IEEE 2030.3, and IEC 61850 with its extension IEC 61850-7-420 for Distributed Energy Resources form the frame in which this evidence becomes sustainable through standardisable exchange and demonstration structures; their grounding thereby reaches beyond a binding to nominal values and function commitments.

The second level concerns the feedback of degradation evidence into decisions. Measurement values on internal resistance, capacity loss, balancing quality, temperature gradients, cell dispersion, and calendar shares are, beyond documentation, input quantities of operational management. They influence which services are still offered, which warranty claims remain defensible, and which augmentation or retrofit decisions impose themselves. Quantitatively, degradation costs can be captured as an energy-quantity-related value erosion: the damage contribution of a partial cycle, valued with the specific replacement or augmentation costs per kilowatt-hour of usable capacity, yields a marginal degradation price per dispatched kilowatt-hour, which can be set in dispatch optimisation as opportunity cost against the expected revenue. In high-performance EMS implementations, this price is dynamically parameterised, depending on temperature, SOC range, and current cell dispersion; it separates degradation-favourable from degradation-expensive revenue opportunities and acts back directly on the dispatch decision.

The third level concerns the temporal coupling between degradation management and commercial strategy. A storage system that is aggressively optimised for maximum revenues in its first years can undermine its lifecycle economics before the later market conditions arrive; an over-conservative operation leaves attractive revenues unused in volatile phases. The task lies in the temporally graded balance between the two poles, which a one-time fixing does not deliver. The parameterisation history, the operating-regime changes, and the associated degradation curves form, across the years, a robust basis for the running adjustment of this balance.

Chapter 12 — Data architecture, evidence, and market integration

12.1 Data as an operational layer of the asset

The data layer of a large-scale BESS has long left the status of an accompanying monitoring. It has become a constitutive layer of the system — part of the operational capability, part of the demonstration capability, and part of the commercial connection capability. The storage system is described not only by its data; it is managed, settled, aged, insured, and, in event cases, reconstructed through them.

The operational relevance shows itself in several time layers at once. At the millisecond-to-second level the data layer carries protection functions, control coupling, and plant safety, with sampling rates in the kilohertz range for PCS-internal control quantities and in the millisecond cadence for GOOSE-based protection communication under IEC 61850. At the second-to-minute level it carries the execution of dispatch, reserve activation, and set-point following; the relevant quantities here — active and reactive power, SOC, cell-voltage extremes, temperature gradients, available power in both directions — are typically logged in the 1- to 10-second cadence. At the hour-to-day level it carries performance balancing, KPI formation, and schedule alignment in the 15-minute cadence. At the week-to-year level it carries degradation management, warranty demonstrations, and lifecycle decisions. These four-time layers act in coupled form; the upper management carries only to the extent that the lower base carries.

The resulting data volume is substantial. A large-scale storage system with several thousand monitored cells, hundreds of temperature sensors, dozens of powers and voltage measurement points, protection events, and state quantities generates in normal operation several gigabytes of raw data per day. The architecture of data holding distributes these volumes typically across three layers: a hot store with time-series database for immediate evaluation and operational dashboards in the time horizon of hours to a few days, a warm store for KPI formation, reference cycles, and warranty demonstrations across months to a few years, and a cold store for long-term documentation across the entire lifetime, typically compressed and at reduced temporal resolution but with an unabridged event trace. For all three layers, data formats, time-stamp convention (UTC with documented precision), measurement quantity definition, and aggregation rules are to be fixed in advance, because later corrections break the chain of evidence.

The quality of the data layer is therefore a property of the system itself. A storage system with patchy, inconsistent, or organisationally separated data paths loses, with progressing operating time, its operational truth and thereby the substance on which refinancing, augmentation, and operator change build.

12.2 KPI system and evidence continuity

A robust KPI system maps the real management geometry of the asset, instead of following a generic dashboard logic. Five classes carry the framework. *Energy-related KPI* capture throughput, efficiencies, and dispatchable energy and can be cast in quantitatively defined quantities: AC–AC round-trip efficiency as the quotient of discharged to charged energy across a defined reference cycle, separately reported DC–DC efficiency, auxiliary energy consumption as a share of gross energy, and standby power in hold and reserve windows. *Power-related KPI* capture ramp behaviour (dP/dt in MW/s or as a percentage of rated power per second), overload capability (permissible overload factor and sustained time), reactive-power provision (delivered Q in MVAr at defined voltage), and response quality (time to reach 90 per cent of set-point, overshoot, settling time). *State-related KPI* capture SOC, SOH, and temperature management including cell dispersion, balancing quality, and temperature gradients between modules. *Reliability-related KPI* capture availability in nominal and functional reading, separated by product and time reference, with MTBF and MTTR values for the essential system components. *Lifecycle-related KPI* capture equivalent full cycles (EFC) across a rainflow count, calendar exposure (temperature-time-SOC integral), and accumulated ageing work.

For the robustness of the KPI, the informational content of a demonstration quantity depends on system boundary, time reference, and operating condition. An efficiency that comprises only the PCS deviates systematically from an efficiency including transformer and auxiliary energy. An availability across the entire year can show good values, while the availability in the valuable hours lies markedly below it. Every KPI definition must therefore precisely fix system boundary, reference conditions, temporal resolution, and exclusion criteria, and carry these durably in the documentation; comparability across time and between assets presupposes this precision.

Decisive is *evidence continuity* — the capability to carry forward the technical history of the asset in a form that supports later diagnosis, warranty, audit, and reconfiguration. Measurement data, event chains, parameterisation histories, firmware states, test results, interventions, module replacements, and performance calculations must lie in a re-constructable structure. Time synchronisation across all system layers forms the basis of this re-constructability: PTP (IEEE 1588) for reaction-critical components with microsecond precision, NTP for the broader plant communication with millisecond precision. Without a consistent time base, event chains between BMS, PCS, protection relays, and EMS cannot be robustly reconstructed in retrospect; a protection trip in the 10-millisecond range is not attributable to EMS dispatch decisions without a synchronous reference. This continuity is the substance against which warranty and operator change are decided.

Breaks in evidence continuity rarely arise catastrophically. They arise in transitions — at the change from commissioning to regular operation, at the change of EMS supplier, in firmware updates, at operator change, at the first major retrofit. Every one of these transition points demands an explicit transfer discipline: versioning of the replaced and new configuration, parallel operation of the data paths over a defined comparison period, migration protocol of the historical data holdings with unambiguous origin labelling, and acceptance protocol with reference cycle before and after the transition. Without this discipline, the transition remains as a silent gap in the system's memory.

12.3 Local operating intelligence and the edge layer

Between field device, station logic, and higher-order corporate IT, an operational intermediate layer emerges that is becoming increasingly constitutive for large-scale electrical infrastructures. The edge layer processes data near the plant, coordinates update’s, performs backups, lowers latency, and supports autonomy in the event of fault. Reaction-critical functions remain closer to the asset; aggregating and fleet-related evaluations can be placed higher.

The task allocation between edge and cloud follows in practice a layering by time-criticality, data volume, and security requirements. At edge level remain protection functions, fast control loops, safety-related shutdown logic, the immediate dispatch execution, local state estimation, and the first stage of data preprocessing with filtering, aggregation, and compression.

At cloud level lie portfolio-wide optimisation, forecasting models for prices and grid states, long-term analysis, KPI reporting, and the integration into higher-order market communication. The edge level must bridge the loss of cloud connectivity without loss of controllability: typically, with a local fallback dispatch under predefined rules, local data buffering across at least several days, and autonomous resumption of communication after restoration of the connection. The storage system thereby becomes a cyber-physical operating object, whose physical behaviour and digital representation carry one another reciprocally.

Communication standards structure the layering. IEC 61850 with MMS, GOOSE, and Sampled Values carries the plant-internal protection and control communication, OPC UA with its role-based security model the cross-application coupling, and MQTT with TLS encryption the fleet-near telemetry. For market communication, IEC 60870-5-104 and increasingly IEC 61850 itself are common, for aggregator interfaces OpenADR 2.0b and 3.0. The choice of standards is a decision about later integrability, operator change, and extension; proprietary paths appear economically attractive in the early phase and act in later lifecycle phases as structural impediments.

The cyber-security of this layer carries structural significance for the entire asset. A BESS that takes on grid- or market-critical functions lies within the scope of the NIS2 Directive in its German implementation and thereby falls under the KRITIS regime of the BSI Act, as soon as it exceeds a threshold of grid-relevant power or becomes part of a qualified portfolio. The relevant technical standards are IEC 62443 for the protection of industrial control systems and ISO/IEC 27001 for the higher-order information security management. The base architecture demands role-based access control with two-factor authentication for remote access, a clear segmentation between OT and IT networks through dedicated firewalls and data diodes for unidirectional data flow into sensitive areas, patch capability with regulated maintenance windows and a staging environment, full logging of all configuration changes and privileged interventions in a tamper-resistant form, intrusion detection at OT level with adapted signatures for industrial protocols, and a robust incident-response structure with a defined reporting chain. The secure restart after a communication incident, a power outage of the control technology, or a detected intervention belongs to the base architecture and reaches beyond a downstream operating phase.

12.4 Digital representation and the digital twin

The digital representation of a BESS gains its precise standing from the capability to bring together physical structure, parameters, states, test results, interventions, and operating data consistently; its reach lies above the mere doubling of the installation in data space. A digital twin in this sense is a means of coherence assurance across design, commissioning, operation, modification, and return.

In its operational form, the twin lays several layers above one another. A *static layer* holds the topological and material structure of the asset: plant hierarchy from overall system through strings, racks, and modules to the cell, serial numbers and batch assignment, cell chemistry and production date, physical arrangement, electrical interconnection, and protection-device assignment. A *parametric layer* holds the currently valid settings: BMS protection limits, PCS control parameters, EMS dispatch rules, prequalification conditions, each with versioning and validity range. A *dynamic layer* holds the state quantities in the relevant time cadences. A *model layer* holds physical or data-driven models for SOC and SOH estimation, thermal behaviour, and ageing forecast, which are calibrated against measurement data and continuously updated in operation. An *event layer* holds the history of interventions: parameterisations, firmware updates, module replacements, maintenance, acceptance tests, protection trips, and alarms.

The practical reach of this representation shows itself in concrete tasks. Under parameter changes, the connection between old and new setting must remain traceable. Under module replacement, the assignment of batches, cell state, and performance history must be preserved, so that the new inventory can be embedded into the existing ageing history rather than overwriting it. Under firmware updates, the behaviour before and after the update must be comparable against a defined reference cycle. Under warranty dispute, the documented operating mode must be attributable to the originally agreed operating regime, including C-rate histogram, DoD distribution, temperature history, and cycle count under the respective conditions. Under augmentation, the relation of old and new inventory in each string must be transparent, because from this follows the ageing heterogeneity of the overall installation.

The digital twin thereby carries the identity of the asset across time. In its absence, the storage system loses its technical legibility after a few years; the measurement data grow without condensing into knowledge. In its presence, the asset remains legible as an object, even where owners, operators, marketers, or market roles change. The coming battery passport logic under the European Battery Regulation strengthens this requirement, because material origin, cell characteristics, lifecycle data, and circular information become continuing components of the object's history; the digital representation thereby becomes simultaneously a regulatory demonstration instrument.

12.5 Balancing capability and operational market integration

As soon as a BESS is operated market- or grid-near, the physical connection capability does not suffice for orderly operation. The asset must be embedded into the formal world of schedule, balance group, measurement, activation, settlement, reserve logic, and accounting. This layer is at once commercial, regulatory, and operating-architectural; it belongs in the base structure of the asset.

Balancing capability designates the capability of the asset to render its behaviour legible, callable, and accountable in standardised energy and capacity regimes. Quarter-hour-precise measurement under the requirements of the German Measurement and Calibration Act, RLM (Registered Power Measurement) meters with certification by a notified body, data acquisition through a smart-meter gateway or comparable certified interface, assignment to a balance group, schedule reporting in defined time windows, and transmission to the responsible market roles do not act separately; a break at one point can endanger the entire claim of a month. The market communication follows standardised processes of the BNetzA: *MaBiS* (Marktregeln für die

Bilanzkreisabrechnung Strom — market rules for balance-group settlement) for balance-group accounting, *GPKE* (Geschäftsprozesse zur Kundenbelieferung mit Elektrizität — business processes for customer supply with electricity) for processes between market roles, *WiM* (Wechselprozesse im Messwesen — switching processes in metering) for the change in metering, and *Redispatch 2.0* for grid-related interventions. Each of these processes defines its own message types (EDIFACT-based), deadlines, and data formats, the observance of which is a precondition of revenue realisation.

The role allocation structures the operational environment. Operator of the technical installation, connection network operator, balance-group responsible party, supplier, marketer, direct marketer, metering point operator, and where relevant transmission system operator as partner for system services operate in formally separated but operationally tightly interlocked roles. For a storage system that commercialises aFRR in Germany, the interfaces to the connection TSO are relevant in real time (activation, actual delivery, feedback), while spot-market participation runs through EPEX Spot with its own cadence and deadlines; both interfaces must be served technically and procedurally simultaneously.

The operational market integration reaches beyond participation in the spot market. It comprises the transition from the physical behaviour of the storage system into an institutional order of time cadences, product definitions, activation logics, data obligations, prequalification regimes, and settlement. The European frame — particularly the Guideline on Capacity Allocation and Congestion Management (CACM) and the Guideline on Electricity Balancing (EB-GL) — sets the regulatory architecture; the European platforms PICASSO for aFRR and MARI for mFRR integrate the national reserve markets increasingly into a Europe-wide balancing architecture, with corresponding effect on the product definitions and the technical requirements on prequalified installations.

From this follows a double insight. The market logic determines operational management down into reserve windows, SOC policy, and response times; the storage system aligns its inner behaviour with the product architecture in which it participates. At the same time, the storage system remains, even under market-based operation, a grid object, because its real effectiveness remains embedded in a physical system with congestions, voltage limits, and safety criteria. The grid-side restrictions act through the market logic; redispatch instructions, network-operator-side congestion-management measures, and local voltage requirements can override market positions ex post and are to be anchored in the dispatch architecture as a priority of their own.

Part IV — Economic operation, safety, and institutional embedding

Chapter 13 — Safety as a property of permitted operation

13.1 Safety as property, not as layer

Safety in the large-scale BESS is a property of permitted operation itself; a supplementary protection layer placed over an otherwise finished design does not carry this task. Its substance lies in the system's capability to remain in controllable states, or to be brought into them, under deviation, fault, ageing, partial unavailability, and incomplete information. Fire detection and extinguishing technology, access rules, and protection relays are instruments within an architecture whose quality lies in the coherence of the permitted state space.

A safe storage system is, in this reading, fault-tolerant in the sense that fault evolutions can be bounded, detected, isolated, and transferred into a controlled residual state. This reading shifts the attention from the notion of an event-free operation to the question of which behaviour remains controllable under which deviation. In this shift, the difference between a formally erected and an infrastructurally robust storage system marks itself.

Methodologically the established procedures of functional safety from electrical plant engineering apply. IEC 61508 as overarching foundational standard with its derived sector-specific standards defines the procedure: hazard analysis in structured form (HAZOP, FMEA, or equivalent), assignment of a Safety Integrity Level (SIL) to each safety-related function on the basis of quantified risk measures, technical implementation of these functions under defined reliability requirements, and documented demonstration across the full lifecycle. For the BESS this means that protection functions are to be qualified not only with respect to their effectiveness, but also with respect to their failure probability, diagnostic depth, and proof-test intervals.

With growing voltage level, power density, and commercialisation intensity the importance of this management logic rises. Market opportunity, aggressive state utilisation, and high technical compactness widen the distance between nominal capability and safely controllable operation, where this management is missing. Safety begins, then, with the determination of the state space in which the asset may be run at all; fire protection and access control build on this determination.

13.2 The permitted state space as a safety object

The permitted state space introduced in Chapter 8 as an operational geometry gains in safety-related light a second contour. Its boundaries are condensed safety decisions, not pure operating specifications. SOC windows, temperature windows, C-rate limits, isolation conditions, short-term current limits, communication requirements, and intervention rules carry safety logic; their violation is an endangerment, not only a performance problem.

This order is graduated. The *hard limits* are physically anchored and can be cast in concrete values: cell-voltage maxima typically at 3.65 V for LFP and 4.2 V for NMC, cell-voltage minima at 2.5 V and 3.0 V respectively, cell-temperature upper thresholds in the range of 55 to 60 °C for operation and 65 to 70 °C for protection action, and insulation resistance against earth typically above several hundred ohms per volt of rated voltage under the requirements from IEC 62477-1 and the relevant grid connection rules. Their violation triggers automatic protection reactions and withdraws itself from operational weighing. The *soft limits* are operationally set and lie at a safety distance within the hard limits: reserve windows, maintenance states, alarm corridors, degradation-related derating thresholds. Their violation generates no immediate endangerment, but reduces the distance to the hard limits. Between the two lies a *precautionary window* in which the system remains controllable and interventions can take effect without protection trip.

The verification logic of this layered state space operates in several time horizons. At sub-second level, hardware interlocks and protection trips from the BMS act independently of the higher-order control logic when hard limits are exceeded. At second-to-minute level, rule-bound permissives in the PCS and event logics in the EMS monitor soft limits and trigger preventive deratings. At hour-to-day level, periodic state checks (BIT, background diagnostics, balancing quality) evaluate the cumulative distance to the limits and prepare maintenance decisions. At month-to-year level, proof tests of the safety functions themselves run, typically as Factory- and Site-Acceptance-near repetition tests with defined test coverage under IEC 61508.

The safety-related quality of the state space arises from the discipline with which the layering is held. A storage system that regularly enters the soft limits under operational pressure loses its precautionary window and moves gradually closer to the hard limits. The alarm frequency rises, the reserve between normal and safety-relevant state melts away, and in event cases the system reacts more tightly and less forgivingly. The dissolution of safety lies in this case not in the event but in the running habituation to reduced distances.

13.3 Safety under degradation and regime change

The safety characteristic of a BESS is not static. It shifts with ageing, with interventions on the system, with regulatory regime changes, and with changed market roles. An installation that was qualified as safe under initial parameters may, under later operating conditions, require tighter limits; this follows from the changed state position of the asset and not from a deterioration.

Ageing acts in this in several directions at once. The cell internal resistance rises typically in the order of 20 to 50 per cent across the specified lifetime, in extreme cases markedly above; this reinforces the warming under given current loading and changes the thermal limit curve. Uneven loading generates increasing voltage drift between cells, so that the system reaches the cell-voltage limit of the weakest cell earlier at the same mean SOC; the balancing load increases, and the effectively usable capacity is determined by the worst cells. Temperature courses become more inhomogeneous, because ageing gradients within modules and strings reinforce heat generation locally. The reaction to high C-rates becomes more sensitive, because the combination of raised internal resistance and reduced capacity raises the specific loading per kilowatt-hour of usable energy. The hard safety limits remain at the same place, but the asset reaches them more quickly under the same operating modes. The protection philosophy must respond to this — through adaptive limits coupled to the current SOH, through tighter reserve windows, through more careful thermal management, and through a revision of the propagation assumptions after defined ageing intervals.

Regime changes sharpen this dynamic. An installation that was qualified for diurnal arbitrage operation behaves differently under a new reserve service with high cycle frequency; the thermal input rises, cell dispersion increases more quickly, the maintenance interval shifts. An installation that was set under grid-following and is now to run grid-forming experiences different electrical loading: the power electronics is operated in voltage control instead of current control, the short-circuit current contribution changes, fault ride-through profiles run differently, and the thermal loading of the semiconductors under more frequent transient peaks rises. Structural interventions — module replacement, extension, firmware update — generate a new demonstration space in which the existing safety commitment does not automatically remain valid; the inserted components bring their own ageing histories or new parameter ranges, and the coherence of the protection grading must be newly aligned under the changed conditions.

Safety is therefore a property to be confirmed continuously. Its validity hangs on the evidence of its being carried forward; a one-time issued demonstration does not sustain it across the lifetime. Technically, this carrying forward requires that every parameterisation, firmware, or configuration change is examined and documented in a change-management process against the existing safety analysis, and that safety-related metrics — alarm frequencies, derating time-shares, isolation drift, cell dispersion, temperature gradients — remain visible as running quantities in operational management.

13.4 Cyber-physical safety and the digital fault chain

Safety in the BESS does not run only along physical paths. Communication faults, false state pictures, disturbed data paths, and faulty automation reactions can reinforce or first trigger physical fault paths. The safety characteristic of the system is therefore a coupling of material and digital evolution logic.

This insight has practical depth. A falsified SOC value from a sensor or communication problem can move the EMS to charging commands that, in combination with the actual cell position, become dangerous. A delayed telegram can block a protection action long enough for the local fault to escalate; in GOOSE-based protection paths under IEC 61850, watchdog and test-counter mechanisms are foreseen for these cases, the correct parameterisation of which belongs to the safety function itself. An uncoordinated firmware update can shift alarm semantics, change threshold values, or redefine event codes without operational management noticing this in real time. The edge layer carries here a safety-critical role; it holds local controllability even when higher layers are disturbed.

Cyber-attacks enter as an additional class of deviations into this order. The relevant attack vectors for BESS control technology comprise manipulation of state values through compromised sensors or bus communication (false data injection), targeted shutdown or slowing of protection functions through denial-of-service on the control technology networks, injection of manipulated set-points through compromised EMS or VPP interfaces, unauthorised firmware updates with altered protection parameters, and access to remote-maintenance accesses with takeover of administrative roles. The consequences range from coordinated mis-trips across a portfolio of prequalified installations through to deliberately generated cell states with propagation potential. Large batteries in grid-critical functions — frequency holding, voltage support, grid-forming behaviour — therefore fall under the scope of the NIS2 Directive in its German implementation and the associated KRITIS requirements of the BSI. The normative basis is formed by IEC 62443 with its seven Foundational Requirements (identification and authentication, use control, system integrity, data confidentiality, restricted data flow, timely response to events, and availability of resources) and the Security

Levels SL 1 to SL 4 derived from them, the application of which to the BESS is to be differentiated across the architecture levels of the control technology.

13.5 Safety demonstration and institutional validity

Safety must be established and held demonstrably. For the BESS this gives rise to a second order of safety architecture — the evidence of its own validity. Tests, parameterisation states, firmware changes, interventions in limit values, alarm histories, maintenance events, fault analyses, and repetition tests together form the documented safety reality of the asset.

The normative bases of this demonstration are differentiated. At cell level, transport tests under UN 38.3 and cell safety tests under IEC 62619 apply. At system level, the safety requirements for stationary storage systems under IEC 62933-5-2 and the requirements for grid connection under IEC 62477-1 and the relevant VDE application rules 4110 (LV/MV), 4120 (HV), and 4140 (EHV) are effective. The propagation test under UL 9540A examines the thermal propagation behaviour under defined conditions and supplies the data basis for the spacing and installation logic under NFPA 855 and the European and German requirements built on it; in Germany, building law, TA Lärm, the VdS guidelines for lithium-ion storage systems, and where relevant the Major Accident Ordinance (Störfallverordnung) depending on installation size and substance inventory are additionally to be considered. The DIN VDE 0510 series and the emerging standards of IEC TC 120 carry these requirements further.

The institutional reach of the evidence is considerable. Insurability, creditworthiness, operator liability, warranty enforcement, and the assessment after events depend on the quality of the carried-forward safety history. Under augmentation, retrofit, operator change, hybridisation, or changed market role, the importance of this history rises strongly. A parameter set is releasable only where it remains permissible safety-side, operations-side, and contractually; each of these three dimensions presupposes its own examination, and the release is sustainable only when all three converge in a documented connection.

The safety architecture of a BESS is therefore doubly laid out. It is a real protection and management structure and at the same time a documented, auditable, reconstructible structure of validity. Only their cooperation makes the asset durably robust. The evidence continuity from Chapter 12 gains at this point its safety-related depth; it carries not only the performance and warranty history, but the validity of safety itself.

Chapter 14 — Fault states, propagation, and protection concepts

14.1 Fault states in the coupled system

The safety characteristic of the BESS arises from the coupling of electrical, thermal, chemical, and control-related processes. Local deviations rarely remain local. Temperature rises change internal resistances and current distributions; uneven loading accelerates ageing and voltage drift; faulty parameterisation can shift charge and discharge limits into ranges in which electrochemical and thermal instabilities reinforce one another. From individual disturbances arise fault paths.

The classes of relevant fault states can be ordered along the physical layers of action. At the *electrical level* act short circuits between poles, between modules, or against earth; insulation breakdowns following mechanical damage or moisture ingress; earth faults with the special challenges in the IT (isolated terra) network of the DC side (the absence of earthing of the storage voltage requires continuous insulation monitoring instead of earth-fault relays); overcurrent situations through short circuit or load-distribution faults; connection degradation at bolted or welded joints with local resistance rise and warming; and uncontrolled voltage deviations through control faults or sensor failure. At the *electrochemical level* act lithium plating during low-temperature charging, SEI decomposition under overheating, internal short circuits through dendrite formation or mechanically induced separator damage, electrolyte breakdown with gas formation, and mechanical cell damage through pressure build-up or external impact. At the *thermal level* act cooling failure through pump, chiller, or HVAC defect; uncontrolled exotherms from advanced cell ageing or maloperation; inhomogeneous temperature distribution with local hotspots; and the transition into thermal runaway. At the *control-related level* act faulty sensor data through calibration drift, cable break, or EMC ingress; control oscillations through faulty parameterisation or coupling between controllers; lost communication between BMS, PCS, and EMS; and misinterpretations by EMS or BMS under ambiguous state data.

Faults migrate between these levels. A beginning internal short circuit at cell level generates heat that overwhelms the thermal management; the BMS registers deviations and intervenes, but can no longer catch up with the chemical dynamic once self-heating exceeds a threshold value. A cooling disturbance raises the operating temperature, accelerates the electrochemical degradation, and the rising internal resistance reinforces the heat generation in turn. A misreport in the BMS lets the EMS issue an arithmetically permissible charging command that, at the actual cell position, leads to over-voltage and electrolyte decomposition. A single fault class is therefore insufficient as an assessment frame; the quality of the safety architecture shows itself in the capability to recognise and interrupt transitions between levels before a self-reinforcing path establishes itself.

14.2 Thermal runaway and propagation logic

The thermal runaway of a lithium-ion cell is the most thoroughly studied failure scenario of the BESS, and at the same time the one that shapes the architecture of the overall system most strongly. It describes a self-reinforcing process in which exothermic decomposition reactions generate heat faster than it can be dissipated.

The kinetic sequence runs chemistry-specifically differently, but follows a recurring pattern. At around 80 to 120 °C, SEI decomposition at the anode begins with moderate heat generation. Between 120 and 180 °C, the reaction between intercalated lithium and electrolyte sets in. Between 130 and 200 °C, the polyolefin separator fails mechanically (shutdown function and subsequent meltdown), which can lead to internal short circuits. Above 180 to 250 °C, cathode decomposition begins, for NMC under oxygen release from the crystal lattice, for LFP markedly later and with lower oxygen activity. Electrolyte ignition occurs as soon as the released combustible gases meet oxygen and an ignition source. The resulting energy release differs substantially: LFP cells show in runaway onset temperatures above 230 °C, a markedly slower temperature rise rate, and an energy release that typically lies a factor of two to three below comparable NMC cells. This difference shapes the entire safety philosophy — the dominant use of LFP in current large-scale storage systems follows substantially from this propagation friendliness.

For the system level, the propagation is decisive, not the runaway of the individual cell. A lost cell is a technical event; a propagated chain of cells is a fire event; a propagated module block is a container event; a propagated container block is a plant event. The quality of the architecture shows itself in the interruption of each of these escalation stages.

Propagation logic demands decisions at several levels at once. At the *cell level* act material choice (LFP vs. NMC vs. emerging chemistries such as LMFP or sodium-ion), separator design with ceramic coating for temperature stability, form factor (prismatic, cylindrical, pouch), and integrated pressure-relief elements. At the *module level* act cell spacings, thermal barriers from aerogel or mica materials between cells, pressure-relief paths, and structured gas conduction outwards. At the *rack and string level* act dividing walls, isolation distances, ventilation, and electrical separability of individual strings through DC contactors with load-switching capability. At the *container level* act fire compartments under the requirements of German building law and the relevant fire-protection concepts, distances between containers, gas detection, and explosion relief. At the *plant level* act area arrangement, access routes, fire-water provision, and surroundings.

Relevant test and demonstration standards give this fabric a robust order. UL 9540A defines a four-level test procedure: cell level (thermal runaway of a cell), module level (propagation within a module), unit level (propagation between modules within a unit), and installation level (propagation onto adjacent units). The test results supply the empirical basis for the spacing assessment under NFPA 855 and the European and German concepts derived from it. Typical minimum distances between containers lie in the order of 2.4 to 3.0 metres, where the UL 9540A test at unit level has shown no propagation; under demonstrated propagation limitation, distances can fall lower; under absent demonstration, correspondingly larger. In Germany, VdS 3103 (lithium-ion energy storage), the Workplace Ordinance (Arbeitsstättenverordnung), the building-permission practice of the federal states, and where relevant the Major Accident Ordinance (Störfallverordnung) on exceedance of the quantity thresholds of the 12. BImSchV (Zwölfte Bundes-Immissionsschutzverordnung, the Twelfth Federal Immission Control Ordinance) flank the safety-technical classification.

14.3 Gas management, detection, and early intervention

Before thermal runaway in the strict sense, gases typically arise. The composition comprises electrolyte vapours from dimethyl carbonate, ethyl carbonate, and ethyl methyl carbonate; hydrogen as a product of electrolyte decomposition; carbon monoxide and carbon dioxide; methane and other low-molecular-weight hydrocarbons; and fluorine-containing decomposition products from the conducting salt $LiPF_6$, including hydrogen fluoride with considerable toxic effect. This gas phase carries the most important early-warning time of the system and makes the difference between a detected, bounded event and an escalating fire or explosion event.

The safety-technical reach of this early-warning time is considerable. The detection technology combines several principles of action, because no individual sensor covers the entire spectrum with sufficient reaction speed. Hydrogen detection through catalytic or electrochemical sensors responds early, before flammable concentrations are reached (lower explosion limit around 4 vol-%, detection thresholds typically at 10 to 25 per cent of the LEL). VOC measurement through photoionisation detectors captures the electrolyte vapours before the actual smoke development. Smoke detectors in different principles of action (photoelectric, ionisation, aspirating smoke detectors with high sensitivity) carry the conventional fire detection. CO sensors react to the decomposition phase. Temperature-difference sensing and infrared monitoring detect local hotspots before these transition into thermal runaway. Fibre-optic temperature measurement along module rows enables a spatially resolved monitoring with point resolution in the centimetre range.

The intervention is multi-stage. A first stage consists in the reduction or shutdown of individual rack strings as soon as first gas indications occur; the affected inventory is electrically disconnected without losing the entire container. A second stage activates additional ventilation in order to hold the gas concentration below the explosion limits; the ventilation design follows the expected gas release from the determining runaway scenario. A third stage disconnects the entire container unit from the grid and transfers it into a defined rest state. The explosion problem arises where gases accumulate in closed containers before ignition sources become effective; deflagration relief through defined pressure-relief surfaces under NFPA 68, active ventilation under trigger thresholds, and the spatial separation of electrical switching components and gas-carrying areas belong to the base architecture of the container. Fire-fighting with water has, in lithium-ion events, a cooling function and bounds the propagation, but does not stop a running runaway; the required fire-water quantities lie, depending on installation size, in the range of several hundred cubic metres across the deployment duration, which is to flow into the design of the fire-water reserve and the retention.

Coordination with the local fire brigades and disaster-protection structures belongs in the preparation of operation. Intervention guidelines, access routes, information paths on battery chemistry and plant state, fire-water reserves, retention volumes for contaminated fire-water, restricted areas, and smoke-propagation models are operational parts of the safety architecture. A technically safe BESS whose external response forces are not capable of action in the event case remains institutionally incomplete. Regular exercises with the local fire brigade, embedded in the alarm and emergency-response planning, belong in the operating regime.

14.4 Protection concepts, shutdown logic, and selectivity

Protection concepts are often described along individual protection devices — overcurrent protection, earth-fault protection, differential protection, voltage protection, frequency protection. For the BESS this view is insufficient. Relevant is the interaction of limit-value logic, protection grading, shutdown behaviour, isolation strategy, auxiliary-system reaction, and restart condition, as shown in Figure 4. These quantities form a common regime.

The protection structure extends across several voltage levels. At the *cell and module level*, the cell-monitoring ICs of the BMS work with shutdown thresholds for voltage and temperature in the millisecond range, coupled with pyrotechnic or electromechanical cell disconnections in individual cell formats. At the *string and rack level*, DC contactors and high-current fuses act to separate the string from the busbar system, complemented by insulation monitoring with thresholds in the range of 100 to 500 ohms per volt of rated voltage. At the *PCS level*, the internal protection functions of the power electronics act — IGBT short-circuit protection in the microsecond range, overcurrent shutdown, voltage and frequency protection, and anti-islanding detection. At the *medium-voltage level*, classical protection relays under IEC 61850 work with differential protection for transformer and busbars, distance protection for the connecting line, and overcurrent and earth-fault protection with time grading. At the *grid connection level*, the protection devices of the network operator work under the requirements from VDE-AR-N 4110 and 4120, including decoupling protection and certified NA protection devices (Netz- und Anlagenschutz).

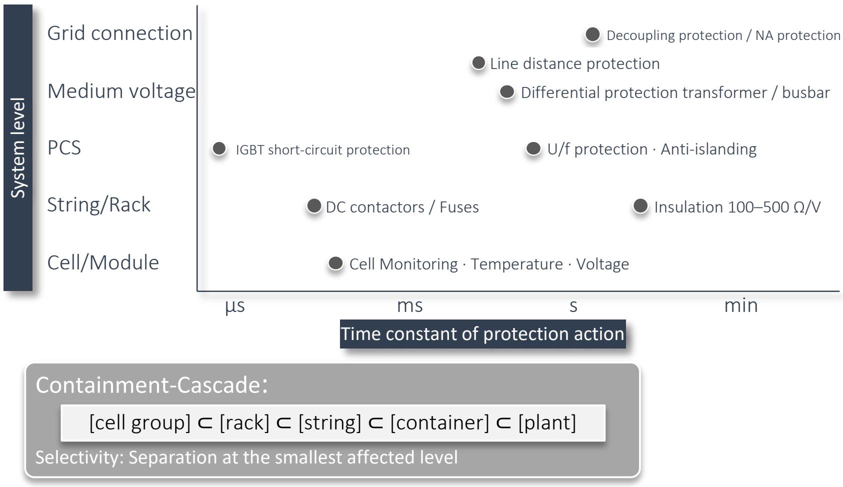

Figure 4. Protection and selectivity grading of a BESS across system and time levels. Selective protection coordination bounds the fault to the smallest effective unit and complements classical overcurrent logics in converter-coupled installations.

Selectivity is the central quality property of the protection concept. It describes the capability to disconnect only that part of the system that is really affected. A selective concept isolates an individual cell group without losing the rack; isolates a rack without losing the string; isolates a string without losing the container; isolates a container without losing the plant. Each stage of selectivity preserves a residual function, technically as well as commercially. A coarsely graded concept that, in case of doubt, trips the entire plant, purchases apparent safety with a high degree of availability loss.

Converter-coupled installations demand a special protection coordination. The short-circuit current contribution of a grid-following inverter is bounded to 1.0 to 1.2 times the rated current, that of a grid-forming inverter to 1.1 to 1.5 times; both values lie markedly below the short-circuit current contribution of a synchronous generator of equal power. Classical overcurrent grading, which is based on short-circuit currents in the range of 10 to 20 times the rated current, loses its selective action under these conditions. The protection coordination therefore works with complementary principles: distance protection for line sections, differential protection for zones with unambiguous current balance, directional protection in both feed-in directions, direction-dependent earth-fault protection, and communication between protection devices through GOOSE under IEC 61850 for zone coordination. The grid-forming-specific fault-current limitation must be considered in the parameterisation of the grid-connection protection, because converters change their behaviour on reaching the current limit and transition into a voltage-controlled limitation.

The protection grading organises the temporal order of the interventions. Preventive limitations — for instance power reduction near a temperature limit, C-rate reduction under cell dispersion, reactive-power reduction under voltage violation — act before hard protection trips, where this is safely possible. Fast protection trips act where the dynamics of the fault leave no time for softer interventions. This grading demands a clear coordination between BMS, EMS, PCS, and higher-order protection logic; an inconsistently coordinated grading produces shutdowns that come either too early or too late.

The *isolation strategy* determines how a separated system part is transferred into a safe residual state. Mere contact separation does not suffice, where residual energy remains in the inventory, auxiliary systems continue running, or the cooling regime changes through the separation. A clean isolation strategy holds the separated segment in a state in which diagnosis, intervention, and controlled resumption are possible, without generating secondary hazards: residual voltage is dissipated across defined discharge paths or held in controlled form, the cooling regime is switched into a hold mode, monitoring remains active, and the electrical separation is secured by visible isolating distance or corresponding interlocking.

14.5 Restart and residual function

Every shutdown changes the system state. It shifts load and voltage relations, influences temperature courses, can take auxiliary systems out of the foreseen regime, and generates new requirements on diagnosis and release. A protection concept is therefore sustainable only where it orders the transition into the safe state and the transition out of it equally.

The *restart* demands a documented release chain and may be neither accelerated by economic pressure nor delayed by organisational ambiguity. The chain typically comprises the following steps: diagnosis of the event through evaluation of the event histories from BMS, PCS, EMS, and protection relays under use of the synchronised time base; assessment of the cause on the basis of the combined signals and comparison with known fault scenarios; physical inspection of the affected segments, where relevant with thermographic examination and insulation-resistance measurement under IEC 61557; verification of the parameterisation against unintended changes during the event; test run under reduced loading with stepwise load increase and observation of the relevant state quantities; release by the responsible authority after comparison with the safety documentation; staged return into regular operation. A storage system without a disciplined restart regime remains incomplete in safety terms, even where its shutdown chains work technically cleanly. Safety does not end in the trip; it continues in diagnosis, release, and controlled resumption.

The *residual function* complements the selectivity. It describes in what extent the asset can deliver services after a partial shutdown. A container in an installation of twelve containers can remain disconnected without safety disadvantage, while the remaining eleven deliver a reduced but clearly defined power. The residual-function modes must be foreseen in the plant design and mapped in the EMS logic: the remaining units operate under a reduced rated power, the reserve windows are adjusted to the lower capacity, prequalified services are renewed in their dimensioning or paused, and the grid interface reports the reduced availability in the relevant time cadences. This capability for residual function is economically valuable, because it preserves revenues during repair or examination, and safety-relievingly effective, because it reduces the organisational pressure on a fast full restart.

At this point safety touches resilience. Resilience demands that restoration is a structured part of the operating order, not an improvised return. A BESS whose restart logic, residual-function modes, and release criteria are clearly determined in advance remains manageable even under event pressure. A BESS that improvises this order only in the event case loses its manageability in the moment in which it is most needed.

Chapter 15 — Economic operation and revenue architecture

15.1 Economic viability as a consequence of technical coherence

Economic viability is, in the BESS environment, often shortened to arbitrage spreads, control reserve prices, and CAPEX curves. For the management reality of a large-scale storage system this shortening does not suffice. *Economic manageability* — in the systems-engineering reading the running steerability of a system under changing boundary conditions — designates the capability to hold the asset across its useful life in a form that is technically, contractually, and operationally compatible with itself, and that does not play revenues off against state truth, safety, demonstration capability, or service life.

This manageability arises from technical coherence. Efficiency, reserve holding, degradation discipline, grid restrictions, auxiliary-system consumption, maintenance windows, ramp behaviour, prequalification conditions, safety regime, and data quality act immediately on revenue reality. A storage system with high theoretical value creation but weak state management or unclear restriction regimes remains economically fragile.

From this follows a disciplined perspective on the revenue logic. A usable spread is not automatically really usable; a technically possible service is not automatically commercialisable in parallel; a higher revenue extraction improves asset value only where it does not at the same time generate accelerated ageing, raised residual uncertainty, or sinking demonstration capability. Economic manageability begins with the question which technical and operational truth of the system is durably compatible with which revenue architecture.

15.2 Earnings contribution and the structure of revenues

The periodic earnings contribution of a BESS is composed of several revenue and cost streams that run differently in direction, time scale, and volatility. The revenue streams comprise energy-related revenues from arbitrage on day-ahead and intraday markets, reserve-related revenues from FCR, aFRR, and mFRR (each split into capacity and energy prices), grid- and system-near revenues from redispatch, reactive-power provision, congestion management and specific system services, and where relevant capacity payments from capacity mechanisms. The cost streams comprise energy procurement costs for charging energy including grid charges and levies in the extent permitted by the regulatory classification, operating costs (O&M, insurance, marketer commission, balance-group fees), degradation-related capital-erosion costs, and capital costs on the bound asset value.

Arbitrage forms in many German commercialisation models the volume-strongest revenue path. It arises from the difference between purchase and sale price across time, corrected for efficiency losses and parasitic loads. For a 2-hour storage system on the day-ahead market, the daily spread between the lowest and highest 2-hour block in Germany has, in recent past, lain in the range of €30 to €80 per MWh in quiet periods and €100 to over €200 per MWh in volatile periods; intraday spreads typically lie higher, with stronger temporal dispersion. The substance of this path hangs on price volatility, the temporal structure of the spreads, the forecast quality, and the asset's capability to translate the operating mode in fact within the relevant time windows. With growing storage capacity in the market, spreads tend to flatten — the cannibalisation effect is markedly visible in the more recent market analyses and shapes the long-term expectation, with declines in the specific arbitrage revenues of the order of 20 to 40 per cent for a doubling of the aggregated storage capacity in the control area.

Reserve holding is composed of capacity prices for the reservation and energy prices for the actual call. FCR pays exclusively capacity; aFRR and mFRR pay capacity and energy separately. In Germany, FCR capacity prices lay in the years of high market tension at well over €5,000 per MW per week, but have reduced and stabilised markedly with growing market participation of battery-based providers. With the integration into the European platform PICASSO for aFRR and MARI for mFRR, the national price structures are increasingly approaching a Europe-wide level. The earnings logic of these services differs structurally from arbitrage; it rewards availability and reliability, not throughput. A storage system that primarily runs reserve ages differently, binds SOC differently, and forgoes arbitrage opportunities that it would otherwise theoretically have. The combination of both revenue paths is economically attractive but demands the reserve-holding discipline treated in Chapter 10.

The empirical market observation of these revenue paths condenses, in the German context, in the *ISEA Battery Revenue Index* of RWTH Aachen, which links the spot-market segments (day-ahead auction, intraday auction, intraday continuous) with the control reserve markets (FCR, aFRR) in a unified reference. It makes the revenue potentials systematically comparable across the relevant trading time scales and forms thereby the currently most robust publicly accessible orientation quantity for the commercialisation of German large-scale storage systems. Trading practice moves increasingly towards cross-market operation, in which FCR holding, aFRR prequalification, and intraday trading are run not as alternatives but as coordinated revenue layers; the M5BAT research installation operated at ISEA delivers an operational reference, in which cross-market optimisation between FCR and intraday is tested under real market conditions. The earnings basis of a concrete asset arises from the capability to coordinate these layers robustly in correspondence with ramp capability, response time, SOC availability, and contractual embedding.

To these are added *system-near services* — reactive-power provision, voltage support, congestion management, instantaneous reserve, and grid-forming-near contributions. Their revenue paths are in Germany at present predominantly contractually and regionally structured, frequently bound to specific prequalifications. With growing share of power-electronically coupled generation and falling systemic inertia, the role of the BESS shifts. It remains a storage system in the strict sense and emerges in grid operation increasingly as a management-capable power-electronic system. In strongly converter-dominated grids the temporal shifting of active power or the punctual reactive-power provision no longer suffices; required becomes an operating behaviour that brings frequency and voltage support, fault ride-through capability, controlled transitions between operating regimes, and a robust interaction with protection and reclosing logics together. The relevant international works therefore classify the BESS not only as an arbitrage or reserve object, but also as a carrier of voltage control, frequency support, oscillation damping, power-quality functions, and resilience contributions reaching as far as black-start capabilities.

The distinction between grid-following and grid-forming embedding gains in this perspective economic relevance. Grid-following logics presuppose a sufficiently stable grid, on whose voltage and frequency they orient themselves; the weaker the grid, the lower the rotating inertia, and the stronger the coupling through converters, the less is this precondition given. In such situations, an operating mode gains importance in which the power electronics deliver electrical management contributions themselves. For the BESS this means that its converter and control architecture is not to be assessed solely on stationary active-power provision, but on whether it can support an ordered voltage and frequency management in disturbed or reconstructive grid states.

This becomes particularly visible in the black-start and restoration context. In strongly converterised systems, restoration is a question of sequence, of energisation-path logic, and of management of transient effects. A common black-start and restoration scheme becomes sustainable, under high power-electronics penetration, only where grid-forming-capable converters, BESS for voltage and frequency support, and a strategy for safe re-energisation work together. The BESS serves as an energy store, for the smoothing of volatile feed-in, for voltage excitation, for active frequency balancing, and for relief in critical transition states. Equally relevant is *soft energisation*: lines and equipment must not be reclosed in such a way that overloads, voltage oscillations, or transformer saturation are triggered. The restoration capability of a BESS is thereby the result of regulation, protection coordination, energy reserve, state management, and systemic embedding, not an isolated product property.

For the classification in project and operating practice, an enlarged assessment logic follows. A BESS that claims system-near services is to be assessed beyond its prequalification for market-based reserve products on whether its control and protection architecture remains robust under weak-grid conditions, ride-through requirements, ramp specifications, power-quality requirements, and restoration logics. The relevant Grid Codes and Interconnection Rules — particularly for voltage and frequency ride-through, ramp rate control, protection parameterisation, and interoperability — set the frame within which these capabilities are institutionally recognised. The technical maturity-tier of a BESS rises in the measure to which it not only offers services but also remains predictably manageable in varying grid situations.

15.3 Value stacking, blocking, and the economics of coordination

The attractiveness of the BESS rests substantially on the capability to deliver several services from the same technical inventory. This capability is referred to as *value stacking*. Its real reach is bounded by *operational blocking* — the technical fact that the same SOC window, the same power ramp, and the same control attention cannot simultaneously be fully available to several services.

Technically, the bindings can be cast in concrete quantities. A symmetric FCR holding of 1 MW binds in arithmetic terms 0.25 MWh of energy per direction and a symmetric SOC corridor that maps full activation in both directions over 15 minutes; parallel aFRR holding in the same direction adds the corresponding reserve quantities under product specification (typically five minutes sustained capability). The available PCS power is divided between the services, and the simultaneous delivery of several fast services demands that the control logic resolves the priority between competing set-points unambiguously. The sum of contractually committed services must therefore lie regularly below the sum of theoretically possible services.

The economic substance of value stacking hangs on the quality of this coordination, less on the number of theoretically stackable services. An installation that holds FCR and at the same time skims intraday spreads must place SOC windows so that neither reserve failures nor missed arbitrage windows arise. An installation that offers system services to a TSO can use arbitrage opportunities only in those time windows in which it is not bound in call-readiness. The difference between theoretically possible and contractually committed services is the price of operational robustness; it prevents reserve commitments failing in the event case, prevents penalties falling due, prevents prequalifications being lost, and prevents warranty claims being devalued. An undisciplined multi-use can generate revenues that are already consumed by the follow-on costs of a single reserve-failure penalty.

To this is added the perfection assumption of many economic stacking expectations. Retrospective arbitrage and optimisation calculations frequently assume complete knowledge of future prices and call situations. Real operational management does not have this knowledge. Real revenues therefore lie systematically below the results of perfect foresight; academic estimates name shortfalls of the order of ten to twenty per cent. For the BESS it follows that stacking hangs on physical availability, on forecast quality, on decision discipline, and on robustness against forecast errors at the same time. What appears in retrospect as elegantly stackable can fail in real operation already because price windows, call probabilities, and state reserves are not present with the same certainty.

The economic attractiveness of multi-use is to be distinguished from operational over-booking. A storage system can be assigned to several revenue layers and still not deliver each of them simultaneously in full hardness. The real boundary arises where the same SOC window, the same ramp capability, the same thermal reserve, and the same control attention are pledged multiple times. Value stacking is thus determined not by the number of services but by the capability to cut commitments so that safety reserves and control-reserve discipline remain preserved even in the disturbance or call case. Precisely because stacking is highly case-specific, it does not serve as a flat-rate viability lever but only as a result of a disciplined coordination architecture.

Degradation appears in this coordination as an economically effective limit quantity. Cyclic ageing is a technical lifetime effect and influences the operating mode immediately. An additionally discharged megawatt-hour deployment is economically rational only where its revenue exceeds the implicit costs of value reduction of the remaining system. This connection can be mapped through marginal charging or discharging costs that enter dispatch optimisation as opportunity costs. Arithmetically, the marginal degradation costs arise from the damage contribution of a partial cycle, valued with the specific replacement or augmentation costs per kilowatt-hour of usable capacity; at augmentation costs of €150 per kWh and a cycle endurance of 6,000 full cycles under reference conditions, the nominal degradation costs amount to around €25 per MWh of throughput, with this value varying strongly depending on temperature, SOC, and DoD. A robust stacking approach optimises not against an abstract full utilisation but against a bounded state and lifetime body, the substance of which is co-loaded by every deployment.

The discipline of stacking shows itself in clear priority rules. Safety stands above regulatory obligation; regulatory obligation stands above contractual commitment; contractual commitment stands above opportunistic revenue maximisation. The priority hierarchy introduced in Chapter 8 translates here into the economics of multi-use. An installation that holds this hierarchy operationally stable realises stacking returns in robust form; an installation whose priorities remain negotiable in the individual case loses, with time, the basis of its participation.

15.4 Revenue restrictions through grid and state

The theoretically available revenue basis of a BESS is cut by several restriction classes that have been developed in the preceding chapters. Their economic effect is robustly quantified.

The merchant reference case introduced in Chapter 10, with 10 MW and 20 MWh, shows the orders of magnitude. In the unrestricted case, annual revenues lay at around €2.1 million, that is around €210,000 per MW. A symmetric power limitation from 10 MW to 8 MW reduced revenues by around 11 per cent; an asymmetric limitation of charging power to 6 MW by around 16 per cent; a nightly blocking window from 19:00 to 06:00 by around 39 per cent. The figures are to be read as a structural statement, not as a universal magnitude: the distance between nominal capacity and really achievable revenue hangs strongly on restrictions that do not appear in the headline figures.

The grid-side restrictions — export limits, import limits, temporary redispatch requirements, voltage and frequency band limitations, local congestion management under Redispatch 2.0 — act immediately on the revenue architecture. A storage system at a grid-technically bounded node cannot, despite optimal market signals, translate its operating mode in full breadth. The frequency and severity of these restrictions hang on the concrete connection point, the regional grid topology, and the build-out state of the surrounding grid; in southern German regions with a low regional generation surplus they appear differently from northern German regions with high wind power and bounded transmission capacity to the south. Curtailment shares for PV and wind installations in the single-digit per cent range are already established in certain grid regions; for storage systems analogous restrictions can occur as grid-side dispatch specifications, with corresponding effect on revenue reality.

The state-side restrictions — temperature windows, degradation discipline, warranty conditions, maintenance windows — act in the same direction. An installation that, in high summer at ambient temperatures above 35 °C, must temperature-related reduce its charging power loses revenues in precisely those hours in which typically high PV feed-in and thereby favourable charging conditions coincide; the correlation between restriction and revenue chance is negative, and an averaging approach systematically underestimates the resulting losses. Warranty conditions set upper bounds for cycles per year, mean DoD, and average cell temperature; the exceedance of these bounds shifts the warranty status and is to be carried in the earnings contribution as a risk class of its own. Maintenance windows reduce annual availability typically by one to three per cent, depending on maintenance strategy and augmentation planning.

15.5 Capital value, degradation costs, and lifecycle economics

The overall economics of a BESS condense in the *capital value* — in the difference between the discounted inflows from revenues and the discounted outflows from CAPEX, OPEX, augmentation, decommissioning, and degradation costs. The structure of the capital-value calculation follows the established conventions: $NPV = \Sigma\,(CashFlow_t / (1 + r)^t)$, with a WACC that maps the project's capital structure and the project-specific risk. For infrastructure projects of this class, the nominal WACC values in Germany currently lie typically in the range of 6 to 9 per cent, depending on project structure, debt share, and revenue stability.

The investment and system-cost side differs structurally between the storage segments, and market observation shows a robust pattern. The ISEA-RWTH market data show, for large-scale storage systems in Germany, system prices in the range of €310 to €465 per kWh, with substantial dispersion by size class, system duration, and equipment depth. Commercial storage systems lie in a higher price class of €580 to €710 per kWh, home storage systems markedly above that, at around €1,200 per kWh. The degression of specific costs with growing system size thereby forms a substantial structural advantage of the large-scale storage segment, which carries the economic plausibility of utility-scale projects against decentralised alternatives. The price bands are not to be read as point forecasts but as market bandwidths within which project-specific factors such as system duration, grid-forming capability, safety equipment, connection body, and contractual embedding generate substantial variance. For the lifecycle economics of concrete assets they form the frame in which investment and later augmentation decisions are to be located.

A complementary metric is the *Levelised Cost of Storage (LCOS)* as discounted total costs per dispatched kilowatt-hour across the lifetime; for large-scale storage systems with four hours' duration, current LCOS values lie in the range of €70 to €120 per MWh, depending on capital costs, cycle count, and lifetime assumption. The metric allows comparison between projects and with alternative flexibility options; it does not, however, map the revenue side and does not replace the NPV consideration.

The degradation costs deserve their own treatment. They are a running energy-quantity-related value erosion, not a downstream balance-sheet write-down. Every cycle choice, every temperature exposure, every aggressive operating mode inscribes value reduction into the asset. The marginal degradation costs per delivered MWh are a management-relevant quantity: where they exceed the spread between purchase and sale, the cycle is economically destructive, regardless of its balance-sheet appearance. Their quantitative determination follows the connection introduced in 15.3 between cycle endurance, replacement or augmentation costs, and depth-dependent damage weighting; in high-performance EMS implementations, this price is dynamically parameterised and flows as opportunity cost into every dispatch decision.

The lifecycle economics include augmentation, retrofit, possible hybridisation, decommissioning, and potential second-life use. Augmentation becomes relevant as a rule after five to ten operating years, when the combination of capacity loss and changed cell price makes a retrofit economically attractive against the existing inventory. The decision hangs on the documented ageing history of the inventory, the cell prices reachable at the decision point, and the compatibility conditions between old and new inventory in a common string; with growing ageing heterogeneity between old and new cells, the effort for adapted parameterisation, cell monitoring, and dispatch logic rises. The long-term perspective of the installation is therefore to be anchored already in the design: in the mechanical and electrical accessibility of the racks, in the modular interface structure between BMS, PCS, and EMS, in the reserve of area and connection capacity, and in the carrying-forward of the evidence trail on which later decisions on augmentation, retrofit, partial decommissioning, or structural continuation build. The economic manageability of a BESS arises from the coherence between technical state truth, operational discipline, revenue architecture, and lifecycle perspective. Each of these four dimensions hangs on the sustainability of the remaining three.

Chapter 16 — Connection charges, grid charges, and infrastructure costs

16.1 Grid access as part of the value architecture

The costs of grid connection form, for large-scale batteries, a part of the value architecture of the asset; their reading as a mere entry fee does not carry the project economics. Connection charges, connection conditions, grid-charge systematics, potential special regimes, and grid-side restrictions shape the viability of the system earlier and more strongly than the later dispatch model. This statement gains in the German development situation particular sharpness.

The shift has two causes. Grid access itself has become a scarce resource; the connection competition described in Chapter 3 moves the allocation question to the centre. At the same time, the regulatory treatment of stationary storage systems develops dynamically further. The Federal Court of Justice has, with its decision of 15 July 2025 (case reference EnVR 1/24), determined that electrical storage installations are to be treated, in the sense of grid access law, as final consumers in the sense of § 3 No. 25 EnWG, and accordingly that connection charges fall due under § 11 NAV and the relevant provisions of StromNEV. This clarification anchors storage systems institutionally in the same cost frame as procurement customers and changes the economic basic position of the asset relative to the grid.

For the project and operating perspective, a sober insight follows. The value architecture of a storage system begins in the cost structure of access to the grid; the grid coupling point marks a later place of its unfolding. Site choice, connection power, voltage level, and local grid situation condense to a capital burden that fundamentally shapes the later earnings calculation. A connection at the extra-high voltage level (220 or 380 kV) with the requirement of own substation displaces the infrastructure cost structure into a different order of magnitude from a connection at the high-voltage level (110 kV) or in the medium-voltage grid; the specific costs per connected MW differ between these voltage levels by a factor of two to five, depending on the concrete grid situation and required primary technology.

16.2 Connection charges as a threshold mechanism

The connection charge (Baukostenzuschuss, BKZ) acts in the German system as a *threshold mechanism*. It binds site choice, connection power, voltage level, and grid situation to an early, substantial capital burden. For BESS, this binding is particularly effective, because the economic attractiveness of high power and fast response capability regularly coincides with grid-side relevant connection dimensions. The greater the proximity to grid-technically valuable nodes, the greater can at the same time be the infrastructural cost sharpness of access become.

The dimensioning of the BKZ follows in practice the cost-reimbursement logic under § 11 NAV and the supplementary determinations of the network operators. The concrete level per megawatt hangs on the voltage level, the distance to the next suitable feed-in point, the necessity of own switchgear bays and transformers, and the loading situation of the surrounding grid. For high-voltage connections, BKZ values currently move in projects typically in the range of €60,000 to €120,000 per MW of connection power; for extra-high voltage connections, in a markedly higher order of magnitude; the actual project values disperse considerably, because the cost-causing grid expansion measures fall out differently locally. For a 100-MW storage system, the BKZ burden lies thereby in the single-digit to lower double-digit million range and reaches an order of magnitude that, in relation to the cell costs of the system, is no longer subordinate.

The maturity-tier process of the German transmission system operators, in force since 1 April 2026, has additionally structured the handling of the BKZ. The charge is levied in three tranches, typically 30 to 50 per cent at reservation, 30 to 50 per cent at a defined milestone, and 20 per cent at final connection. This tranching distributes the capital burden temporally and reduces the early loading of cash flow; at the same time it couples the charge payments to project progress and maturity-tier evidence. The application fee of €50,000 and the realisation deposit of €1,500 per MW, also anchored in the maturity-tier process, act as additional disciplining mechanisms. They reduce the number of speculative connection requests and enforce an early commitment to project realisation.

The consequences for project design are considerable. The connection power becomes a decision variable of first rank; a power dimensioning oriented towards short-term revenue maximisation can generate BKZ burdens that the later additional revenues do not justify. The choice between higher and lower connection power forms a capital-structure question and reaches beyond a technical fine calibration. An overly conservative design at the same time excludes later expansion and hybridisation options that would themselves carry value. In practice, the decoupling of connection power and internal system power gains importance: a storage system can be operated with a connection power of 80 MW at an installed PCS power of 100 MW, where the operating strategy does not pass the internal peaks through to the grid node; this decoupling reduces the BKZ burden under bounded revenue disadvantages and demands an EMS that runs the connection limit as an active restriction in every dispatch decision.

16.3 Grid charges between cost coverage and behavioural signal

Grid charges were, in the original design of the system, an instrument of cost allocation; they were to distribute the actually arising grid costs to users in a polluter-pays manner. In the current debate, function and expectation shift. Grid charges are increasingly understood as *behavioural signals* that are to influence site choice, deployment logic, and market reactions. This loading is institutionally consequential.

The structure of grid charges under StromNEV combines a power-related and an energy-related share. The power price is referenced to the annual peak load measured in a settlement year and falls due independently of the actual energy withdrawal; it lies, depending on grid level and network operator, in the range of €70 to €140 per kW per year. The energy price is referenced to the energy taken and lies in a comparable order of magnitude in the range of 1 to 3 cents per kWh. For a storage system with high power and comparatively low annual work, the power price can make up an over-proportional share of the charge burden and become the economically dominant quantity.

For stationary storage systems, the grid-charge question has a particular sharpness. A storage system takes energy for charging and feeds it back later. Where the procurement side is fully loaded with grid charges, a double burden arises; the storage system acts grid-relievingly but bears the same charges as a pure consumer. Germany has foreseen, for stationary storage systems, corresponding relief and exemption regimes, including via § 118 paragraph 6 EnWG for installations with a defined commissioning window and via § 19 StromNEV for certain special forms of grid use. The exact configuration of these provisions has changed several times across time, and the BGH decision of 15 July 2025 has further sharpened the reach of these reliefs. The future viability of these provisions hangs on the systemic debate about which storage contributions are recognised as grid-supportive.

The Aurora study on dynamic grid charges sharpens this connection in unusual clarity. It examines whether regionally differentiated and temporally resolved charges can promote grid-supportive behaviour, and arrives at a differentiated result. Already at a spread of ±€10 per MWh, generation shifts of 54 TWh are reported for the year 2037, against 19 TWh of redispatch in the base scenario without dynamic grid charges. At the same time, generation from renewable energies sinks, thermal generation rises, electricity prices rise, and CO_2 emissions increase.

The structural reading of these figures carries further than their punctual order of magnitude. An instrument that is to trigger grid-supportive behaviour can, under coarse regional aggregation and temporal forward-shifting of the signals, generate new inefficiencies. Market value and system value can be actively shifted against one another by the same control signal. For the BESS it follows that the revenue effect of dynamic grid charges is to be assessed by their concrete effect on the actual dispatch reality of the asset; the political intention carries this assessment only to a limited extent.

16.4 Local signals, granularity, and institutional misdirection

The Aurora study works with 22 regions and makes visible that the insufficient granularity of regional signals frequently does not adequately map local congestions. Installations without a local congestion can be unnecessarily loaded, while installations actually loading the grid are not addressed precisely. To this are added forecast errors, because the signals are fixed one day in advance.

The mechanics of this misdirection can be precisely described. Dynamic grid charges are typically formed on the basis of a day-ahead load-flow forecast for a defined regional schema; the resulting regional and temporal surcharges or discounts are published the day before for the quarter-hours of the following day. The physical grid reality deviates from this forecast in several dimensions: local congestions occur at a grid node or line-section level finer than the regional boundaries; the actual load flows follow the current feed-in and load patterns, which differ from the day-ahead forecast; unplanned outages and switching states change the congestion situation intraday. A storage system that reacts to a regionally averaged day signal can thereby systematically operate past the actual congestions.

For the BESS this insight has two immediate consequences. The robustness of the project architecture against signal quality becomes a planning quantity. An asset whose economic viability hangs on a precise, regionally accurate signal remains fragile, because signal architecture and physical grid reality rarely coincide in practice; a robust project economics takes inaccuracy of fit already into the design. The diversity of the earnings basis becomes the second planning quantity. An installation that is viable under a single institutional regime and becomes fragile under a different variant stands under the risk of regulatory shifts; this risk is particularly pronounced under dynamic instruments, because their configuration is readjusted in short cycles. The revenue architecture described in Chapter 15 thereby gains an additional dimension; its robustness against regulatory regime changes becomes part of its quality.

The discussion of local signals points beyond itself to a structural observation. Institutional misdirection forms a real risk of project economics and reaches beyond a marginal edge phenomenon. A storage system that reacts wrongly to a coarse-granular signal loses the potential incentive and at the same time causes system costs for which it is not remunerated and which, in case of doubt, are transferred into other institutional mechanisms — redispatch, congestion management, curtailment.

16.5 Infrastructure costs as part of the capital structure

The cost elements treated so far — connection charges, grid charges, potential special regimes — act with the classical CAPEX and OPEX positions to a total capital structure that carries the value architecture of the asset. The capital structure of a BESS comprises today at least five classes: the cell and system costs, the costs of the connection body from substation, switchgear, and routing, the connection charges and other connection levies, the grid charges across the lifetime, and the costs of augmentation and decommissioning.

The relative weighting of these classes has shifted in the past years. The cell costs have fallen considerably; lithium-ion large-scale storage systems have developed from a capital-intensive exceptional investment into a scaled infrastructure class. The connection and infrastructure costs have risen or become more visible. The share of non-battery costs in the total investment, often referred to as *balance of system (BoS)*, shapes the project economics more strongly than before. For 4-hour systems, the BoS shares lie, depending on site and design, typically in the range of 35 to 55 per cent of the total investment. Within the BoS, the largest positions fall to the medium-voltage transformer and the medium-voltage switchgear, to the substation with primary technology under IEC 62271 and secondary equipment under IEC 61850, to the routing between storage field and substation with the corresponding cable cross-sections and laying costs, to the civil works (foundations, drainage, fencing, access roads), and to HVAC, fire-protection, security, and IT infrastructure. The PCS costs lie crosswise to this classification but are carried separately in many project calculations. This shift has a strategic implication. The project gain has displaced from cell selection and OEM negotiation to factors above the cell level; differentiation arises in access to the grid, in the capital structure of the connection, in regulatory understanding, and in project maturity.

For the project and operating perspective, a clear management task follows. The infrastructure costs of grid connection are to be run as design parameters of the project; their treatment as a fixed entry price does not carry the project economics. Connection power, voltage level, site choice, tranching strategy, coordination with the network operator, load profile management, the decoupling of connection and PCS power through active load limitation, and the deliberate choice of a substation architecture that can absorb later augmentations and hybridisations form levers that influence the capital structure. A project that disciplines these levers early earns its value already in the design of its entry into the grid; dispatch acts on this basis but does not first establish it.

Chapter 17 — System services, grid-supportive contribution, and external effects

17.1 The systemic value beyond the market price

The systemic value of a large-scale battery storage system comprises market-price smoothing and short-term flexibility provision as one part of its effect; to these are added forecast-error compensation, reserve holding, congestion situations, network losses, redispatch costs, voltage management, and the temporal quality of generation and load adjustment. The multi-layered character of these effects explains why the earnings basis of the asset is mapped only partially in the institutional mechanisms of the electricity system. One part of this value is visible in the market. The storage system reacts to price signals and product definitions organised in spot markets, control reserve tenders, and contractual individual relationships. Another part remains external. Grid effects, system-service contributions, and spatially and temporally differentiated system qualities appear incompletely in the current remuneration regimes. The difference between achieved and caused value is a structural property of the present storage economics, not an edge observation.

This falling-apart has practical consequences. The same dispatch decision can be commercially rational and grid-loading; a grid-relieving behaviour can remain economically under-incentivised. An assessment that measures the storage system exclusively against the trading result captures only one part of its effect; an assessment that measures it exclusively against the grid contribution captures the economic logic of its management only in sections. The robust reading runs both levels at once.

17.2 System-supportive and grid-supportive contribution — two related terms

The terms *system-supportive contribution* and *grid-supportive contribution* are frequently used synonymously in the debate on large-scale batteries. For a technically and ordo-politically robust classification their distinction is required.

System-supportive contribution designates the positive contribution of the BESS to the function and cost structure of the electricity system as a whole; it comprises — as developed in 2.3 — market-side effects (price moderation in scarcity hours, reduction of redispatch effort, capacity replacement), grid-side effects (congestion relief, voltage support, provision of grid-forming properties), and system-stabilising effects (substitution of instantaneous reserve, black-start capability). In an economic-overall view these effects condense to an assessment that brings together generation costs, congestion management, other system-service costs, security of supply, emissions, and local grid effects. A system-supportive storage system holds several cost and benefit quantities in a defensible balance.

Grid-supportive contribution is narrower. It refers to the effect of the storage behaviour on grid costs, congestions, voltage maintenance, and grid-related system services. It is measurable in physically grid-side quantities: avoided redispatch in GWh and euros per settlement period, reduced curtailment of renewable feed-in in MWh per year, change in voltage profiles at the connection point and at neighbouring nodes in per cent of rated voltage, relief of individual line sections in MW and hours per year, reduction of (n−1) violations in critical grid situations. Its quantification is technically concrete; the data basis is formed by the network operators in their grid analyses, by the settlement data of the Redispatch 2.0 processes, and by the load-flow calculations of the network development planning. The economic assessment remains difficult, because grid-supportive contributions are only partially remunerated in the existing market regimes.

The analytical difficulty lies in the boundary between market and grid added value running fluidly. Grid-supportive contribution is to be read within the frame of economic-overall system-supportive contribution. A grid-relieving operating mode is economically-overall superior only where it does not destroy market flexibility with high system value. The exclusive assessment against wholesale spreads, price volatility, or reserve prices remains accordingly insufficient. A storage system is always effective in market and grid simultaneously in the electricity system.

17.3 Classes of system services

The family of system services in which a BESS can participate can be ordered along physical time scales. Their regulatory configuration differs strongly between the classes.

At the millisecond-to-second level, instantaneous reserve and grid-stabilising contributions act. With the dismantling of conventional power plants, their rotating mass moves into the background; power-electronically coupled installations are increasingly asked to deliver comparable contributions in a control-engineering manner. Grid-forming properties — the voltage-imposing behaviour, the inherent reaction to frequency deviations and phase steps, the capability for black-start-near system support — shift the standing of the BESS from a market-following to a system-carrying role. The relevant technical reference frames of the European network operators, in particular the ENTSO-E High Penetration Power Electronic Interfaced Power Systems (HPoPEIPS) Technical Reports and the requirement documents derived from them, define for this purpose properties such as inertia response (virtual inertia typically between 2 and 6 seconds), fast frequency response with response times below 500 milliseconds, synchronisation capability in grids with low short-circuit power (SCR below 2), and defined behaviour during phase steps. In Germany, the regulatory differentiation of these contributions is not yet completed; their technical importance grows with the rising share of converter-coupled generation.

At the second-to-minute level, frequency control reserves act — FCR, aFRR, and mFRR. These products are regulatorily and commercially differentiated, the participation conditions are standardised, and remuneration takes place through market tenders. The prequalification under the relevant criteria of the transmission system operators examines, among other things, full activation within 30 seconds for FCR, within 5 minutes for aFRR, and within 12.5 minutes for mFRR; the sustained capability across the product-specific periods; the accuracy of set-point tracking; the time-stamp quality of the measurement data; and the behaviour in marginal areas of the state space. With the integration into the European platforms PICASSO for aFRR and MARI for mFRR, price formation becomes increasingly Europe-wide. For large-scale batteries, these products are today the volume-strongest system-service paths; their monetisation demands the SOC and reserve policy developed in Chapter 10 in coordination with the arbitrage side.

At the minute-to-hour level, congestion management, redispatch, and grid-relieving operating modes act. These services were until now primarily reserved to conventional generation units; battery storage systems appear here as actors with a specific profile. The congestion management volume in Germany lay in 2022 in the low 30,000-GWh range with around €4.2 billion of system costs. The order of magnitude shows the value potential; the institutional embedding of battery storage systems in Redispatch 2.0 structures is technically and regulatorily demanding, because dispatch and redispatch have different optimisation logics and settlement times, and because storage systems take, in the attribution logic of failure and replacement work, a position of their own that is not fully mapped in the balancing rules.

A special form is presented by the *grid booster*: large-scale battery storage systems that, under the responsibility of the transmission system operator, serve grid operation exclusively, without participating in the electricity or control reserve markets. The Network Development Plans 2035 and 2037 foresee several such installations, including the TenneT projects Audorf/Süd and Ottenhofen with 100 MW and 100 MWh each, and the TransnetBW project Kupferzell with 250 MW and 250 MWh. Their function consists in the short-term relief of thermally overloaded line sections and in the reduction of preventive redispatch by allowing lines to be operated under (n−0) conditions at maximum permissible utilisation, with the storage system stepping in immediately on the occurrence of an (n−1) outage. Regulatorily, grid boosters are classified as grid equipment and are refinanced via grid charges. The institutional separation between grid boosters and commercially marketed storage systems follows the difficulty of running an installation simultaneously as grid equipment and as a market actor without endangering the network operator's neutrality towards competition.

To these are added voltage support and reactive-power provision, whose remuneration is predominantly contractually regulated between installation operator and network operator. The technical requirements follow the relevant grid connection rules VDE-AR-N 4110 for the low and medium voltage grid and VDE-AR-N 4120 for the high voltage grid, which require, for the static and dynamic reactive-power range, typically a power factor cos ϕ between 0.90 underexcited and 0.95 overexcited, where applicable with a voltage-dependent Q(U) characteristic. The economic attractiveness hangs on the concrete configuration of the respective contracts and on the grid situation of the site; at weak grid nodes with high reactive-power demand, contractual remuneration can form a relevant revenue position, while at strong nodes reactive-power provision is required as a mandatory service without separate remuneration.

17.4 External effects and the gap between market value and system value

The economic value of a BESS arises in a cascade that is not everywhere fully internalised. At the first stage, the storage system reacts to price signals of the wholesale market; here the remuneration is market-based and transparent. At the second stage, it contributes to reserve holding; here too, the remuneration is market-based, although via separate products. At the third stage, it acts on grid congestions, redispatch volumes, curtailment of renewable feed-in, and voltage management; here the most relevant external effects arise.

The mechanics of the internalisation gap can be described precisely. The wholesale price reflects the national or zonal generation cost structure and the cross-border interconnection capacities, but does not map intra-German grid congestions; the German price zone remains, despite regular discussion, undivided. Redispatch interventions of the network operators correct the market-based dispatch decisions afterwards, and the arising costs are passed on through the grid charges to the grid users; the market-based dispatch causer does not bear them. A storage system that chooses an operating mode at the spot market that is grid-loading at its location receives the market price for it and at the same time causes system costs that are not attributed to it. A storage system that chooses a grid-relieving operating mode forgoes the market price without being separately remunerated for its grid-supportive contribution. The asymmetry acts in both directions.

The more recent studies on large-scale battery storage systems in Germany show that the effect on redispatch and congestion management can be substantial, but is mapped only partially in operator revenues. The Aurora study on dynamic grid charges addressed in Chapter 16 additionally shows that institutional attempts to close this gap can, under coarse regional granularity, generate new misdirections.

This gap between market value and system value carries a structural property of the current remuneration regime and reaches beyond a transitional phenomenon. It will be closed step by step — through differentiated system-service products, through more local price signals, through capacity mechanisms, through contractual grid-service models, possibly through a debate about regional price zones. For the management of a concrete asset it follows that the systemic value of its operating mode regularly lies higher than its market revenue, and that attractive market revenues coincide with high system value only partially. A BESS can be economically-overall sensible and at the same time enterprise-economically fragile; it can be enterprise-economically attractive and at the same time systemically tension-laden. The robustness of a storage project is measured by how tightly its earnings basis is coupled to a single regime. An installation that remains viable under several plausible regime developments is strategically more robust than an installation whose viability is bound to a specific institutional configuration.

17.5 Capacity mechanisms and the institutional embedding of flexibility

The institutional embedding of the BESS is currently shifting beyond classical energy-only logics. The German debate on an integrated capacity market, the European further development of capacity mechanisms, and the differentiated treatment of storage and load flexibility change the institutional space in which large-scale batteries are economically and technically run.

A first German manifestation of this institutional shift is the Innovation Tenders (Innovationsausschreibungen) under §§ 39j ff. EEG and the Innovation Tender Ordinance. They award market premiums for combinations of renewable generation and battery storage, provided that the storage system is charged exclusively from the directly connected RE installation and reaches a minimum power of one third of the RE rated power. The 105 projects awarded in the first five auction rounds together comprise around 479 MWh of battery energy and 309 MW of power, with energy-to-power ratios between around 25 minutes in the first round (oriented to the aFRR test-protocol minimum duration) and two hours in the later rounds (oriented to the principle of minimum duration required for limited energy storage). The institutional construction anchors BESS in the RE support logic and at the same time excludes, through the charging restriction to the directly connected installation, a substantial part of the possible commercialisation roles — FCR, negative aFRR, and spot purchase are not accessible under the current rules. The 20-year term of the market premium additionally demands a lifetime guaranteed across this period, which in some projects leads to deliberately restrained operating modes for ageing minimisation. The Innovation Tenders show thereby exemplarily how strongly the interplay of support instrument, charging restriction, and lifetime guarantee shapes the operating logic of the storage systems; future capacity and flexibility mechanisms will, in their configuration, generate comparable feedbacks on design and commercialisability.

For BESS, several developments are relevant. The possible integration into availability obligations remunerates not energy or reserve power in the strict sense, but the capability to provide power on demand. The development of de-rating factors assesses the contribution-capable power of a BESS taking into account its bounded duration and availability characteristic; in reference designs of other European capacity markets, the de-rating factors for 1-hour systems lie typically in the range of 20 to 35 per cent of rated power, for 2-hour systems at 40 to 60 per cent, and for 4-hour systems at 70 to 90 per cent, with the concrete values derived from the analysis of loss-of-load expectation in critical system states. The aggregation rules decide whether smaller and medium-sized storage systems can participate via aggregators. The prequalification requirements define the technical and operational preconditions for participation and thereby substantially determine access.

The configuration of these elements is not neutral. De-rating factors decide whether 4-hour systems, 2-hour systems, or in perspective systems with duration below one hour are recognised as a capacity contribution; they thereby act immediately back on the economically sensible design of system duration. Prequalification rules determine the effort of participation and the economic reachability for different project sizes. Aggregation rules decide on the accessible actor structure. The auction design decides whether storage systems must be competitive against conventional capacities, against demand response, or against other storage systems; demand-response resources with low capital costs and low annual activation frequency can, in purely clearing-price-based auctions, push prices below the level viable for storage systems where no technology-differentiating elements are foreseen.

For the BESS an enlarged reading role follows: electricity-economic classification, capacity-market classification, and system-service-near classification enter the assessment simultaneously. The economic architecture remains robust against different configurations of these mechanisms where it can carry several plausible regime states. An installation whose viability is bound to a specific manifestation of a not-yet-introduced capacity mechanism remains structurally fragile. The architectural decisions that carry this robustness concern the system duration and its later extensibility, the interface openness for different commercialisation structures, the prequalification capability for several products in parallel, and the contractual configuration that absorbs operator changes and altered commercialisation roles.

Chapter 18 — Contractual capacity, governance, and institutional embedding

18.1 Contractual capacity as a derived property

As soon as a BESS is operated market- or grid-near, a complex contractual landscape arises around the asset. Grid connection and construction contracts, operating-management models, optimisation and commercialisation agreements, balancing and settlement relations, guarantees and warranties, data and communication responsibility, insurance conditions, and finance-related reporting obligations act together and determine jointly whether the technical capability of the asset is translated into an institutionally robust performance.

The quality of this contractual landscape hangs immediately on the clarity of the technical and operational system body. Contractual capacity thereby forms a property derived from technical and organisational determinacy; its reading as a legal afterthought does not carry the institutional robustness. Where control authority, data authority, release rights, intervention rights, responsibility for parameterisation, safety responsibility, and degradation risks remain unclear, contractual grey zones arise that act back later immediately on operation, liability, and revenue reality.

The reach of this insight shows itself particularly clearly in event cases. After a disturbance, a reserve violation, a degradation-related power drop, or a safety incident, several contracting parties appear simultaneously on the scene — installation operator, EMS provider, marketer, network operator, insurer, financier, manufacturer. The attribution of responsibility succeeds where technical layers, data responsibilities, and intervention rights have been ordered in advance; it fails on the retrospective reconstruction of responsibilities that have remained indeterminate in running operation.

18.2 Role logic and the decoupling of technical layers

The technical architecture of the BESS — BMS, EMS, PCS, SCADA, higher-order optimisation layer — is in practice rarely operated from a single hand. Customary is a division of labour between asset owner, technical operating manager, EMS provider, marketer, and where relevant aggregator. This division of labour permits specialised competence at each layer; it demands a precise role logic that draws interfaces technically, legally, and operationally clearly.

The central question is *control authority*: who may set which set-point, reserve or release reserve windows, change a protection threshold, trigger a restart after a shutdown? Technically this authority is mapped in the architecture of the control technology. Safety-relevant parameters (protection trip thresholds, temperature and cell-voltage limits, insulation monitoring) lie in the BMS and are typically protected by password levels, hardware keys, or signed firmware; their change is reserved to the asset owner and the manufacturer. Grid-relevant parameters (ramps, reactive-power ranges, fault-ride-through profiles) lie in the PCS and follow the specifications of the network operator within the frame of the connection conditions; their change demands a documented coordination. Dispatch parameters (schedules, reserve windows, opportunity reservations) lie in the EMS and are coordinated between operator and marketer according to a defined command hierarchy; higher-order commercialisation set-points are checked through the EMS layer of the individual installation against the local restrictions before they are passed on to the PCS. An installation with graded control authority — local priority for safety, EMS priority for operating mode, marketer priority for market decisions — holds its management capability also in conflict situations.

The second question concerns *data authority*. Who owns the raw and movement data of the system, who may process them, who may pass them on, who receives access in real time, who only in aggregates, who only on request? The contractual ordering of these accesses must work together with the architecture of the data holding from Chapter 12: the raw data lie in the ideal case on an infrastructure that belongs to the asset owner or is contractually permanently assigned to them; the EMS provider receives access within the frame of their operating tasks under preservation of data availability on provider change or insolvency; the marketer receives the data points necessary for their products in the necessary temporal resolution; the network operator receives the grid-side relevant signals according to the relevant reporting obligations. An installation with data authority remaining contractually with the asset owner preserves its substance also under provider changes.

The third question concerns *intervention and release rights*. Structural changes — parameterisations, firmware updates, retrofit, augmentation — change the demonstration space of the asset. Who may commission, release, document such interventions? The release chain runs typically in three stages: a technical examination by the operating manager or manufacturer, a release by the asset owner on the basis of a documented change justification, and a downstream documentation in the evidence layer with timestamp, responsibility, and demarcation against the previous state. The contractual ordering of these rights decides whether the asset remains under a coherent technical history or whether it becomes, after a few years, an object with a fragmented responsibility history.

18.3 Warranty, liability, and risk distribution

The warranty logic of a BESS is in the early phase usually clearer than in later operation. Manufacturers give performance guarantees at cell level, integrators give system commitments, EPC partners give construction-execution guarantees. The commitments are bound to clearly defined operating windows: temperature ranges (typically cell temperature between 15 and

35 °C on average), maximum C-rates (frequently 0.5 C charging and 1.0 C discharging for LFP systems), SOC corridors (frequently 10 to 90 per cent in regular operation), cycle profiles (typically 365 to 550 equivalent full cycles per year), maintenance intervals with defined test protocols. Violations of these windows can let warranty claims lapse.

The examination mechanics of the warranty operate via periodic capacity tests and throughput measurements. A typical warranty regime foresees an annual or semi-annual reference cycle, in which the system performs a discharge with full capacity measurement under defined temperature, SOC, and C-rate conditions; the result is compared against a guarantee curve (capacity retention vs. cycles and time). Where the measured value lies below the curve, the warranty applies in the form of replacement delivery or compensation; where it lies above, the cycle is documented as contract-conforming. The validity of this examination hangs on the observance of the reference conditions, on the calibration of the measurement systems, and on the gap-free documentation of the operating history since the last examination.

With progressive ageing and under the conditions of real operation, the edges of this order shift. The marketer is interested in revenue maximisation; the asset owner in the observance of the warranty conditions; the manufacturer in a restrictive reading of these conditions; the insurer in the completeness of the demonstrations. These interests are not congruent; their balancing is the subject of running governance, not of a one-time contractual setting.

The risk distribution between the parties is to be ordered in several dimensions. Technical risk concerns failures, degradation beyond expectation, malfunctions of the power electronics, and thermal events; it is covered primarily through manufacturer warranties and insurances. Operational risk concerns operating modes that do not demonstrably correspond to the agreed conditions; it lies with the operator and, depending on the operating-management model, with the operating manager. Market risk concerns revenue shortfalls through volatile prices or regulatory shifts; it is distributed, depending on commercialisation structure, between owner and marketer. Regulatory risk concerns regime changes in grid charges, capacity mechanisms, reserve products, or prequalification requirements; it lies as a rule with the owner but can be partially shifted to offtakers or marketers via change-in-law clauses. Financing risk concerns the value sustainability of the asset under changing return expectations.

The dynamic grid charges discussed in Chapters 16 and 17 and the developing capacity mechanisms sharpen this point. Regulatory instruments that generate regional, hour-precise, and partly counter-running signals raise the requirements on risk distribution. Two attribution questions move into the foreground: the attribution of the risk of erroneous or coarse-granular signals, and the responsibility for the asset's reaction to contradictory market, grid, and charge impulses. The contractual clarification of these attributions belongs in the architecture of the framework contract, because in the event case it is no longer negotiable.

18.4 Operating models and the structure of the value chain

The operating structures around a BESS can be ordered along several basic forms. Each form distributes risks, revenue chances, and operational responsibility differently, and each has its own requirements on contractual capacity and governance.

In the *merchant model*, the asset owner bears the full market and revenue risk. Commercialisation takes place via a specialised marketer or via an own trading function. The revenues fluctuate with price volatility, control-energy prices, and system-service remuneration. The operational management can be internal or outsourced. Contractually the model demands a commercialisation contract with clear regulations on settlement models (fixed fee, pure performance component, or combination), minimum utilisation, performance KPI, notice periods, and handover mechanisms on marketer change; the typical remuneration structure lies in the range of 10 to 20 per cent of net revenues with a combination of a monthly base remuneration and a revenue-dependent component. The model is at once revenue-open and volatile; it sets high requirements on risk management and contractual robustness against market shocks.

In the *tolling model*, the asset owner transfers the rights of use over capacity and flexibility of the storage system for a defined period to a tolling partner against a fixed payment. The tolling partner bears the revenue risk; the asset owner receives plannable income. Typical tolling contracts run over five to fifteen years and remunerate the installation availability per MW per month, frequently complemented by performance components for cycle endurance and response times. The contract architecture must precisely map the technical limits of the asset — power, energy, lifetime budget, maximum cycle count per year, temperature and SOC windows, safety restrictions — so that misuse or unintended overloading is excluded. The role of data authority is here particularly important, because it forms the basis of every later dispute resolution; typical is a mirroring of all dispatch and state data in real time to the asset owner alongside the tolling partner.

In the *PPA-near model*, a part of the storage use is bound to an offtaker — for instance an industrial customer who purchases peak-load smoothing and security of supply, or a generator who smoothes their feed-in profile. These models are structurally hybrid; they combine fixed payments with conditional components and demand a careful ordering of which use the offtaker may call up with priority. The prioritisation is typically mapped in the EMS logic as a permissive layer: the offtaker call has priority over opportunistic commercialisation and is bounded by contractually defined maximum sizes (maximum calls per day, maximum energy per month, advance notice periods), so that the asset owner can commercialise the residual capacity in a calculable way.

In *hybrid portfolio models*, the BESS is integrated into a composite with other assets — PV, wind, electrolysis, industrial loads. The value chain becomes denser, the institutional complexity rises. At the same time new degrees of freedom arise: internal settlements, coordinated dispatch logics, shared connections, common system services. The contractual structures range from common grid connection with internal load-flow optimisation, through co-location agreements, to integrated energy-hub models with central optimisation. These models are deepened in Chapter 20 in connection with augmentation, retrofit, and hybridisation.

The four models each carry their own profiles; a superiority of one model as such does not exist. The choice hangs on risk preference, capital structure, strategic position, and market environment. The chosen model remains viable where it is run coherently in its technical, contractual, and organisational execution, and where regime changes at its edges do not immediately undermine its substance.

18.5 Governance of the interfaces

The institutional robustness of the BESS arises from a governance of the interfaces. Technical clarity must be translated into role logic, role logic into contractual robustness, and contractual robustness into operational routines that take decisions in realistic times and with reconstructible documentation.

The quality of this governance shows itself in marginal situations. A contractual landscape that functions under normal operation and falls apart on a reserve violation, a degradation discussion, or a safety incident does not carry the institutional robustness of the asset. A governance order that remains workable also under event pressure secures the value of the asset precisely in those moments in which it is endangered.

The practical reach of this insight is considerable. Insurers examine, before underwriting the property and business-interruption insurance, alongside the installation technology, the governance structure of the asset: safety concept under the relevant standards, documented operating and maintenance history, demonstration of regular state checks, clear reporting paths in the event case, alarm and emergency-response plan in coordination with the local response forces. Financiers assess the contractual architecture as part of the default risk and demand, typically, a contractually secured minimum revenue structure, reporting obligations in defined time grids, consent reservations for substantial changes to commercialisation or operating-management contracts, and reserves for augmentation and maintenance. Operator changes are possible at an asset with cleanly ordered governance without evidence, warranty structure, or contractual substance being lost. Structural interventions such as augmentation and hybridisation presuppose a clear order of who may trigger, release, and document them.

The governance of the interfaces forms a constitutive layer of the asset; its reading as an administrative side phenomenon does not carry the institutional quality. It belongs to the seven layers introduced in Chapter 1 as an analytical grid, and its quality shows itself in whether the other six layers are held in coherent form across time.

Part V — Lifecycle and structural further development

Chapter 19 — Asset management across time

19.1 The asset as a temporal object

A battery storage system is, across its lifetime, a temporally variable asset. Its change sets in with the first day of operation and follows a structural transformation under the influence of cycles, temperature, operating regime, software states, spare-parts policy, component changes, regulatory shifts, and altered system roles. The reading as an initially erected installation thereby misses the actual subject; what matters is the capability to hold the asset across time in a legible form.

This reading shifts the subject of asset management. The BESS of year one, the BESS of year six, and the BESS of year twelve form, in commercial and technical respect, different objects. State, performance capability, evidential position, and market value develop partly linearly, partly in jumps. A professional asset management accepts this dynamic as an object of management; its treatment as an unwanted deviation from an ideal initial state does not carry the lifecycle perspective.

The ordering under ISO 55000 condenses this conceptually. Asset management designates the coordinated activity of an enterprise to generate value from assets; the concept of value is here to be read technically, economically, safety-related, regulatorily, and institutionally. The derived standards ISO 55001 (requirements) and ISO 55002 (application guideline) define the structure of an asset management system with the elements of mission, strategy, target system, risk management, performance measurement, continuous improvement, and documented information chain. For the BESS this means the transfer of systems engineering and operations into a permanent management structure that reaches beyond project closure and commissioning and that holds the value sources of the asset open across its lifetime.

19.2 Performance tracking and condition monitoring

The basis of asset management is the disciplined observation of real behaviour. Performance tracking captures how the asset delivers its agreed and possible performances: realised revenues against potentials, availability against targets, reserve delivery against commitments, efficiencies against design values. The KPI structure introduced in Chapter 12 supplies the framework; performance tracking becomes a living management quantity in asset management only by setting the KPI in recurring evaluation cycles against reference curves, portfolio comparisons, and model forecasts.

Condition monitoring reaches deeper. It captures the quantities from which performance arises. At cell level these are cell temperatures, cell voltages under load and at rest, internal resistances from pulse responses, capacity losses from partial-cycle inference or periodic reference cycles, balancing currents and times. At module and string level these are temperature gradients, voltage spreads, insulation resistances against earth, contact temperatures at current collectors. At PCS level these are semiconductor temperatures, filter parameters, control quality (overshoot, settling time, residual deviation), harmonics and THD values. At transformer and switchgear level these are oil temperature and dissolved gases under DGA (dissolved gas analysis), partial discharge measurements, SF_6 densities for GIS variants.

The informational content of these measured values arises from their development across time; the individual value carries it only in exceptional cases. A rising voltage spread in a rack does not in itself form an alarm; it becomes a warning signal where it grows faster than in comparable systems or faster than in its own past. The evaluation therefore works with trend analyses, anomaly detection on the basis of statistical references, and peer-group comparisons within modules, strings, or installations of a portfolio.

From the combination of performance tracking and condition monitoring arises the diagnostic capability that recognises deviations and at the same time reconstructs their causes. A revenue decline can be market-related; it can follow from a creeping power reduction that arises from a creeping warming, from a creeping degradation of certain modules, from an inconspicuous parameter drift, or from a combination of these factors. The diagnostic capability is measured by whether such conclusions are robustly possible. The evidence continuity from Chapter 12 gains here its actual reach. In its absence — under patchy measurement data across the years, uncertain parameterisation and firmware history, unclear assignment of interventions to time points — diagnosis remains restricted to the present state. In its presence, diagnosis becomes historical reconstruction and thereby the basis for decisions on warranty, retrofit, and augmentation.

19.3 Maintenance regime and upkeep

The maintenance logic of a BESS follows three fundamentally different approaches, which mix in practice. The *time-based maintenance* works according to intervals. Typical intervals comprise semi-annual visual inspections with control of the HVAC filters, cleaning of the ventilation openings, visual examination of the cell modules for deformation or electrolyte leakage, and functional testing of the fire and gas detection; annual calibrations of the measurement systems, thermography of the switchgear and contacts under load, insulation tests under IEC 61557, oil samples with DGA at transformers; biennial to four-yearly proof tests of the safety-related protection functions under IEC 61508 with defined test coverage. The time-based maintenance fulfils regulatory minimum requirements and the warranty obligations of the manufacturers, but captures neither early-warning signals nor installation-specific particularities.

The *condition-based maintenance* works according to measured values and thresholds. Cooling-system maintenance is triggered when differential temperatures between flow and return leave a defined corridor; balancing analysis when voltage spreads between cells of a group exceed a threshold of typically 20 to 50 millivolts; transformer testing when DGA values under IEC 60599 enter the warning zones for individual gases (hydrogen, methane, ethylene, acetylene) or the associated ratios; PCS maintenance when fan temperatures, capacitor currents, or filter temperatures deviate systematically from the reference curves. The *predictive maintenance* works with models that derive expected failure or degradation time points from historical data and trigger interventions before the damage case; it uses typically remaining-useful-life estimators on the basis of combined physical and data-driven models.

For large-scale batteries a combination of all three approaches is sensible. Time-based maintenance carries the regulatory mandatory examinations and the plannable basic work. Condition-based maintenance carries the event-near reaction to identified deviations. Predictive maintenance carries the strategic foresight for components whose failure generates high follow-on costs: PCS, transformers, protection equipment, control technology.

The operational reach of the maintenance regime is measured by its effect on functional availability; the number of carried-out interventions carries the assessment only to a limited extent. An installation with high maintenance frequency and weak diagnostic capability reaches the same availability as an installation with lower frequency and high diagnostic capability, at higher costs. The economic viability of upkeep lies in the precision of the intervention.

Particular attention is due to the *spare-parts logic*. Lithium-ion cells, modules, and racks change between production years in form factor, chemistry, balancing logic, and communication protocol; an exact 1:1 replacement is, after three to five years, frequently no longer possible. The holding of reserve inventory — typically one to three per cent of the installed modules, temperature-controlled stored — or alternatively the contractual securing of supply capability across defined periods forms a decision that shapes the controllability of the asset in the medium term. The contractual securing comprises typically a guaranteed supply capability for modules of the same or compatible specification across five to ten years with a defined maximum delivery time and an agreed price development. An installation without secured spare-parts paths remains structurally risky in its second lifetime half.

19.4 Ageing management and degradation budget

The degradation management developed in Chapter 11 enters in asset management into a long-term perspective. Ageing is a running operational quantity and at the same time a balance; it inscribes itself in SOH values, capacity losses, and power reductions, and forces, on reaching certain thresholds, decisions.

The concept of the *degradation budget* orders this balance. It describes the remaining permissible ageing until the reaching of a defined threshold value, for instance 80 per cent SOH, from which augmentation, retrofit, or second-life use is economically or contractually indicated. Formally, the budget can be grasped as a two-dimensional quantity: a time-dependent calendar share that follows from the accumulated temperature-SOC-time integration, and a throughput-dependent cyclic share that arises from the rainflow-weighted sum of the partial cycles. Both shares add in practice with an empirically adjusted interaction that takes into account the combination of temperature and cycle regime. The marginal degradation costs per delivered MWh, introduced in Chapter 15 as a management quantity, are the economic correlate of the budget consumption.

Strategic ageing management demands decisions on the allocation of the budget across time. An aggressive use in the first years generates early revenues, consumes the budget over-proportionally, and forces earlier augmentation; a conservative use preserves the budget longer and leaves revenues unused in volatile phases. The optimal allocation hangs on market development, interest situation, augmentation cost expectations, and strategic considerations on the asset role. Under high spreads and high interest rates, the optimum shifts towards early revenue realisation; under expected flattening of the spreads and falling cell prices, it shifts towards a more restrained use with later augmentation under more favourable cost conditions.

The ageing forecast demands a connection of deterministic models and data-based methods. The Arrhenius and Wöhler approaches from Chapter 11 carry the physical framework; the actual ageing history of a concrete asset contains deviations from the model that disclose themselves only from the operating data. In practice, hybrid approaches establish themselves: a physics-based model with Arrhenius temperature dependence and depth-dependent cyclic damage forms the frame; the parameters are adjusted continuously through recursive estimation procedures (Kalman filters, Bayesian updates) to the observed capacity losses and resistance rises; data-based methods (Gaussian processes, neural networks on reference cycles) complement the forecast with installation-specific patterns that the physical model does not capture. The resulting forecasts deliver, across the lifetime, more robust results than any single method on its own; their quality hangs immediately on the quality of the reference cycles and the consistency of the operating data.

19.5 Decision moments and the strategic management of the asset

The lifetime curve of a BESS contains a manageable number of strategic decision moments. They mark transitions at which state, market, regulation, and financing converge, and at which the future trajectory of the asset is substantially co-decided.

The *first decision moment* is the transition from commissioning into regular operation. It marks, from the asset-management perspective, the change from the EPC-centred to the operational management logic. Technically it consists in the acceptance (Site Acceptance Test with defined test protocol, punch-list working-off, demonstration of the prequalification conditions, establishment of the warranty reference curves), the contractual handover (transition of operating responsibility, commencement of warranty periods, activation of insurance coverage), and the data-related handover (securing the commissioning data as a reference point, establishment of the running data pipeline, takeover of the evidence structure). An incompletely verified system generates uncertainties across years that inscribe themselves in warranty disputes, performance deficiencies, and operational frictions.

The *second decision moment* lies typically after three to five operating years and concerns the systematic examination of the performance guarantees. The procedure follows the warranty contract: performance of a reference cycle under the contractually defined conditions (typically at 25 °C cell temperature, 80 per cent DoD, 0.5 C charge and 1.0 C discharge rate), determination of the current capacity and the internal resistance, comparison with the contractual guarantee curve, documentation of the result in a test report. The quality of the evidence continuity is here immediately effective: where data are missing or are inconsistent, the warranty enforcement becomes a tough dispute; where clean data are available, it becomes an orderly contractual procedure.

The *third decision moment* concerns the augmentation or retrofit decision. It poses itself typically between the sixth and tenth operating year, when the SOH loss exceeds an economic threshold. The decision is complex because it couples several variables at once: current state, remaining useful life, augmentation costs, market development, regulatory environment, alternatives such as second-life use or sale.

The *fourth decision moment* concerns the end of the original use phase. It puts on the agenda the questions of further operation under reduced power, second-life use in less demanding applications, decommissioning, or sale. The quality of the evidence that the asset has gathered across the years determines the room for action. The residual-value corridor of an asset with gap-free history reaches well beyond the scrap value of an asset without robust history.

The management of a BESS across its lifetime takes place via a carried-forward technical identity that arises from the temporal linking of operating data, diagnostics, state estimates, event histories, maintenance measures, parameterisation changes, and the decisions derived from them. Condition monitoring in the sense of the relevant asset-management literature aims at the reduction of unplanned failures, the timely recognition of state changes, the better use of the asset, and where applicable its lifetime extension. Its effectiveness hangs on the punctual diagnosis and on the capability to manage state information across time in such a form that changes become recognisable and action-effective. For the BESS this demands the management of SOH estimates, temperature and load profiles, protection events, cell or string anomalies, availability losses, and interventions as a coherent evidence chain. Under this condition, the operating-data collection carries the actual controllability of the infrastructure asset.

Asset management in this understanding forms the continuation of systems engineering under temporal conditions. It carries the coherence of the asset across those transitions at which technology, economy, and institution converge; in this performance it is decided whether a storage system merely runs out its lifetime or actively manages it.

Chapter 20 — Augmentation, retrofit, and hybridisation

20.1 Structural interventions as the normal case of the lifetime

Large-scale storage systems belong to the few energy infrastructures in which substantial interventions during the use phase are the normal case. Modules, strings, cell inventory, PCS components, HVAC systems, control technology, EMS logics, sensor systems, and protection parameters are replaced, complemented, or fundamentally altered across the lifetime. The technical particularity of the battery — its ageing, its modular construction, its bounded lifetime in comparison with the surrounding infrastructure — makes such interventions unavoidable.

Three terms structure the field. *Augmentation* designates the addition of energy or power inventory within an existing frame: additional modules, new strings, complementary racks for the compensation of lost capacity or for the provision of additional power. *Retrofit* designates the technical upgrading or modernisation of existing subsystems: new PCS generations, updated EMS logics, upgraded protection equipment, modernised control technology. *Reconfiguration* designates an altered functional assignment of the system: new market roles, changed reserve policy, hybridisation with other assets, adapted grid requirements.

The difficulty lies in the arising heterogeneity. An asset in which different cell batches, various ageing states, new PCS generations, or altered EMS logics are brought together becomes larger or newer and at the same time becomes a different storage system. Its performance characteristic, ageing dynamic, fault sensitivity, warranty situation, and demonstration structure shift with every larger intervention. Every substantial intervention therefore demands the treatment as a structural change of the system body.

20.2 Augmentation as a decision under several variables

The augmentation decision marks the third of the four strategic decision moments from Chapter 19. It poses itself typically between the sixth and tenth operating year, when the SOH loss exceeds an economic or contractual threshold. The decision couples several variables: current state, remaining useful life, augmentation costs, market development, regulatory environment, alternatives such as second-life use or sale.

The first question concerns the state. An augmentation carries economically where the existing infrastructure — site, connection, transformer, switchgear, building, auxiliary systems — exhibits a residual useful life that lies markedly above that of the added battery inventories. This residual useful life decides on the classification as sustainable intervention or as short-term repair. In practice, before an augmentation decision, transformer and switchgear are examined via DGA and partial-discharge measurement, the control technology is assessed for remaining manufacturer-support windows and communication capability with current hardware, and the civil infrastructure (foundations, drainage, access roads, fire-protection concept) is examined for suitability for additional inventory capacity.

The second question concerns compatibility. New cells of a newer product generation differ from the original cells in internal resistance, capacity, rest-voltage characteristic, ageing behaviour, and performance characteristic. Where they are integrated into the same strings, currents distribute themselves inversely proportional to the internal resistances; the aged cells with higher *R*$_i$ carry a smaller share of the load and age thereby more slowly, while the new cells are over-proportionally loaded and fall faster to the level of the old cells. Thermally, additionally, inhomogeneous heat distributions arise, because the different power losses per cell must be dissipated in the same cooling zone. The clean solution separates old and new inventories into separate strings, connected in parallel via the PCS or via separate PCS units, but managed homogeneously within themselves. This architecture demands already in the original design a sufficient modularity of the primary and secondary technology: a sufficient number of independent DC inputs at the PCS or space for additional PCS units, separated DC switchgear, independent BMS communication paths, extensibility of the EMS logic by additional, in themselves homogeneous cell groups with separated SOH tracking and adapted derating curves.

The third question concerns the economics. Augmentation costs per kWh lie typically below new-build costs per kWh, because site, connection, structural infrastructure, substation, and a part of the control technology are dropped. The saved balance-of-system shares lie, depending on the project, at 35 to 55 per cent of the original project costs. For the decision, less the absolute cell price at the decision point is relevant than the ratio between augmentation costs and the expected revenue path of the remaining useful life.

The fourth question concerns the warranty and contractual landscape. Augmentation can affect the warranty order of the original asset, through explicit triggering clauses or through implicit boundary shifts. The coordination with the original manufacturer, the integrator, and the insurer is a substantial precondition. An augmentation that devalues warranty claims on the original inventory without creating an adequate replacement can be economically destructive; the typical contractual safeguarding comprises a consent declaration of the original manufacturer to the augmentation, a clear demarcation of the warranty periods for old and new inventory, and an adjustment of the insurance coverage to the altered system configuration.

20.3 Retrofit and the modernisation of subsystems

Retrofit differs from augmentation in that not the battery inventory but other subsystems are upgraded. Typical retrofit objects are power electronics, EMS logics, protection equipment, HVAC systems, fire-protection technology, control technology, and communication interfaces.

The *PCS renewal* is the economically most significant retrofit case. PCS generations develop quickly further. New semiconductor technologies — particularly silicon carbide (SiC) and in part areas gallium nitride (GaN) — enable higher switching frequencies (10 to 50 kHz instead of 2 to 5 kHz under classical IGBT topologies), higher efficiencies (typically one to one and a half percentage points against established Si-IGBT modules), lower losses in the partial-load range, and more compact construction through smaller filter components. New control logics extend the possibilities of grid support: grid-forming functionality, more precise fault-ride-through profiles under updated grid connection rules, extended reactive-power ranges, oscillation damping. A PCS retrofit becomes economically attractive after seven to ten years, when the original unit has aged or the requirements have grown. The decision hangs on the costs, on the compatibility with the existing DC and AC infrastructure — DC voltage windows, transformer transformation ratio, protection coordination — and on the regulatory requirements on the future installation behaviour.

The *EMS renewal* is operationally important, technically often simpler, and regulatorily less burdened than the PCS renewal. Newer EMS generations integrate predictive algorithms, improved forecast models, differentiated multi-use optimisation, model-predictive-control procedures, stochastic scenario optimisation, and tighter coupling with commercialisation platforms. Their introduction demands a disciplined transition planning: parallel operation of old and new systems across several weeks with identical input data and comparison of the dispatch decisions, validation of the results against documented reference scenarios, stepwise migration across individual products or individual time windows, clean cut-over with documented rollback option. The evidence continuity from Chapter 12 is immediately effective; an EMS change without transition planning generates a gap in the memory of the asset.

The upgrading of protection and safety equipment follows often regulatory shifts: new requirements on fire protection, on gas management, on access control, on cyber-security. A typical object list comprises the retrofitting of hydrogen and VOC sensor systems, the extension of the propagation barriers in correspondence with updated UL 9540A test results, the complementing of the control technology by IEC 62443-conforming segmentation and logging functions, the adjustment of the alarm and emergency-response plan to the current state of the local response forces. These retrofits form preconditions for the preservation of insurability and operating permission; their treatment as optional measures does not carry the institutional embedding of the asset. Their economic assessment differs structurally from augmentation; they form preservation investments, not extension investments.

20.4 Hybridisation and the coupling with other assets

With growing market maturity, the role of the BESS shifts from a mono-functional storage system towards a more frequently coupled system object. Co-location with wind and PV installations, the embedding at former power-station sites, the proximity to industrial clusters, data centres, charging infrastructure, or in perspective electrolysers changes the use and the system body itself. Hybridisation forms in this context the expression of an altered site and function logic and reaches beyond a fashionable complement.

The technical architecture of hybridisation can be designed to different depths. In the *loose coupling*, the assets share the grid connection. PV inverters and PCS work independently on the AC side; commercialisation takes place separately or coordinated via a higher-order layer. In the *tight AC coupling*, the assets share transformers, medium-voltage switchgear, and possibly the control technology; commercialisation is coordinated, and the connection power is actively managed as a common restriction. In the *DC coupling*, PV arrays and battery share a common DC bus, which saves a conversion step between PV-DC and grid-AC and improves the overall efficiency by one to three percentage points, and at the same time raises the technical and regulatory complexity. The dimensioning of the common DC bus, the voltage windows of the PV strings and the battery must be coordinated; the PCS must manage the combined power and clearly demarcate the behaviour in case of faults in the PV part or in the battery part; the protection coordination must take into account the DC currents of both sources. The relevant CIGRE works on DC coupling have systematically described this configuration.

Hybridisation shifts the system body in several respects. It shifts connection and protection logic, because several feed-in units are coordinated and the protection grading must map the different short-circuit contributions and fault profiles. It shifts the SOC policy, because the PV feed-in shapes the charging window of the storage system in time-of-day correlation and a reserve holding for aFRR in the afternoon hours with high PV feed-in is to be placed differently than on overcast days. It shifts the reserve windows, because forecast errors of the generation installation can be mapped in the reserve requirements of the storage system. It shifts the safety relations, because thermal and electrical events can wander between the systems and the fire-compartment formation must include the neighbouring assets. It shifts the revenue and blocking structure, because the optimisation takes place across several services and several assets simultaneously and the assignment of individual MWh to individual assets must be clearly regulated in balancing and tax terms. And it shifts the later development option of the system, because augmentation and decommissioning are possible only in coordination with the partner assets.

These changes are to be carried along technically, organisationally, and evidentially. Hybridisation shifts the system role itself and thereby suspends the notion of a complement on an unchanged ground. An asset that maps this shift in its technical and institutional identity preserves its legibility and thereby its controllability; a mapping that persists in the original identity does not carry the new system role.

20.5 Power-station-adjacent sites and the economics of existing infrastructure

In the German transformation space, power-station-adjacent sites gain a quality of their own. Former coal and nuclear power station areas have at their disposal existing connection infrastructure: high-voltage or extra-high-voltage installations, transformers, switchgear bays, buildings, access roads, cooling-water infrastructure, specialist personnel on site. This pre-shaping fundamentally changes the economic calculation of a BESS.

The connection question from Chapter 3 poses itself differently at power-station-adjacent sites. The connection capacity already exists in the order of magnitude of the originally installed power-station power, the permitting situation is established, the routing is in place. The BKZ burden can fall lower, because the decommissioned power station has carried the original connection costs and the grid expansion measures to be financed for the storage connection fall smaller. The planning and realisation times are shorter, the project risks lower; typical maturity-tier advantages arise from the existing site control, the known grid connection conditions, and the established coordination with the network operator.

The Fraunhofer ISE works on the role of stationary storage systems at former power-station sites describe this logic systematically. The value of the existing infrastructure lies in the time savings and risk reductions made accessible by it; the infrastructure itself forms the material carrier of this value, but is not identical with it. A power-station-adjacent BESS site goes into operation two to four years earlier than a greenfield project of comparable size; this time advantage translates immediately into revenues that remain inaccessible to a later project. The electrical embedding into the existing switchgear demands an examination of the residual useful life of the primary technology, the compatibility of the secondary equipment with current IEC 61850 requirements, and the protection coordination between the new storage field and the remaining feed-in points.

For the hybridisation logic a particular dynamic results. Power-station-adjacent sites offer good connection conditions and at the same time space for further assets: electrolysers, data centres, industrial loads. The BESS can become the core of a hybrid infrastructure landscape that carries several forms of value creation simultaneously. The NEP supporting documents make visible that the real grid space consists of lines and of switchgear, transformers, infeeds, and customer connections; nodes that are already pronounced gain, against this background, strategic significance.

The actual quality lies in the clean management of the arising complexity. Augmentation, retrofit, and hybridisation form concrete decisions that change the identity of the asset. Their robustness hangs on whether the asset can still be read despite the change: whether its technical history, its parameterisations, its ageing dynamic, and its contractual landscape remain available in carried-forward form.

Chapter 21 — Battery passport, digital identity, and circularity

21.1 The digital identity as continuation of the asset

The digital representation of a BESS, introduced in Chapter 12 as an instrument of coherence assurance, gains in the lifecycle context an additional dimension. It carries a continued identity in which technical, material, and lifetime-related information run together. This identity arises with design, procurement, commissioning, and ongoing operating management, and accompanies the asset across all owner, operator, market, and regime changes.

The reach of this identity shows itself in concrete tasks. Component and material assignment remain reconstructible across years. State and performance histories are carried forward. Augmentation and reconfiguration order themselves into the existing history. Later circularity and second-life decisions stand on a robust data basis. The residual-value curve of an asset with gap-free identity reaches markedly beyond that of an asset with fragmented history, and carries this difference more strongly than any individual measure of upkeep.

21.2 The European Battery Regulation and the battery passport

Regulation (EU) 2023/1542 — the European Battery Regulation — institutionally re-orders the lifecycle of batteries. It obliges manufacturers, integrators, and operators to a continued documentation that reaches markedly beyond the previous product labelling. For stationary battery storage systems with a capacity above 2 kWh, the battery passport becomes a legal obligation from 18 August 2027.

The contents of the battery passport are structured into several information classes. The *identification level* comprises unambiguous labelling, manufacturer, production date, chemical composition, and weight specifications. The *performance and durability level* comprises capacity, power, efficiency, cycle capability, and expected lifetime. The *sustainability level* comprises CO_2 footprint, shares of recycled raw materials, origin of critical materials, and due-diligence obligations in the supply chain. The *lifecycle level* comprises repair, disassembly, and recycling instructions, as well as information on reusability. The *conformity level* comprises certificates, test reports, and regulatory documentation.

The technical implementation takes place as a Digital Product Passport (DPP) with standardised data models, which are concretised in the running standardisation works of CEN/CENELEC JTC 24 and the associated delegated legal acts of the Commission. Access takes place typically via a QR code or a machine-readable identifier on the asset, which refers to a distributed or central data holding; the communication protocols orient themselves to established web standards (REST/JSON with signed data for manipulation security). The access control distinguishes several levels: publicly accessible are basic safety and environmental information; confidential remain detailed performance and state data; regulatorily accessible are data for market-surveillance authorities, recyclers, and authorised maintenance enterprises. This access tiering is a design feature that is to carry the data value across time.

The structuring is consequential for the asset management. It shifts the responsibility for data contents from the individual operator to a distributed order of manufacturer, integrator, operator, and recycler; each party carries a part of the data, and each makes its partial history available in a form that runs together interoperably. The data authority discussed in Chapter 18 thereby gains a regulatory anchoring and is no longer only a contractual question.

21.3 Traceability and the object history

Traceability has several functions at once. It enables the cause analysis after events by tracing a defective module back to production batch, supplier, and operating conditions. It enables the quality tracking across the supply chain, because systematic deviations of certain production batches are identified and addressed. It enables the value preservation under later further use: the second-life value of a module with known history reaches well beyond the recycling value of a module without history. It enables the circularity by allowing recyclers to determine the material composition without renewed analysis and to select process routes in a targeted manner.

The technical implementation takes place via hierarchical labelling systems. Each cell carries an unambiguous serial number, typically under DIN 91252 (Cell Identification Code) or a comparable manufacturer system with specifications on manufacturer, production site, production date, batch number, and running cell number; the labelling takes place physically as a data-matrix code or QR code on the cell and electronically in the BMS communication. Each module aggregates the serial numbers of its cells and carries an own module identifier; each rack aggregates the serial numbers of its modules; the overall system aggregates the rack identities. This hierarchy remains consistent across the entire lifecycle — from production through integration, commissioning, augmentation, possible second-life use, to recycling. A module replacement without corresponding upkeep of the identity hierarchy generates a gap that is later no longer to be closed; the updating of the hierarchy belongs in the change-management process of every intervention.

The object history arising from this traceability forms the operative substance of the battery passport. It transfers the battery from the status of an anonymous product into the status of an identified infrastructure object with legal, economic, and systemic consequences. The second lifetime half of an asset with robust history carries structurally more soundly than that of an asset with patchy continuation — an experience from older infrastructure classes that, in battery technology, is institutionally anchored in advance.

21.4 Second-life use and the re-assessment of the inventory

Second-life use, in international usage referred to as second life, describes the deployment of aged battery modules in applications that are less demanding than their original use. A module that has, after ten years of stationary deployment, fallen below its 80-per-cent SOH threshold continues to possess value in other applications: as buffer battery in charging infrastructure, as energy storage in decentralised PV systems, as backup storage in telecommunications installations, or as reserve capacity in less cycle-intensive applications.

The economic basic idea is intuitive: the remaining storage capability of an aged module has a market value that lies above the scrap value and below the new value. Second-life use makes this value difference accessible. The practical implementation demands technical, legal, and economic clarifications that are not yet fully matured in the current market phase.

The technical assessment of an aged module is more demanding than that of a new one. The dispersion between individual modules of an originally homogeneous batch grows with the ageing; internal resistances drift apart, capacities develop differently, thermal properties vary, the cell dispersion within a module grows. A sensible second-life use therefore demands a careful measurement and classification. The typical examination procedure comprises a complete capacity measurement under defined reference conditions, an internal-resistance measurement via HPPC pulse tests, a cell-voltage and temperature homogeneity examination under load, an insulation measurement, and a visual examination for mechanical damage. On the basis of these measured values the modules are classified into grade classes (typically A, B, C with different tolerance bands for remaining capacity and internal resistance); the recombination into groups as homogeneous as possible within the second-life application follows the classification results. The reconditioning costs form a substantial part of the second-life costs and lie in the order of magnitude of €20 to €40 per kWh of usable residual capacity.

The legal classification is complex. A second-life module occupies a category of its own between new product in the sense of product liability and recycling product; it carries the complete assignment to neither of the two established classes. Warranty, product liability, operator liability, and insurability are to be re-ordered; the placer-on-the-market of the second-life application typically takes on the role of a new manufacturer with corresponding obligations under the General Product Safety Regulation, the Battery Regulation, and the Low Voltage Directive. The European Battery Regulation has created framework rules; the practical configuration still varies markedly between member states and application classes. The battery passport forms a central instrument; in its absence, legal second-life use remains difficult, because origin, chemistry, operating history, and safety state must be demonstrated for the new conformity assessment.

The economic viability of second-life use hangs on the price gap between new modules and second-life modules, as well as on the reconditioning costs. Under fast-falling new prices, the second-life window narrows, because the absolute price difference between new and second-life module shrinks and the reconditioning costs take on a larger relative share. Under stagnating or rising new prices, the window opens. The strategic assessment of second-life use takes this dynamic into account and does not derive itself statically from current conditions.

21.5 Decommissioning, recycling, and circularity

The decommissioning of a BESS transfers the asset into a new material and responsibility order; its reading as a final administrative act does not carry this transfer. Safety, logistics, disassembly capability, data availability, residual energy, and material separation belong in a common order. In this order quality it is decided whether the asset remains, in its material substance, circular-capable, or ends as a cost-intensive disposal case.

The operative mechanics of decommissioning begin with the electrical discharge of the cells to a safe voltage level. The discharge to below 30 per cent SOC reduces the energy quantity in the fault case; a deeper discharge to 0 V is required for individual recycling pathways but must take place in a controlled and chemistry-specific manner, because an uncontrolled deep discharge under some chemistries triggers copper dissolution and thereby raised short-circuit risks during disassembly. The disassembly follows a hierarchical logic from rack through module to cell level, with thermographic monitoring and redundant insulation measurement at every step. The transport is subject to the regulations for dangerous goods under ADR: used lithium batteries fall under UN 3480 (lithium-ion batteries) or UN 3481 (packed in equipment); damaged or defective batteries fall under the sharper regulations for UN 3536 (batteries transported in vehicles) with specific requirements on packaging, labelling, and accompanying documentation.

The European Battery Regulation sets quantitative requirements. By 2027, at least 65 per cent of the weight of lithium-ion batteries must be materially recovered; by 2031 the requirement rises to 70 per cent. Minimum recovery rates for individual raw materials apply additionally: 50 per cent for lithium and 90 per cent for cobalt, copper, lead, and nickel by 2027; 80 per cent for lithium and 95 per cent for the remaining substances by 2031. These quotas are demanding and drive the industrial development of the recycling infrastructure in Europe strongly.

The technical recycling pathways differ structurally. *Pyrometallurgical procedures* smelt the batteries in blast furnaces and recover metals from the slags; they are robust against different input materials, energy-intensive, and they lose lithium to the slag and the electrolyte as a volatile phase, which complicates the lithium recovery rate of the regulation. *Hydrometallurgical procedures* dissolve the active materials, after mechanical pretreatment (shredding, sorting of the black-mass fraction), chemically in acid baths and recover individual elements through selective precipitation or solvent extraction in high purity; the achievable yields for lithium, nickel, and cobalt typically lie above 90 per cent; the requirements on pretreatment and wastewater treatment are high. *Direct recycling* aims to recover the cathode active materials without complete chemical decomposition and to deploy them again after a reactivation; the procedures are technically promising but still stand in an early development phase and presuppose a precise knowledge of the original cell chemistry, which without battery passport is not given.

Circularity arises in separability, component identity, data depth, safety information, and documented degradation history already in the design; its retrospective production at the end of the use phase does not carry this quality. Three preconditions carry the operative side of this property: violence-free disassembly capability of the modules, unambiguous documentation of the cell chemistry for the targeted assignment to the suitable recycling pathway, and robust information about the safety state at the end of use, in order to bound logistical risks and additional costs in decommissioning.

The European perspective on battery production and battery economics takes up these connections systematically. Supply chains, material intensity, CO_2 footprint, industrial sovereignty, and raw-material governance are discussed jointly with the circular economy. The assessment of a BESS project treats its circularity as a quality feature of the system architecture; its reduction to a compliance dimension does not carry the strategic significance. The economic measurability of the difference between circular-capable and non-circular-capable assets will rise in the coming years — through raw-material prices, disposal costs, reputational effects, and regulatory reliefs.

Chapter 22 — European value creation and industrial embedding

22.1 The asset as part of an industrial order

The BESS is an operating object across the lifetime of the system and at the same time part of a European production and site economy. Which storage systems are produced, integrated, operated, modernised, and recycled in Europe acts into the lifecycle of the individual assets. Spare-parts paths, augmentation options, OEM stability, component homogeneity, and re-investment capability hang on the industrial depth of the value creation.

This binding is frequently underestimated. A storage system whose operator relies on cell and system suppliers that produce far outside Europe and whose availability hangs on geopolitical shifts carries a risk that is rarely fully mapped in the project calculation. A storage system in a denser European ecosystem profits from redundancies in the supply chain, from shorter routes, from compatible regulatory framework conditions, and from a more stable spare-parts and augmentation logic.

The industrial embedding of the BESS forms a structural property of the asset; its reading as a macroeconomic background condition does not carry its reach. It shapes capital structure, operational manageability, and strategic resilience across the entire useful life. The assessment of a storage project under inclusion of the robustness of its industrial environment carries the subject completely.

22.2 European production in the global comparison

The global battery cell production is currently dominated by Asian manufacturers. China holds an estimated 75 to 80 per cent of the worldwide cell production capacity; Japan and South Korea together hold a further 10 to 15 per cent. The European production shares in upstream-near segments — cathode materials, anode materials, electrolyte production, cell production — lie markedly behind these values. For cathode materials, the European share of the global production in 2024 lay below 10 per cent; for cell production at around 10 to 15 per cent, with a rising tendency.

The dynamic is shifting. European and North American cell production is growing under the influence of industrial-policy initiatives. The European Battery Alliance (since 2017), the US Inflation Reduction Act (2022), the European Critical Raw Materials Act (2024), and the Net-Zero Industry Act (2024) drive the build-up of local production capacities. The announced European cell production capacities for 2030 sum to around 1,200 GWh per year; the actually realised capacities will, according to current forecasts, lie between 600 and 900 GWh per year.

The industrial structure behind these figures is heterogeneous. Scandinavian projects concentrate on integrated production with low CO_2 footprint through access to low-emission electricity supply; German and French projects orient themselves to automotive-near sites with connection to existing OEM clusters; Eastern and Southern European projects increasingly arise as joint ventures with Asian partner enterprises that contribute technology and capital, while the local production and labour structure is borne European. The realisation rates of these projects differ considerably; delays through permitting procedures, financing difficulties, volatile cell prices, and demand uncertainties in the automotive sector have in past years led to shifts and withdrawals of individual announcements.

For the BESS perspective, the order of magnitude is more significant than the detail figure. The European cell production will, by 2030, presumably suffice to cover a substantial part of the European demand from automotive sector and stationary storage segment locally. The availability will improve structurally, even if a complete independence from Asian suppliers is not reachable in the coming ten years. In parallel, a growing layer of European system integrators and EPC actors is developing, which anchors the integrative part of the value creation — container construction, EMS development, project realisation, operating management — more strongly in Europe. This integration level is immediately relevant for the lifecycle stewardship of a concrete BESS, because it carries the competence for commissioning, augmentation, retrofit, and warranty handling on site.

22.3 Raw materials, supply chains, and critical materials

The raw-material basis of lithium-ion batteries is structurally vulnerable. Lithium, cobalt, nickel, graphite, and copper form the core assortment; each of these raw materials carries its own supply risks. Lithium production concentrates on Australia (hard-rock spodumene), Chile and Argentina (brines from salt lakes), and China (both pathways); the refinery stage that recovers from the raw material lithium in battery-grade quality lies for over 60 per cent in China. Cobalt comes for over 70 per cent from the Democratic Republic of Congo; the refinery for over 70 per cent from China; the concentration in conflict regions is a continuing topic of the due-diligence obligations under the European Battery Regulation. High-grade natural graphite is predominantly mined and processed in China; synthetic graphite is likewise predominantly produced in China. Nickel has, with Indonesia, Russia, and Australia, a broader but geopolitically sensitive basis; battery-grade nickel sulphates come, in a growing share, from Indonesian projects with specific CO_2 and environmental debates. Copper has the broadest geographical basis and carries the lowest supply risk.

The European answer to this situation pursues several pathways at once. The Critical Raw Materials Act sets for 2030 the targets that 10 per cent of the strategic raw materials are mined in Europe, 40 per cent processed in Europe, and 25 per cent recovered from European recycling. Additionally, no third country is to deliver more than 65 per cent of an individual strategic raw material.

The targets are ambitious and will in part not be fully reached; they shift the strategic orientation of the supply chains markedly. The processing side develops faster than the mining side; European lithium refinery projects in Germany, France, and Finland stand in different implementation phases and will, insofar as realised, reduce the dependence on Chinese refinery markedly.

The dynamic is influenced by chemistry variants that have different raw-material profiles. *LFP cells* (lithium iron phosphate) require neither cobalt nor nickel and are markedly less raw-material-critical than NMC cells. For stationary large-scale storage systems, LFP has gained a growing market share in past years; according to current projections, by 2030 more than 70 per cent of all newly installed stationary BESS will work on an LFP basis. This shift reduces the raw-material exposure of the stationary storage segment against the mobility sector, which remains more strongly dependent on NMC and its variants. Alongside LFP, further raw-material-low variants develop. *LMFP* (lithium manganese iron phosphate) raises the energy density against LFP at a similar raw-material basis. *Sodium-ion cells* structurally change the raw-material logic: sodium is recovered from marine brines and terrestrial rock-salt deposits that are geographically broadly distributed and European-accessible; cobalt and nickel drop out completely, lithium likewise. The cathode materials are based predominantly on manganese, iron, and sodium iron phosphate compounds; the anodes frequently on hard carbon instead of graphite. Thereby the exposure to the most critical supply chains of the lithium world reduces considerably, at, however, lower energy density than LFP. *Solid-state cells*, finally, fundamentally change the electrolyte chemistry and lead, depending on the pathway — oxide-based, sulphide-based, or polymer-based — to different new raw-material requirements, for instance lanthanum or zirconium in garnet-structured oxide solid electrolytes, germanium or phosphorus in sulphide-based systems. The question when these pathways gain stationary relevance is open; sodium-ion cells have in the past two years reached the threshold to commercial availability, but stand in stationary applications still at the beginning of market penetration.

The significance for the BESS asset is double. The LFP orientation reduces the structural supply criticality; a 4-hour system on LFP basis carries the raw-material procurement more robustly than comparable NMC systems. New chemistry pathways are emerging on the horizon and belong in strategic technology observation, less in the immediate project decision of the coming years.

22.4 Recycling infrastructure as second industry

The recycling quotas of the European Battery Regulation discussed in Chapter 21 require the build-up of a self-standing recycling industry. Currently the European recycling capacity for lithium-ion batteries is designed for around 100,000 tonnes per year; for 2030, capacities between 400,000 and 700,000 tonnes per year are projected. This dimension is economically substantial and corresponds to an investment volume of several billion euros across the coming decade.

The recycling infrastructure acts as an industrial value-creation stage that exists independently of primary production and at the same time supports primary production. For the BESS this second industry has several functions. It supplies primary production with reconditioned raw materials and reduces the import dependence. It creates a residual-value market for aged battery systems and stabilises the lifecycle economics. It reduces the disposal costs and thereby the total burden across the useful life. And it creates regulatory conformity that becomes increasingly relevant for the operating permission in Europe.

The geographic distribution of the recycling infrastructure concentrates on sites with existing metallurgical competence. Belgium contributes with historically grown non-ferrous-metal hydrometallurgy and integrated recycling installations; France develops pyrometallurgical and hydrometallurgical capacities in the context of its automotive industry; Germany builds up recycling capacities in the industrial Ruhr area and in eastern Germany, typically in coupling with the cell production sites; Norway and Sweden use the existing metallurgy and favourable electricity prices; Spain and Poland develop sites in connection with growing vehicle and storage production. The process routes differ between the sites, which is relevant for the assignment of individual end-of-life batteries to recovery pathways: pyrometallurgical sites take in mixed input streams with broad chemistry variance; hydrometallurgical sites demand a prior chemistry separation via the battery passport or via sorting technology.

The proximity between cell production, application markets, and recycling sites changes the logistics and the cost structure of the entire value chain. A stationary BESS whose modules are reconditioned at the end of their use within a few hundred kilometres' distance has a different decommissioning economics from one whose end-of-life modules are transported across continents. The transport costs for dangerous-goods class 9 under ADR lie for end-of-life batteries typically at €100 to €250 per tonne in Europe; they rise markedly for intercontinental sea freight with sharper packaging and monitoring requirements.

22.5 Industrial sovereignty and strategic infrastructure policy

The European debate on industrial sovereignty has in the past years reached a depth that classifies the BESS as a self-standing infrastructure object. The combination of energy transition, geopolitical reordering, and industrial-policy rediscovery has lifted stationary battery infrastructure out of the niche of a specialist discussion into the rank of a strategic infrastructure class. The shift shows itself in three parallel developments. *Security of supply* demands strengthened reserves of locally available flexibility, in order to remain capable of action under market disturbances, grid congestions, or generation fluctuations. The *system integration of renewable energies* demands structurally growing storage capacities, in order to balance the temporal and spatial difference between generation and consumption. The *industrial policy* demands local value creation, in order to reduce dependencies and secure high-quality jobs. All three developments raise the strategic importance of stationary storage systems in the European infrastructure structure.

The empirical dimension of this shift is visible. According to the market reports of the European industry association Energy Storage Europe (EMMES 9.0 and 9.5), the European battery storage market reached at the end of 2024 an installed total capacity of around 89 GW and grew in 2025 to about 100 GW; for the period through 2030, build-out rates in the order of 115 per cent are expected. These figures comprise home, commercial, and large-scale storage jointly and lag in their methodological granularity behind the German market statistics. They show nevertheless that the European storage space has structurally left the phase of niche market and is developing into an infrastructure layer whose order of magnitude becomes comparable with other infrastructure classes of the energy system. Germany, the United Kingdom, Italy, and Ireland lead this development in their respective market constellations; the European diffusion follows different revenue regimes, from merchant-oriented British models to capacity-market-integrated Irish and Italian structures. The heterogeneity of the market accesses generates a differentiated landscape in which the industrial and institutional embedding of stationary battery storage in the European space takes place differently depending on market structure.

The role of the BESS has thereby structurally changed. From a complementary technology for individual application problems, a flexibility and arbitrage instrument under market volatility arose, and from this instrument arises a strategic infrastructure class whose availability, controllability, and industrial embedding flow into the national and European capacity to act. The development acts structurally and reaches beyond a rhetorical figure: it shapes regulation, financing, site decisions, and the quality of the contractual and governance architecture from the preceding chapters.

For project practice, the shift has several concrete consequences. Permitting procedures tend to be relieved when stationary storage systems are recognised as strategic infrastructure; in the German implementation, this is reflected in shortened planning and permitting periods under the Energy Industry Act and the building and planning law, insofar as the classification as infrastructure of overriding public interest applies. Financing instruments open up when public funding or guarantees are coupled to industrial-policy targets; IPCEI frameworks (Important Projects of Common European Interest), EIB financings, and national funding instruments under the innovation-engine approach can carry parts of the project financing. Local value-creation requirements increasingly enter public procurement and favour the selection of European suppliers; the Resilience Criteria of the European Commission, foreseen for public procurement in the area of net-zero technologies, act complementarily. Insurance and credit conditions improve when the strategic relevance of the asset flows into the risk assessment. The effects appear regionally and project-dependent in different intensity, but gain increasingly in visibility and belong in strategic project development.

The European perspective condenses the long-term structure of the BESS in an order of its own. Degradation, availability, residual performance capability, augmentation, warranty transitions, hybridisation, battery passport, circularity, and industrial embedding form the substance that carries the asset across its lifetime. A BESS confirms its original plausibility only in the course of its lifetime; the day of commissioning marks the beginning of this confirmation, not yet its conclusion. The confirmation succeeds to the extent that change is carried in legible form.

Part VI — Principles and conclusion

Chapter 23 — Seven principles of infrastructural controllability

Seven principles condense the carrying insights into a manageable form. Each draws from several chapters and remains action-guiding in planning, construction, operation, and further development. *Controllability* designates in this connection the capability to hold a BESS as an infrastructure object across its lifetime in technically, operationally, and institutionally legible states. The concept reaches beyond the pure protection and safety semantics with which it is frequently narrowly led in electrical plant engineering. It includes the active management of the permitted state space, the coordination between the technical layers, the consistent carrying-forward of evidence, the governance of the contractual interfaces, and the preservation of connectivity to changing regulatory and market regimes. A BESS is, in this sense, controllable when it can be operated under changing conditions and at the same time held across time in its technical, operational, and institutional identity.

23.1 Controllability begins before operation

The quality of a BESS is decided in the prehistory of its realisation. Dispatch acts on this basis and does not first establish it. Site dependence, connection body, substation, cell and component availability, conversion chain, and supply path lay down the real form of the later asset. Where these elements are brought together with discipline, a project arises that holds its plausibility also under shifted market conditions. Where they are treated as separate work streams, an asset arises whose lifetime is burdened early by silent incoherences.

The connection competition particularly pronounced in the German context, from Chapter 3, sharpens this principle. A storage project can appear economically plausible, chemically available, and technologically in principle controllable, and nevertheless remain infrastructurally immature where its early architecture is not robust. Project maturity forms in this reading the first form of the later system body; its reading as a formal pre-stage of construction does not carry this reach.

23.2 The state space is the actual quantity of management

A BESS is determined by the state space in which it may in fact move. SOC windows, temperature limits, C-rate limits, reserve commitments, safety limits, and grid-side restrictions together form an operational geometry that circumscribes the effective capacity of the asset. What lies outside this space exists only as a nameplate specification.

The management of this space forms the central discipline of operation. It transfers technical capability into deliverable performance and decides whether headline figures become robust system services. The difference between nameplate and reality, between nominal and functional availability, between geometric and economic duration carries the entire relation between design, operation, and assessment.

23.3 Safety is a property of permitted operation

Safety in the BESS is a property of the state space itself. It arises from the capability of the system to remain in controllable states or to be transferred into them under deviation, fault, ageing, and incomplete information. Fire-detection and extinguishing technology, access rules, and protection relays form instruments within this architecture. The quality of safety shows itself in the selectivity with which events are bounded. A storage system that loses an individual cell remains in the area of a technical event;

a storage system that lets a cascade run through exceeds this boundary and becomes a fire event. Propagation logic, protection grading, selective isolation, and disciplined restart form the structure in which from potential endangerment operative controllability arises and safety remains legible across the lifetime.

23.4 Evidence carries the temporality of the asset

The BESS changes from the first day of operation. This change remains institutionally legible only when it is recorded along in a consistent data and evidence structure. Measurement data, event chains, parameterisation histories, firmware states, test results, interventions, warranty references, and degradation indicators belong in a reconstructible structure. Without it, the asset loses, with growing operating time, its operational truth.

The reach of this principle exceeds the administrative reading. Insurability, creditworthiness, warranty enforcement, operator change, augmentation, hybridisation, battery passport, and second-life use hang on the quality of the carried-forward history. The residual-value corridor of an asset with gap-free evidence carries its institutional robustness; a fragmented history cancels this corridor irretrievably.

23.5 Market and grid are always jointly effective

A storage system is rarely fully carried by a single institutional mechanism. Its earnings basis is composed of arbitrage, reserve holding, system services, and increasingly from capacity and availability mechanisms. These mechanisms overlap, partly block one another, and are bounded by grid-side restrictions. The isolated optimisation of an individual revenue path overlooks the coupling that shapes the reality of operation.

The isolated consideration of the market overlooks at the same time that the storage system remains a grid object. Dispatch decisions that are market-rational can act grid-loadingly; grid-relieving operating modes can remain economically under-incentivised. The gap between market value and system value forms a structural property of storage economics; its reading as an anomaly of the present regime does not carry the institutional reality. A project that remains viable under several plausible regime developments carries structurally more robustly than a project whose viability is bound to an individual configuration of institutional mechanisms.

23.6 Institutional clarity is part of the technical quality

The technical architecture of a BESS is in practice rarely operated from a single hand. Asset owner, technical operating manager, EMS provider, marketer, aggregator, network operator, insurer, and financier form a contractual landscape whose quality acts immediately back on the controllability of the asset. Where control authority, data authority, release rights, and safety responsibility remain contractually unclear, friction can arise in operation and paralysis in the event case.

The governance of the interfaces proves its quality in marginal situations: after a reserve violation, a degradation-related power drop, a safety incident, an operator change, or an augmentation. A contractual landscape that remains workable under event pressure secures the value of the asset precisely in those moments in which it is endangered. A governance that is established only in the event case loses its robustness in this very moment.

23.7 The asset must be able to tell itself across time

The lifetime of a BESS contains a manageable number of strategic decision moments: the transition from commissioning into regular operation, the first performance examination, the augmentation and retrofit decision, the transition into second-life use or decommissioning. At each of these moments, the previous history of the asset enters into a new assessment. Its quality is decided by the capability to explain itself; the original specification carries this assessment only to a limited extent.

This capability arises from the coherence with which the asset carries its technical, material, operational, and institutional layers across time. A storage system remains legible as an infrastructure object when its architecture remains clear also after ten years, when its data and evidence structure holds changes reconstructible, when its augmentations and retrofits can be classified in the object history, when its connectivity to new regulatory and market regimes is preserved, and when its material and digital identity prepares decommissioning, second-life use, and circularity. In the absence of this coherence, the asset becomes an installation without a history; with the loss of its legibility, the basis of its value falls away.

Chapter 24 — Concluding consideration

The BESS appears in the transformed electricity system as a coupled infrastructure object; its reading as an enlarged battery or as a mere arbitrage instrument does not carry the reality of this object class. The value of the asset arises and preserves itself where technical architecture, state management, safety logic, market and grid integration, evidence continuity, and lifetime of the system act together in robust form. The coherence of the system under changing electrical, thermal, market, and regulatory conditions carries thereby further than the installed megawatts and megawatt-hours. The seven principles of the preceding chapter condense this insight into a manageable form and remain action-guiding for planning, construction, operation, and further development of the asset.

Several developments will shift the reading of the BESS in the coming years and at the same time preserve the basic architecture of the controllability developed here. Alternative cell chemistries — sodium-ion batteries, solid-state cells, LMFP — will shift the raw-material basis, the safety characteristic, and the lifecycle economics of the stationary storage segment; the core statements of the compendium relate to the BESS as a coupled system object and remain therefore independent of the specific chemistry. Grid-forming functions are at present moving from the debate into technical differentiation; regulatory recognition, market remuneration, and technical certification will gain in profile in the coming years, and storage systems with grid-forming-qualified power electronics will take on a more strongly carrying role than the current market logic makes visible. The institutional embedding of flexibility — capacity mechanisms, local grid charges, system-service markets, the role of the aggregators — will undergo substantial shifts in the coming decade; which mechanisms prevail remains open, and the robustness of a storage project is measured by its capability to remain viable under several plausible regime developments. The BESS enters increasingly into a denser composite with electrolysers, data centres, industrial loads, charging infrastructures, offshore connections, and heat systems; each of these couplings changes the system body and demands a discipline of management of its own, the depth of which remains to be developed in future, self-standing works.

The BESS has, in the past fifteen years, travelled a path that has rarely been observed in the energy industry at comparable speed. From a technology that was understood at the beginning of the 2010s as a niche complement, an infrastructural object class has arisen that co-carries the transformed electricity system in Germany and Europe. This development follows from the interaction of accelerated variable generation, growing grid congestions, differentiated commercialisation and system-service roles, digitally guided operating logics, and the insight that stationary storage systems form an object class of their own that reaches beyond the concepts of the battery and the market instrument. The role that the BESS will take on in the coming decade is foreseeable in its basic features and open in its details: it will carry security of supply, support renewable integration, mitigate grid restrictions, and serve industrial-policy targets. It will enter into denser connections with other infrastructure classes and operate in growing institutional complexity. It will run through its own ageing and reach repeatedly into decision moments at which its previous history enters into a new assessment.

Through all these transitions, the asset carries to the extent that its controllability has been held with discipline in every phase. The conditions of this controllability are technical, economic, safety-related, institutional, and temporal at once. They reach across every individual layer and well beyond a one-time setting. They arise in the discipline with which the asset is held across years — through project maturity, through operation, through evidence, through governance, and through structural further development. The installations that are erected in the coming years will show across their lifetime whether this controllability was held.

Appendix I – Formal models, derivations, and analytical depth

I.1 Purpose, scope, and model boundaries

Appendix I gathers the formal, analytical, and dimensional relations whose relevance the main text establishes and whose full derivation is reserved for the annex. Its purpose is the formal underpinning of the central relations that govern the planning, design, operation, assessment, and long-term development of large-scale BESS. It replaces neither a chemistry-specific cell record nor a complete digital twin. It supplies an analytical core deep enough to describe operative restrictions, loss paths, ageing, and grid-side bindings in a load-bearing manner, and at the same time reduced enough not to dissolve technical and economic decisions into unnecessary detail. The annex thereby carries the formal conditions under which system controllability, in the sense of the main text, becomes operatively measurable.

The scope covers stationary, grid-connected battery storage systems at the system level. The focus rests on the LSS segment in the sense of the MaStR and Battery Charts classification introduced in the main text. The formal relations carry largely also for ISS, while HSS — owing to its reduced restriction spaces and simpler conversion chain — is not treated separately. The derivations relate to the BESS as a coupled object made up of electrochemical inventory, conversion chain, auxiliary systems, state management, evidence structure, and market- and grid-side embedding.

The relations are chemistry-open in the sense that the structural equations — energy balance, restriction spaces, ageing logic, economic and grid binding — are formulated largely independently of the chosen cell chemistry. The parametric calibration carries for the currently dominant lithium-ion family, including the LMFP variant identified in Chapter 5 as the first deployment-ready extension path. For sodium-ion cells the structural equations apply analogously, with differing parameterisation of voltage range, temperature window, and cyclic ageing. Solid-state cells, on the thresholds referenced in Chapter 5, are not at present a deployment-ready alternative for stationary large-scale applications; their formal treatment lies outside the core of this appendix.

I.2 Symbols, units, and basic conventions

I.2.1 Symbology

A consistent symbology is used. The quantities below appear throughout Appendix I; subscripts denote reference, position, or state of the quantity in each case.

Base quantities.

- P: power $[\mathrm{kW}]$, $[\mathrm{MW}]$
- E: energy $[\mathrm{kWh}]$, $[\mathrm{MWh}]$
- U: voltage $[\mathrm{V}]$
- I: current $[\mathrm{A}]$
- Q: electric charge $[\mathrm{Ah}]$, $[\mathrm{C}]$
- R: resistence $[\Omega]$
- t: time $[\mathrm{s}]$, $[\mathrm{h}]$
- DoD: depth of discharge per cycle [-]
- SOC: state of charge [-]
- SOH: state of health [-]
- η: efficiency [-]
- T: temperature $[{}^{\circ}\mathrm{C}$, $[\mathrm{K}]$

Derived and composite quantities.

- *C-Rate*: ratio of power to usable energy content $[\mathrm{h}^{-1}]$
- f_{SOC}: usable SOC range, $f_{SOC} = SOC_{max} - SOC_{min}$ [–]
- ΔSOC_{op}: operative SOC window [–]
- ΔSOH: change in state of health [–]
- N_{EFC}: equivalent full cycles [–]
- P_{avail}: real available power [kW, MW]
- P_{res}: power bound to reserve [kW, MW]
- P_{aux}: parasitic auxiliary load [kW]
- P_{loss}: loss power [kW]
- P_{cool}: active cooling power [kW]
- S: apparent power [kVA, MVA]

Efficiencies, thermal and kinetic constants.

- η_c, η_d: charge and discharge efficiency [–]
- η_{cell}, η_{DC}, η_{PCS}, η_{trafo}: efficiencies of individual conversion stages [–]
- η_{conv}: combined conversion efficiency [–]
- η_{tot}: total conversion efficiency [–]
- C_{th}: thermal capacity [J/K]
- R_{th}: thermal resistance [K/W]
- λ: self-discharge rate $[\mathrm{s}^{-1}]$
- k: kinetic relaxation constant $[\mathrm{s}^{-1}]$

Market quantities.

- π: energy price [€/MWh]
- ρ: capacity price for reserve holding [€/MW·h]
- α: activation price [€/MWh]

Subscripts follow a uniform convention: *nom* (nominal), *usable, dispatch* (dispatchable), *res* (reserved), *ref* (reference). The subscripts *E* and *P* denote the energy- or power-related reference of a quantity, in particular for SOH; *AC* and *DC* denote the current type; *ch* and *dis* denote charging and discharging processes; *max* and *min* denote upper and lower bounds; *avail* and *bound* denote the kinetic availability status of the charge in the sense of the decomposition in I.4.3. Spatial and temporal indices *x* and *t* denote site and time dependence.

I.2.2 Basic relations

Power describes a point-in-time capability; energy, an integrated content over time. The fundamental relation is

$$E = \int P(t)\ dt$$

In the discrete case,

$$E \approx \sum_i P_i \cdot \Delta t_i$$

This elementary relation remains central for BESS because many misinterpretations arise precisely where nominal power, nominal energy, and the real available operating space are not cleanly separated.

I.2.3 System boundaries

For a BESS at least four system boundaries must be distinguished:

- cell or DC boundary: the electrochemical inventory up to the DC busbar
- conversion boundary: PCS and further conversion stages
- AC plant boundary: transformer, auxiliary systems, and electrical ancillaries
- operational or market-system boundary: in addition, reserve binding, peripherals, restrictions, and time-related operating conditions

The chosen system boundary determines what is meant by energy, power, efficiency, or duration in any given statement.

I.3 Energy balance of the BESS: from nominal energy to dispatchable AC energy

I.3.1 Basic decomposition

The energy of a BESS that is relevant for operation is the energy content dispatchable under boundary conditions; the nominal energy carries that function only under idealised assumptions. A purposive decomposition reads

$$E_{\text{nom}} \rightarrow E_{\text{usable}} \rightarrow E_{\text{reserved}} \rightarrow E_{\text{dispatch,AC}}$$

with

- E_{nom}: nominal energy
- E_{usable}: energy usable within released SOC bounds
- E_{reserved}: energy bound for reserve, safety, or operating discipline
- $E_{\text{dispatch,AC}}$: energy actually dispatchable on the AC side

A simple basic form reads

$$E_{\text{usable}} = E_{\text{nom}} \cdot f_{SOC} \quad \text{with} \quad f_{SOC} = SOC_{max} - SOC_{min}$$

Taking into account the reserve binding E_{res}, the conversion efficiency η_{conv}, and the parasitic AC-side auxiliary load energy E_{aux}, one obtains

$$E_{\text{dispatch,AC}} = (E_{\text{usable}} - E_{\text{res}}) \cdot \eta_{\text{conv}} - E_{\text{aux}}$$

I.3.2 Duration as a derived quantity

Storage duration follows from the ratio of dispatchable energy to available power; it is therefore a derived rather than fixed system property:

$$t = \frac{E_{\text{dispatch,AC}}}{P_{\text{AC}}}$$

A 2 h or 4 h system is one only on a nominal basis. Under reserve binding, auxiliary loads, temperature restrictions, and reduced AC power, the real duration shifts.

I.3.3 Reference values

For large-scale BESS, 1 h, 2 h, and 4 h systems carry at present as the most important reference classes. A 10 MW / 20 MWh system is a typical 2 h system on a nominal basis and serves below as the annex-internal reference example.

The revenue figures referred to several times in this appendix rest on a model-based calculation under the following assumptions: the German day-ahead spot market of a representative market year as the sole revenue source, an AC-side round-trip efficiency of 90 per cent, at most two full cycles per day, perfect foresight of the daily price profile, and no consideration of reserve binding, ageing, or parasitic loads. Under these assumptions, annual revenues in the unrestricted case amount to around €2.1 million, that is, around €210,000 per MW. Under symmetric power restriction from 10 MW to 8 MW, revenues fall by around 11 per cent; under asymmetric restriction of charging power to 6 MW, by around 16 per cent; under a nightly blocking window from 19:00 to 06:00, by around 39 per cent.

The order of magnitude of the three restriction effects shows that available power and available time are the operative core variables of economic operation. The formal effect of these restrictions is taken up in I.5.3 (asymmetric power limits) and I.8.6 (grid-side restrictions and blocking windows), without restating the revenue figures.

I.4 State management and SOC modelling

I.4.1 The state of charge as an operative state variable

The state of charge SOC carries as an operative state variable; a reading as a fill-level indicator falls short. Formally, in a first instance,

$$SOC(t) = \frac{Q(t)}{Q_{\text{ref}}}$$

where $Q(t)$ is the charge available at a time point and Q_{ref} is a reference quantity of the system. Its choice is non-trivial. Depending on viewpoint, it can mean nominal charge, currently estimated usable charge, or age-corrected reference charge. For stationary large-scale storage, the same SOC value carries different operative content under varying C-rates, temperature states, reserve bindings, or ageing states.

I.4.2 Coulomb counting

A basic approach to determining the state of charge is current integration:

$Q_{t+\Delta t} = Q_t + \eta_{\text{c}} I_t \Delta t$ (charging) and $Q_{t+\Delta t} = Q_t - \frac{1}{\eta_{\text{d}}} I_t \Delta t$ (discharging).

In dimensionless form,

$SOC_{t+\Delta t} = SOC_t + \frac{\eta_{\text{c}} I_t \Delta t}{Q_{\text{ref}}}$ and $SOC_{t+\Delta t} = SOC_t - \frac{I_t \Delta t}{\eta_{\text{d}} Q_{\text{ref}}}$

The approach is simple and useful, but reflects drift, inertia, self-discharge, temperature, and ageing only incompletely.

I.4.3 Kinetic extension

A purposive extension lies in the distinction between an immediately available and a delayed-mobilisable charge fraction:

$$Q(t) = Q_{\text{avail}}(t) + Q_{\text{bound}}(t)$$

$\frac{dQ_{\text{avail}}}{dt} = -I(t) + k(Q_{\text{bound}} - Q_{\text{avail}})$ and $\frac{dQ_{\text{bound}}}{dt} = -k(Q_{\text{bound}} - Q_{\text{avail}})$

with k as relaxation or exchange constant. The decomposition makes formally visible that not every nominally available energy content is immediately accessible under every power demand.

I.4.4 SOC window

In operation, the SOC is never used over the full geometric range from 0 to 100 per cent. The relation

$SOC_{min} \leq SOC(t) \leq SOC_{max}$ with $\Delta SOC_{\text{op}} = SOC_{max} - SOC_{min}$

holds. The operative window is the result of technical, safety-related, ageing-related, and service-specific decisions. For BESS, at least three window types are relevant:

- the technical window
- the operating-strategic window
- the situational window

I.4.5 Reserve binding

As soon as a BESS addresses several services, a part of the SOC window is no longer freely dispositional. For reserved energy, in simplified form,

$$SOC_{\text{disp}} = SOC - SOC_{\text{res}}$$

or, asymmetrically,

$SOC_{disp,up} = SOC - SOC_{min,res}$ and $SOC_{\text{disp,down}} = SOC_{max,\text{res}} - SOC$.

This explains why the same nominal capacity carries different free operating spaces depending on the service bundle.

I.4.6 Self-discharge and state drift

For longer holding or backup phases, in addition to active current flow, self-discharge must also be considered:

$$\frac{dQ}{dt} = -I(t) - \lambda Q(t)$$

In the system, the pure self-discharge effect is often less relevant than the combined state drift arising from self-discharge and parasitic peripherals.

I.5 Power capability, C-rate, and operative restriction spaces

I.5.1 Power as the minimum of competing limits

The real power capability of the BESS is state-dependent. It results from competing limits:

$$P_{\text{avail}}(t) = \min\{P_{\text{chem}}(t), P_{\text{therm}}(t), P_{\text{conv}}(t), P_{\text{grid}}(t), P_{\text{ctrl}}(t)\}$$

The narrowest restriction path becomes the effective power limit at each moment.

I.5.2 C-rate and duration

The C-rate describes the ratio of power to usable energy content:

$$C = \frac{P}{E_{\text{usable}}}$$

In the ideal case the inverse holds, $t = \frac{1}{C}$. A 1 h system corresponds approximately to $1C$, a 2 h system to $0{,}5C$, a 4 h system to $0{,}25C$. For real systems, however,

$$t_{\mathrm{real}} = \frac{E_{\mathrm{dispatch,AC}}}{P_{\mathrm{avail}}}$$ is the binding quantity.

I.5.3 Asymmetric power limits

Charging and discharging power are not necessarily symmetric:

$$0 \leq P_{\mathrm{ch}}(t) \leq P_{\mathrm{ch,max}}(t) \quad \text{und} \quad 0 \leq P_{\mathrm{dis}}(t) \leq P_{\mathrm{dis,max}}(t)$$

with, in general,

$$P_{\mathrm{ch,max}}(t) \neq P_{\mathrm{dis,max}}(t)$$

Asymmetries arise from cell-side charging and discharging characteristics, from thermal restrictions, from conversion-side design, and from grid-side connection conditions. Their economic effect is substantial; the annex-internal reference example from I.3.3 shows a revenue reduction of around 16 per cent for an asymmetric restriction of charging power to 6 MW.

I.5.4 Functional power space

The real power capability is purposively described as the set of permissible states:

$$\mathcal{P}(t) = \{P \mid P_{min}(t) \leq P \leq P_{max}(t),\ SOC_{min}(t) \leq SOC(t) \leq SOC_{max}(t),\ T_{min} \leq T(t) \leq T_{max}\}$$

Under grid restrictions,

$$\mathcal{P}_{\mathrm{ra}}(t) = \mathcal{P}(t) \cap \mathcal{P}_{\mathrm{gi}}(t)$$

I.5.5 Ramps

Power is also gradient-limited:

$$\left|\frac{dP}{dt}\right| \leq R_{max} \quad \text{bzw. diskret} \quad |P_{t+\Delta t} - P_t| \leq R_{max}\Delta t$$

Ramp restrictions act on reserve capability, schedule changes, and grid compatibility.

I.6 Temperature, losses, and parasitic systems

I.6.1 Loss layers

The total loss power of the BESS may be written as

$$P_{\mathrm{loss,tot}}(t) = P_{\mathrm{cell}}(t) + P_{\mathrm{ohmic}}(t) + P_{\mathrm{conv}}(t) + P_{\mathrm{therm}}(t) + P_{\mathrm{aux}}(t)$$

with corresponding loss energy

$$E_{\mathrm{loss,tot}} = \int P_{\mathrm{loss,tot}}(t)\ dt$$

I.6.2 Ohmic losses

The elementary relation reads

$$P_{\mathrm{ohmic}} = I^2 R \quad \text{with} \quad E_{\mathrm{ohmic}} = \int I^2(t) R(t)\ dt$$

Since R is temperature- and ageing-dependent, electrical losses couple directly to thermal behaviour and to degradation.

I.6.3 Efficiency chain

For the overall conversion, approximately,

$$\eta_{\mathrm{tot}} = \eta_{\mathrm{cell}} \cdot \eta_{\mathrm{DC}} \cdot \eta_{\mathrm{PCS}} \cdot \eta_{\mathrm{trafo}} \quad \text{and} \quad E_{\mathrm{AC}} = E_{\mathrm{DC,usable}} \cdot \eta_{\mathrm{tot}} - E_{\mathrm{aux}}$$

I.6.4 Thermal model

A simple thermal approximation reads

$$C_{\mathrm{th}} \frac{dT}{dt} = P_{\mathrm{loss,tot}}(t) - \frac{T(t) - T_{\mathrm{amb}}(t)}{R_{\mathrm{th}}} - P_{\mathrm{cool}}(t)$$

with C_{th} as thermal capacity, R_{th} as thermal resistance, and P_{cool} as active cooling power.

I.6.5 Parasitic peripherals

The net power follows from the gross power less the auxiliary loads:

$$P_{\mathrm{net,AC}}(t) = P_{\mathrm{gross,AC}}(t) - P_{\mathrm{aux}}(t) \quad \text{and} \quad E_{\mathrm{net,AC}} = E_{\mathrm{gross,AC}} - \int P_{\mathrm{aux}}(t)\ dt$$

In reserve, stand-by, and holding phases in particular, the relative weight of P_{aux} can become large.

I.6.6 Real duration

With mean parasitic load P_{aux},

$$t_{\mathrm{real}} = \frac{E_{\mathrm{nom}} \cdot f_{SOC} \cdot SOH \cdot \eta_{\mathrm{tot}} - E_{\mathrm{res}}}{P_{\mathrm{load}} + P_{\mathrm{aux}}}$$

I.7 Ageing, degradation, and residual-value logic

I.7.1 Calendar and cyclic ageing

Ageing is composed of calendar and cyclic ageing. At the system level,

$$SOH_E(t) = 1 - \Delta_{\mathrm{cal}}(t) - \Delta_{\mathrm{cyc}}(t) \quad \text{with} \quad \Delta_{\mathrm{cal}} = f_{\mathrm{cal}}(T, SOC_{\mathrm{mean}}, t) \quad \text{and} \quad \Delta_{\mathrm{cyc}} = f_{\mathrm{cyc}}(DoD, C, T, N_{\mathrm{EFC}})$$

The parametric form of the functions f_cal and f_cyc is chemistry-specific. For the lithium-ion family, SEI growth, lithium plating under cold fast-charging, active-material crack formation, and electrolyte decomposition are the dominant drivers; they act with differing weight on SOH_E and SOH_P.
Sodium-ion cells follow structurally similar paths under different temperature and voltage sensitivity. Solid-state cells shift the degradation focus to the electrode–electrolyte interfaces, the mechanical stresses at these interfaces, and the dendritic intrusion of lithium-bearing phases into the solid electrolyte. The thresholds referenced in Chapter 5.2 — separator thickness around 10 µm, solid-electrolyte conductivity above 100 Ωcm², specific charge-transfer resistance below 100 Ωcm², active-material particles below 5 µm — are not reached in the cells demonstrated to date, so the structural equation in the following sections rests on the lithium-ion calibration.

I.7.2 Energy- and power-related SOH

A distinction is to be made between the energy-related and the power-related state of health:

$SOH_E(t) = \frac{E_{\text{ref}}(t)}{E_{\text{ref,0}}}$ and $SOH_P(t) = \frac{P_{\text{ref}}(t)}{P_{\text{ref,0}}}$. In general, $SOH_E(t) \neq SOH_P(t)$

I.7.3 Equivalent full cycles

For equivalent full cycles the relation

$N_{\text{EFC}} = \sum_i \frac{|\Delta E_i|}{2E_{\text{usable,ref}}}$ or, equivavlently $N_{\text{EFC}} = \frac{E_{\text{throughput}}}{2E_{\text{usable,ref}}}$ holds.

I.7.4 Linear first approximation

A linear first approximation reads

$$SOH_E(t) = 1 - a_{\text{cal}}t - a_{\text{cyc}}N_{\text{EFC}}$$

I.7.5 Residual power capability

The real power capability follows approximately as

$$P_{\text{avail}}(t) = P_{\text{nom}} \cdot g_1!\left(SOH_P(t)\right) \cdot g_2!\left(T(t)\right) \cdot g_3!\left(SOC(t)\right) \cdot g_4!\,(\text{restrictions})$$

I.7.6 Ageing costs

For degradation-related value reduction,

$\Delta V_{\text{deg}} = V_{\text{asset}} \cdot \Delta SOH$ and, energy-based $c_{\text{deg}} = \frac{\Delta V_{\text{deg}}}{\Delta E}$

I.7.7 Residual value

The residual value can be described as a function of residual energy, residual power, residual useful life, and future applications:

$$V_{\text{res}}(t) = h!\,(SOH_E(t), SOH_P(t), t_{\text{rem}}, \mathcal{A}_{\text{ftr}})$$

I.8 Economic operation and revenue restrictions

I.8.1 Period contribution

For a periodic profit contribution,

$\Pi_t = R_t - C_t$ with $R_t = R_{t,\text{energy}} + R_{t,\text{reserve}} + R_{t,\text{grid}} + R_{t,\text{other}}$ and $C_t = C_{t,\text{energy}} + C_{t,\text{aux}} + C_{t,\text{deg}} + C_{t,\text{grid}} + C_{t,\text{O\&M}}$

I.8.2 Arbitrage

For energy-related revenues,

$$R_{T,\text{energy}} = \sum_{t\in T}\left(\pi_t^{sell} P_{dis,t} - \pi_t^{buy} P_{ch,t}\right)\Delta t$$

I.8.3 Reserve holding and activation

For reserve revenues,

$$R_{T,\text{reserve}} = \sum_{t\in T} \rho_t P_{res,t}\,\Delta t$$

with an additional activation term

$$R_{T,\text{activation}} = \sum_{t\in T} \alpha_t E_{act,t}.$$

I.8.4 Blocking

As soon as a BESS simultaneously addresses several services — market position, reserve holding, grid-side commitment — each of these services binds a share of the available power. The cumulative power binding carries:

$$P_{market},\ t + P_{reserve},\ t + P_{grid},\ t \leq P_{avail}(t)$$

This binding condition is introduced in the main text under the heading blocking and names the mechanism by which the economic multi-use of a BESS finds its formal limit at the shared power ceiling. An analogous binding holds on the energy side through the SOC-window reservation formulated in I.4.5; together, the power and energy bindings carry the available dispositional space for the revenue architecture from I.8.1.

I.8.5 Losses and auxiliary loads

The energy-side costs include parasitic loads:

$$C_{T,\text{energy}} = \sum_{t\in T} \pi_t^{buy}\left(P_{ch,t} + P_{\text{aux},t}\right)\Delta t$$

I.8.6 Grid restrictions

With an export limit,
$P_{dis,t} \leq P_{\exp,\max,t}$ with a charging limit, $P_{ch,t} \leq P_{\mathrm{imp,max},t}$ and under blocking windows, $P_{dis,t} \leq \chi_t P_{max,t}$, $\chi_t \in \{0,1\}$.
The economic effect of grid-side restrictions is substantial; the annex-internal reference example from I.3.3 shows a revenue reduction of around 39 per cent for a nightly blocking window from 19:00 to 06:00.

I.8.7 Net present value

A simple net-present-value structure reads

$$NPV = -CAPEX_0 + \sum_{t=1}^{T} \frac{\Pi_t}{(1+r)^t} - \sum_k \frac{CAPEX_{\mathrm{re},k}}{(1+r)^{\tau_k}}$$

A calibration of this structure to European market conditions yields concrete orders of magnitude at the LSS segment. For a 10 MW / 20 MWh system, $CAPEX_0$ — following ISEA-RWTH market observation — falls in a range from around €310 to €465 per kWh, in absolute terms between approximately €6.2 and €9.3 million. The annual profit contribution Π_t distributes across the revenue sources decomposed in I.8.1; in the unrestricted merchant case of the reference example introduced in I.3.3, it lay at around €2.1 million annually, and was reduced accordingly under the restriction situations shown there. Reinvestment expenditures $CAPEX_{re,k}$ arise from augmentations that counteract the degradation described in I.7; they act on the net present value through the discounting term.

I.9 Grid-side embedding and systemic boundary conditions

I.9.1 Grid restriction

The grid-permitted power reads

$$P_{\mathrm{grid}}(t) \leq P_{\mathrm{grid,max}}(t)$$

and consequently

$$P_{\mathrm{avail,real}}(t) = \min\{P_{\mathrm{avail}}(t), P_{\mathrm{grid,max}}(t)\}$$

I.9.2 Redispatch

The realisable schedule is

$P_{\mathrm{real}}(t) = P_{\mathrm{sched}}(t) + \Delta P_{\mathrm{redispatch}}(t)$ with, generally $\Delta P_{\mathrm{redispatch}}(t) \neq 0$.

Under congestion conditions, a difference exists between the market-economically optimal schedule and the grid-side permitted operation; this difference is the reference quantity of the redispatch measure. Three empirical magnitudes mark the operative reach.

First, the annual congestion-management volume: according to Federal Network Agency data, 32,772 GWh in 2022, 34,297 GWh in 2023, and 30,304 GWh in 2024. Second, the corresponding system costs: around €4.2 billion in 2022, €3.335 billion in 2023, and €2.776 billion in 2024. Third — on a longer horizon — the model-derived reduction potential through dynamic grid charges. Aurora Energy Research (April 2026, consortium study for EnBW, Engie, LEAG, Onyx Power, RWE, Statkraft, Trianel, Uniper, and Vattenfall) models the effect of regionally differentiated dynamic grid charges for the analysis years 2029 and 2037 — that is, after the entry into force of the grid-charge reform planned for 31 December 2028 within the BNetzA AgNes process. For 2037, the redispatch volume in the base scenario without dynamic grid charges lies at 19 TWh; with dynamic grid charges of $$10 €/MWh, a reduction between 5 and 63 per cent is modelled, while at the same time the price signal triggers a generation shift of 54 TWh that exceeds the redispatch volume many times over and is associated with systemic side effects.

The three quantities carry differing evidential weight. Volume and costs reflect the realised intervention need and are shaped in the short term by weather, fuel prices, and grid-expansion progress. The reduction potential stands under model qualification; its effect depends on the still-open design of the grid-charge reform within the AgNes process.

I.9.3 Uniform price, local restriction

For two sites a and b in the same price zone,

$\pi_a(t) = \pi_b(t) = \pi(t)$, but $P_{\mathrm{grid,max},a}(t) \neq P_{\mathrm{grid,max},b}(t)$, hence $\Pi_a \neq \Pi_b$

I.9.4 Reactive and apparent power

For the apparent-power space,

$S^2 = P^2 + Q^2$ and hence $P^2 + Q^2 \leq S_{max}^2$

When reactive power is supplied, the residual active-power space is reduced.

I.9.5 Ramps

Grid-side ramp limitations read

$\left|\frac{dP}{dt}\right| \leq R_{max}$ or $|P_{t+\Delta t} - P_t| \leq R_{max}\Delta t$

I.9.6 Site-dependent flexibility

At one and the same installation, not every technically available power corresponds to a power that is also dispatchable on the grid side. The site-specific difference between technical availability and the callable flexibility share can be described through a flexibility factor $\phi_x(t)$:

$0 \leq \phi_x(t) \leq 1$ with $P_{\mathrm{flex},x}(t) = \phi_x(t) \cdot P_{\mathrm{avail}}(t)$

$\phi_x(t)$ carries the combined effect of connection capacity, local congestion situation, voltage and reactive-power requirements, and time-varying grid restrictions at the site x. In relation to the site differentiation from I.9.3, $\phi_x(t)$ makes visible that even two nominally identical BESS in the same price zone can carry different dispositional power shares.

I.10 Verification, evidence, and evaluation logic

I.10.1 Stages of system verification

Verification regimes follow a stepped logic:

$$V_1 \to V_2 \to V_3 \to V_4$$

with

- V_1: component verification
- V_2: integration verification
- V_3: system verification
- V_4: operational verification

I.10.2 Parameterisation

A parameter set Θ is release-capable when

$$\Theta \in \mathcal{F}_{\mathrm{sf}} \cap \mathcal{F}_{\mathrm{oe}} \cap \mathcal{F}_{\mathrm{cnrc}}$$

where $\mathcal{F}_{\mathrm{sf}}$, $\mathcal{F}_{\mathrm{oe}}$, and $\mathcal{F}_{\mathrm{cnrc}}$ denote the safety-related, operative-economic, and contractually-normatively required feasibility regions.

I.10.3 Evidence quantities

The evidential weight of a quantity X depends on system boundary, time reference, and operating condition:

$$X = X(\mathcal{G}, t, \mathcal{B})$$

I.10.4 Function-related availability

For a function f over the period T,

$A_f(T) = \frac{t_{\mathrm{avail},f}}{T}$ with, in general $A_{f_1}(T) \neq A_{f_2}(T)$

I.10.5 Test space

A representative test space reads

$\mathcal{T} = \{(SOC_i, T_j, P_k, M_l, R_m)\}$ with states of charge, temperatures, power levels, operating modes, and restriction situations.

I.10.6 Demonstrability through evidence continuity

The demonstrability N of a BESS follows as a function of three input quantities, which in the main text are jointly carried under the heading of evidence continuity:

$$N(t) = g\big(D(t), Q(t), C(t)\big)$$

with $D(t)$ as data continuity in the narrower sense (gap-free, time-synchronous capture of the relevant measurement and state quantities), $Q(t)$ as data quality (accuracy, resolution, freedom from drift, calibrated traceability of the sensor system), and $C(t)$ as consistency (correspondence of the quantities across the system boundaries from I.2.3, in particular between the cell, conversion, and AC plant boundary). Evidence continuity carries not the individual measurement datum but the lifetime-spanning relation of capture, quality assurance, and consistency check. It is the precondition of the function-related availability from I.10.4 and the reference quantity of the intervention logic from I.10.7.

I.10.7 Interventions and the new evidence space

For structural changes,

$\mathcal{N}_{new} = \Phi(\mathcal{N}_{old}, \Delta S)$ with ΔS as a structural intervention into the system.

I.10.8 KPI structure

Five KPI classes are purposive:

- energetic KPIs
- power-related KPIs
- state-related KPIs
- reliability-related KPIs
- lifetime-related KPIs

Appendix II – Definitions of load-bearing concepts

Appendix II consolidates the load-bearing concepts of the work as definitions. It is not a glossary in the comprehensive sense — acronyms and abbreviations are listed in Appendix III. Each entry condenses the use that has been established for the concept across the main text and indicates the chapter at which the concept first carries the argument.

II.1 — System controllability

The capacity to keep a BESS as an infrastructure object in legible states across its lifetime — technically, operationally, and institutionally. System controllability comprises the active stewardship of the permitted state space, the coordination of the technical layers, the consistent maintenance of evidence, the governance of the contractual interfaces, and the preservation of connectivity to changing regulatory and market regimes. (cf. Executive Summary; § 1.3)

II.2 — System body

The architecture of a BESS as a coupled assembly of electrochemical inventory, conversion chain, auxiliary systems, state management, evidence and safety layer, contractual embedding, and grid-side connection structure. The *system body* carries the asset across its seven layers as a single coherent operative unit; system controllability acts on the interplay of these layers rather than on any layer in isolation. (cf. § 1.2)

II.3 — Connection capability

Lifecycle connectivity denotes the capacity of a BESS to remain effective under shifting regulatory, market, and technical regimes without structural rupture. It calls for an architecture whose interfaces, data structures, and contractual forms keep later adaptation open without putting the asset at risk. (cf. § 1.3)

Technical connection capability denotes the capacity of a project to emerge, under real conditions of grid and infrastructure scarcity, as a connectable system that subsequently carries the demands of operation. It covers site, connecting line, switchgear bay availability, substation, protection concept, route planning, and the underlying technical evidence. (cf. § 3.1)

II.4 — Project maturity

The multi-dimensional readiness of a BESS project, drawing together site control, connection commitment, technical concept, permitting status, and applicant capability. Project maturity is the precondition under which a project can be prioritised robustly within the competition for scarce connection resources. (cf. § 4.1)

II.5 — Maturity-tier process

The federally uniform assessment procedure of the four German transmission system operators for the prioritisation of connection requests along the dimensions of site control, technical concept, applicant capability, and grid and system benefit. The procedure entered into force on 1 April 2026 and acts as a quality filter within the connection regime. (cf. § 2.1)

II.6 — System-supportive contribution

The property of a BESS, by virtue of its location, design, and operating mode, of contributing to the functional capacity of the overall system. The contribution becomes effective across load-flow control, frequency holding, voltage holding, security of supply, and resilience, and stands in a differentiated relation to the narrower grid-supportive contribution. (cf. § 2.3)

II.7 — Grid-supportive contribution

The property of a BESS, in a specific grid context, of supplying grid-operational functions such as congestion relief, voltage holding, or reactive-power provision. The grid-supportive contribution is more narrowly defined than the system-supportive contribution and refers to the immediate area of effect of the grid into which the installation injects. (cf. § 2.3)

II.8 — Permitted state space

The set of permissible operating states of a BESS, spanned by state of charge, state of health, C-rate, temperature, reserve commitment, and safety-related limits. The permitted state space is the reference surface of every steering intervention; the asset leaves its controllability the moment that real operating points fall outside the permitted region or are carried through its boundary zone without evidence. (cf. § 8.1)

II.9 — AC/DC system boundary

The dividing reference line between the electrochemical inventory with its DC bus on the one side and the conversion chain with its AC-side connection on the other. The location of this boundary determines whether efficiencies, capacity figures, and availability indicators relate to the cell inventory or to the AC terminal, and it is the precondition of clean power and energy balances. (cf. § 5.4)

II.10 — Functional availability

The property of a BESS, within a defined time window, of actually delivering the contractually committed services — beyond purely technical availability, which only measures the operational readiness of the installation. Functional availability folds in reserve commitments, SOC policy, thermal restrictions, and contractual interfaces. (cf. § 11.4)

II.11 — Blocking

The mechanism by which the simultaneous addressing of several services — market position, reserve holding, grid-side commitment — meets its formal limit at the shared power ceiling and SOC binding of the BESS. Blocking makes visible that multi-use is a question of available dispositional headroom, not a contractual question alone. (cf. § 10.3)

II.12 — Evidence continuity

The property of a BESS that the measurement, state, and event data accruing across its lifetime are carried gap-free, quality-assured, and consistently. Evidence continuity does not rest on the individual datum; it carries the lifetime-spanning relation between recording, quality assurance, and consistency checking. (cf. § 1.1; § 12.2)

II.13 — Grid forming

The operating mode of a grid-connected inverter in which the inverter sets its own voltage and frequency reference and thereby contributes to the formation of the grid itself. Grid-forming-capable BESS carry inertial, black-start, and island-network functions and are becoming, in increasingly converter-dominated systems, a precondition of voltage and frequency stability. (cf. § 2.2; § 6.2)

II.14 — Grid following

The operating mode of a grid-connected inverter in which the inverter synchronises through a phase-locked loop to the existing grid voltage and impresses active and reactive power as a current source. The mode requires a sufficiently stable grid voltage as reference and reaches its systemic limit in weak grid areas. (cf. § 6.2)

II.15 — Economic manageability

The continuous capacity to carry a BESS into a revenue architecture without consuming the remaining dimensions of system controllability. Economic manageability narrows controllability to the area of ongoing economic operation and carries where market, reserve, and grid services together form the objective function. (cf. § 15.1)

II.16 — Precautionary window

The safety-related margin between the soft, operationally set boundaries and the hard, physically anchored boundaries of the permitted state space. Within the precautionary window the system remains controllable, so that interventions can take effect before protective trips engage. (cf. § 13.2)

II.17 — Lifecycle capability

The property of a BESS of holding its place, in every life phase — planning, construction, operation, augmentation, decommissioning, material recovery — as a managed asset. Lifecycle capability requires the continuous connection between technical substance, evidence continuity, and governance across the phase transitions. (cf. § 4.2)

Appendix III – Acronys and abbreviations

Appendix III lists the acronyms and abbreviations used across the work. Entries are grouped by domain — technical, information technology and communications, chemistry and materials, market and regulation, standards and institutions, units and physical quantities — and arranged alphabetically within each group. Where a German regulatory body, statute, or technical term has no settled English short form, the German abbreviation is retained and an English equivalent is given alongside.

Acronym	Name	Description
Technical - system, operation, and control		
AC	Alternating current	The grid-side current form into which the DC battery output is converted by the PCS
AIS	Air-insulated switchgear	The lower-cost, more space-intensive switchgear form; typical where the site offers ample footprint
BESS	Battery Energy Storage System	The subject of the compendium; a coupled electro-technical infrastructure object
BMS	Battery Management System	Protects the electrochemical inventory; monitors cell voltage, cell temperature, and balancing
BoL	Beginning of Life	Reference point of the contractually committed performance at the start of the duty period
C-rate	Charge/discharge rate	The ratio of power to usable energy; 1 C corresponds to a full discharge in one hour
COD	Commercial Operation Date	The transition from EPC responsibility to operative asset management
DC	Direct current	The internal battery current form between the cells and the PCS input
DGA	Dissolved Gas Analysis	Diagnostic procedure for the early indication of thermal and electrical faults in transformers and auxiliary plant
DoD	Depth of Discharge	Driver of cyclic ageing; deep cycles age disproportionately
EFC	Equivalent Full Cycles	Normalised cycle count for the comparability of part-cycles of differing depth
EMS	Energy Management System	Translates market, reserve, and grid demands into set-points for the BMS and the PCS
EoL	End of Life	End of primary use; trigger for second-life deployment, retrofit, or decommissioning
EPC	Engineering, Procurement, Construction	Integrated delivery form under a single general contractor with a fixed performance commitment
EPCM	Engineering, Procurement, Construction Management	Project-management variant without integrated construction-execution responsibility
GIS	Gas-insulated switchgear	Compact switchgear form using SF_6 or alternative gases; small footprint
HSS	Home Storage System	Market segment up to 30 kWh under the MaStR classification; carries around 80 per cent of the installed base in Germany
HV	High voltage	From 60 kV upwards; the transformation and connection level of large-scale BESS
HVAC	Heating, Ventilation and Air Conditioning	Holds the temperature window; a material parasitic load, particularly in heat periods
ISS	Industrial Storage System	Market segment between 30 kWh and 1,000 kWh under the MaStR classification; combines self-consumption optimisation and load management
KPI	Key Performance Indicator	Operative steering variable for performance tracking, maintenance, and asset management
LSS	Large Storage System	Market segment from 1,000 kWh upwards under the MaStR classification; the carrying use case of the compendium
MV	Medium voltage	1 to 60 kV; the typical connection level of mid-sized BESS projects
O&M	Operation and Maintenance	The ongoing OPEX category, comprising time-based, condition-based, and predictive maintenance
OEM	Original Equipment Manufacturer	Manufacturer of cells, modules, or full systems; central addressee of warranty obligations
PCS	Power Conversion System	The grid-side power electronics; the operative heart of the system coupling that mediates between the DC inventory and AC grid behaviour
PLL	Phase-Locked Loop	The synchronisation mechanism of a grid-connected inverter in grid-following mode; presupposes a sufficiently stable grid voltage
PV	Photovoltaics	Most common co-location partner; shapes the charging windows and the SOC policy of the storage
SCADA	Supervisory Control and Data Acquisition	The supervisory and command level above the field devices
SCR	Short-Circuit Ratio	The ratio of grid short-circuit power to installation rating; a measure of grid strength at the connection point
SOC	State of Charge	The scarcest resource of operation; a strategic state variable, not merely a measured one
SOH	State of Health	Aggregated condition relative to the as-new state; the lead indicator for asset management
SOHE	Energy-related state of health	Shapes duration and arbitrage potential; decisive for tolling and spot-driven services
SOHP	Power-related state of health	Shapes ramping and reserve capability; can age more rapidly than SOHE
TDD	Total Demand Distortion	Measure of current harmonics relative to the maximum demand load; relevant for IEEE-519-compliant grid feed-in
THD	Total Harmonic Distortion	Measure of voltage or current harmonics relative to the fundamental; quality criterion of the conversion chain
VPP	Virtual Power Plant	Aggregation of distributed flexibility sources into a jointly marketed unit
Information technology and communications		
DPP	Digital Product Passport	Regulatorily anchored digital identity; mandatory from 18 August 2027 for stationary storage above 2 kWh
IT	Information Technology	Conventional enterprise IT, sitting above the plant boundary; to be separated from the response-critical OT layer
KRITIS	Kritische Infrastrukturen (German regulatory framework for systemically critical installations)	Imposes elevated cybersecurity and reporting obligations; relevant for large grid-critical BESS
MQTT	Message Queuing Telemetry Transport	Lightweight protocol for fleet-near telemetry between edge and cloud
NIS2	Network and Information Security Directive 2	EU directive raising cyber-resilience requirements; tightens reporting and protection obligations for affected BESS operators
OPC UA	Open Platform Communications Unified Architecture	Secure, interoperable OT communication across trades and manufacturers
OT	Operational Technology	The plant-near control layer; locally autonomous if higher layers are disturbed
VOC	Volatile Organic Compounds	Indicator of the gaseous lead-in phase preceding a thermal runaway

Acronym	Name	Description
Chemistry and materials		
GaN	Gallium Nitride	Power-electronics semiconductor of newer PCS generations; high switching frequencies at low losses
LFP	Lithium iron phosphate ($LiFePO_4$)	The dominant chemistry in the stationary segment; high thermal robustness, lower energy density
Li-Ion	Lithium-ion	Overarching technology family of all currently market-relevant large-scale storage
LMFP	Lithium manganese iron phosphate	Extension path of the LFP family with higher energy density; at present the first practically deployment-ready extension path
LTO	Lithium titanate ($Li_4Ti_5O_{12}$)	Very high cycle stability at low energy density; deployed in niche applications
Na-Ion	Sodium-ion	Emerging path with reduced raw-material criticality; industrial maturity is being built
NCA	Nickel-cobalt-aluminium oxide	High-energy cathode chemistry; predominantly mobile-deployed and raw-material-critical
NMC	Nickel-manganese-cobalt oxide	High-energy chemistry; in the stationary segment increasingly displaced by LFP
SEI	Solid Electrolyte Interphase	Central driver of calendar ageing; temperature- and SOC-dependent boundary layer at the anode
SiC	Silicon Carbide	Power-electronics semiconductor of modern PCS; higher efficiency and switching speed than silicon IGBT
SSB	Solid-State Battery	Potential safety and energy-density step-change; stationary industrialisation remains open
Market, regulation, and economics		
ADR	Accord européen relatif au transport international des marchandises Dangereuses par Route	European agreement on the international carriage of dangerous goods by road; UN 3480/3481/3536 cover lithium-ion batteries
aFRR	Automatic Frequency Restoration Reserve	Automatically called following FCR; capacity and energy prices marketed separately
AgNes	Allgemeine Netzentgeltsystematik — German general grid-charge framework	BNetzA decision procedure for the redrafting of the grid-charge system, with planned effect from 31 December 2028
BGH	Bundesgerichtshof (Federal Court of Justice)	Highest-instance clarification of regulatory questions of principle; the EnVR 1/24 ruling on the building-cost contribution obligation
BImSchV	Bundes-Immissionsschutzverordnung (Federal Immission Control Ordinance)	Sub-statutory regulatory framework; the 12th BImSchV (Major Accident Ordinance, Störfallverordnung) applies at hazardous-substance BESS sites
BKZ	Baukostenzuschuss — building-cost contribution	One-off grid-connection cost contribution; for storage made binding by the BGH ruling of July 2025
BNetzA	Federal Network Agency (Bundesnetzagentur)	Regulatory authority for grids, grid charges, and market-organisational decisions
BSI	Bundesamt für Sicherheit in der Informationstechnik (Federal Office for Information Security)	Sets minimum standards and audit regimes for KRITIS installations
CACM	Capacity Allocation and Congestion Management	EU Guideline 2015/1222 on cross-border capacity allocation and congestion management within the internal market
CAPEX	Capital Expenditure	One-off investment cost; in BESS increasingly shaped by balance-of-system
CRMA	Critical Raw Materials Act	EU Regulation 2024/1252; targets for raw-material sovereignty by 2030 (10 per cent extraction, 40 per cent processing, 25 per cent recycling)
EBGL	Electricity Balancing Guideline	EU Guideline 2017/2195; harmonises the balancing markets (FCR, aFRR, mFRR) within the European interconnection
EEG	Erneuerbare-Energien-Gesetz (Renewable Energy Sources Act)	Support framework for renewable generation; §§ 39j ff. govern innovation tenders for storage with direct renewable coupling
EMMES	European Market Monitor on Energy Storage (LCP Delta)	Half-yearly market observation of European storage build-out; around 100 GW of installed capacity at 2025
EnVR	Energie-Verfahren Rechtsbeschwerde — BGH case-reference prefix for energy-industry appeal proceedings	Reference: EnVR 24/23, BGH ruling of 9 July 2025 on the building-cost contribution obligation for stationary storage
EnWG	Energiewirtschaftsgesetz (Energy Industry Act)	Federal statutory framework for grid-bound energy supply; § 3 No. 25 and § 118 (6) are BESS-relevant
EU	European Union	Sets the regulatory ground frame through the Battery Regulation 2023/1542, the CRMA, and the NZIA
FCR	Frequency Containment Reserve	Fast automatic reserve; full activation within 30 seconds
FFR	Fast Frequency Reserve	Sub-second frequency response; predominantly deployed in island grids
GPKE	Geschäftsprozesse zur Kundenbelieferung mit Elektrizität — uniform business processes for electricity supply to customers	Federally uniform process rules for supplier change and supply; flanks MaBiS and WiM
IPCEI	Important Projects of Common European Interest	EU state-aid framework for key industries; carries several funding lines for battery production and energy storage
IRA	Inflation Reduction Act	US industrial policy enacted in 2022; drives the North American cell production build-out through tax credits
LCOE	Levelised Cost of Electricity	Comparative measure across generation forms over the lifetime
LCOS	Levelised Cost of Storage	Counterpart to LCOE for storage; includes cycle and degradation costs
MaBiS	Marktregeln für die Bilanzkreisabrechnung Strom — market rules for electricity balancing-group settlement	Federally uniform settlement rules for balancing groups; precondition for clean storage operation within revenue architectures
MARI	Manually Activated Reserves Initiative	The European platform for the activation and settlement of manually activated balancing reserve
MaStR	Marktstammdatenregister (Market Master Data Register)	BNetzA register of all electricity-generating, -consuming, and -storing installations; mandatory registration for BESS
mFRR	Manual Frequency Restoration Reserve	Manually or partly automatically called following aFRR deployment
NAV	Niederspannungsanschlussverordnung (Low Voltage Connection Ordinance)	Regulation on the general conditions for grid connection; § 11 governs the connection-relevant requirements of small BESS
NEP	Netzentwicklungsplan (Network Development Plan)	Two-yearly framework document for grid expansion produced by the German TSOs
NPV	Net Present Value	Discounted present value of all cash flows over the duty period
NVP	Netzverknüpfungspunkt (grid connection point)	The physical and contractual boundary between the installation and the public grid
NZIA	Net-Zero Industry Act	EU industrial-policy counterpart to the IRA; target of 40 per cent local capacity by 2030
OPEX	Operational Expenditure	Ongoing operating cost, including O&M, grid charges, and augmentation provisions
PICASSO	Platform for the International Coordination of Automated Frequency Restoration	The European platform for the activation of automatically activated secondary balancing reserve
PPA	Power Purchase Agreement	Long-term bilateral supply contract; stabilises revenue flows against merchant risk
ROI	Return on Investment	Coarse comparative measure against alternative asset classes; complements NPV and LCOS
StromNEV	Stromnetzentgeltverordnung (Electricity Grid Charges Ordinance)	Regulation on the charges for access to electricity supply networks; § 19 contains the special provisions for atypical grid users
TSO	Transmission System Operator	The four German transmission system operators — 50Hertz, Amprion, TenneT, TransnetBW — operate the extra-high-voltage level
VDE-AR-N	VDE Application Rules — German technical connection rules	4110 (medium voltage), 4120 (high voltage), 4130 (extra-high voltage), 4140 (low-voltage installations)
VNB	Verteilnetzbetreiber (distribution system operator)	Medium- and low-voltage operators; regionally organised, around 850 entities in Germany
WiM	Wechselprozesse im Messwesen — uniform processes for metering operations	Federally uniform processes for metering-point operation and metering; flanks MaBiS and GPKE

Acronym	Name	Description
Standards, organisiations, and institutes		
BAM	Bundesanstalt für Materialforschung und -prüfung (Federal Institute for Materials Research and Testing), Berlin	Research on material safety, test methods, and battery traceability
BVES	Bundesverband Energiespeicher Systeme (German Energy Storage Association)	German storage-industry body; publishes technical guidelines and represents regulatory interests
CIGRE	Conseil International des Grands Réseaux Électriques (International Council on Large Electric Systems), Paris	International working bodies; Technical Brochures on large-system engineering
Fraunhofer FFB	Fraunhofer Research Institution for Battery Cell Production, Münster	Public pilot lines and the scaling of European cell production
Fraunhofer ISE	Fraunhofer Institute for Solar Energy Systems, Freiburg	Photovoltaic and storage-system research; work on power-station-near BESS sites
Fraunhofer ISI	Fraunhofer Institute for Systems and Innovation Research, Karlsruhe	Battery economics, roadmaps, and market analyses; reference author on chemistry pathways
Fraunhofer IZM	Fraunhofer Institute for Reliability and Microintegration, Berlin	Digital Product Passport, electronics integration, and reliability research
IEA	International Energy Agency, Paris	Global energy and battery-market statistics; annual market and outlook reports
IEC	International Electrotechnical Commission, Geneva	International standardisation of electrical installations; among others, IEC 61850 and the IEC 62933 family
IEEE	Institute of Electrical and Electronics Engineers	Scientific society with its own standards and technical publications
IRENA	International Renewable Energy Agency	Policy analysis and scenarios on renewable energy and storage build-out
ISEA	Institut für Stromrichtertechnik und Elektrische Antriebe (Institute for Power Electronics and Electrical Drives), RWTH Aachen	Chair with a focus on storage technology; publisher of the Battery Charts and the Battery Revenue Index
ISO	International Organization for Standardization, Geneva	Management standards including ISO 55000 (asset management) and ISO 27001 (information security)
M5BAT	Modular Multi-Megawatt Multi-Technology Medium Voltage Battery Storage	Research storage at RWTH Aachen; operative reference for cross-market operation between FCR and intraday
NFPA	National Fire Protection Association, Quincy	US fire-protection standards; NFPA 855 for stationary energy storage systems
NREL	National Renewable Energy Laboratory, Golden, Colorado	US research laboratory; reference source for BESS cost projections and system studies
PEM	Production Engineering of E-Mobility Components, RWTH Aachen	Chair with a focus on battery production engineering and scaling
UL	Underwriters Laboratories, Northbrook	US testing institute; UL 9540A as the reference procedure for propagation tests
VDMA	Verband Deutscher Maschinen- und Anlagenbau (German Mechanical Engineering Industry Association), Frankfurt am Main	Roadmaps for battery production engineering and industrial manufacturing equipment
VdS	VdS Schadenverhütung GmbH, Cologne	German testing institute of the property insurers; VdS 3103 as the reference for lithium-ion storage safety
Units and physical quantities		
A	Ampere — unit of electric current	SI base unit; appears in the BESS context for cell current and switchgear sizing
Ah	Ampere-hour — unit of electric charge (1 Ah = 3,600 C)	Practical unit for cell and module capacity; the basis of coulomb counting
GW	Gigawatt (10^9 W)	Typical magnitude of connection competition and NEP scenarios in Germany
GWh	Gigawatt-hour	Market-wide build-out and demand quantities; congestion-management volumes lie in the TWh range
K	Kelvin — unit of thermodynamic temperature	Basis of the Arrhenius relation; ageing rate roughly doubles every 10 K
kV	Kilovolt	Voltage level of the connection body; from 10 kV to 380 kV depending on project size
kW	Kilowatt	Power magnitude of smaller industrial and commercial storage
kWh	Kilowatt-hour	Specific cost basis (USD/kWh, EUR/kWh) for the comparison of storage concepts
MVA	Megavolt-ampere — unit of apparent power	Transformer rated apparent power; typically 40 to 250 MVA in large-scale storage projects
MVAr	Megavolt-ampere reactive — unit of reactive power	Indicator of voltage support and reactive-power provision at the connection point
MW	Megawatt	Project power magnitude of large stationary storage; from 1 MW to well above 100 MW
MWh	Megawatt-hour	Project energy magnitude; the product of power and duration (e.g., 10 MW × 2 h = 20 MWh)
TWh	Terawatt-hour (10^{12} Wh)	System and redispatch volume at national level (Germany 2022: 33 TWh)
V	Volt — unit of electric voltage	SI unit; cell nominal voltages typically 3.2 V (LFP) or 3.6–3.7 V (NMC)
W	Watt — unit of active power	SI unit of power output; reference quantity for efficiencies and parasitic loads
°C	Degree Celsius	Critical operating windows typically 15–35 °C; Arrhenius-relevant ageing drivers
Ω	Ohm — unit of electrical resistance	Internal resistance as an early indicator of cell ageing; rises with cycles and temperature

Appendix IV - References

Appendix IV brings together the sources that carry the compendium. Entries are arranged by sub-chapter function and, within each sub-chapter, alphabetically by first author or institution. Blanket collective references to bodies of material have been resolved into concretely citable sources. Source titles, publishers, and places are retained in their original form.

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